

SPATIALAB: CAN VISION–LANGUAGE MODELS PERFORM SPATIAL REASONING IN THE WILD?

Anonymous authors
Anonymous submission

ABSTRACT

We make spatial reasoning the central object of evaluation for contemporary vision–language models (VLMs). Although spatial reasoning is fundamental to human cognition, existing VLM benchmarks often rely on synthetic or LLM-generated settings, narrow task designs, and puzzle-like scenes that understate the visual noise, viewpoint variation, and relational diversity of real-world environments. Our first contribution is therefore extbfSPATIALAB, a benchmark for measuring VLM spatial reasoning in realistic, unconstrained visual contexts. As a benchmark contribution, SPATIALAB contains 1,400 visual question–answer pairs organized around six core categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*. Each category is further decomposed into five subcategories, yielding 30 distinct task types, with at least 25 questions per subcategory and at least 200 questions per main category. This design supports two complementary evaluation protocols: multiple-choice evaluation for controlled recognition and open-ended evaluation for generative spatial reasoning. As an empirical contribution, we evaluate a broad set of state-of-the-art VLMs, including open-source, closed-source, reasoning-focused, and spatially specialized models, and compare them with human performance. The results reveal a large and persistent spatial reasoning gap: in the multiple-choice setting, InternVL3.5-72B reaches 54.93% accuracy while humans reach 87.57%; in the open-ended setting, all models drop by roughly 10–25%, with GPT-5-mini achieving the best model score at 40.93% compared with 64.93% for humans. These findings localize failures in complex spatial relationships, depth perception, navigation, and 3D geometry. Together, the benchmark, dual-format evaluation, and diagnostic results make SPATIALAB a reviewer-facing testbed for studying where VLM spatial reasoning succeeds, where it fails, and which future methods may close the gap to human-aligned spatial understanding. SPATIALAB will be released upon acceptance.

1 INTRODUCTION

Spatial reasoning is both a core human cognitive capacity and a practical requirement for embodied artificial agents. It involves understanding spatial layouts, imagining and manipulating objects, and navigating environments, and it underpins applications in robotics, autonomous driving, augmented and virtual reality, geospatial analytics, computer graphics, human–computer interaction, and education (Huang et al., 2024b; Qi et al., 2024; 2025). For agents that must perceive and act in dynamic scenes, spatial competence requires more than object recognition: it requires geometric relations, depth cues, occlusion patterns, scale changes, and agent-centric perspectives to be interpreted under visual clutter and structural complexity (Bar-Anan et al., 2006; Cai et al., 2025; Trope & Liberman, 2010; Ramalho et al., 2018). Recent vision–language models (VLMs) have made strong progress in multimodal representation and language grounding, but their spatial judgments remain fragile in realistic environments (Chen et al., 2025; Kosoy et al., 2025; Pothiraj et al., 2025). This paper contributes a benchmark and analysis designed to make that fragility explicit.

The evaluation gap is structural: many existing spatial benchmarks are narrow, simplified, or visually sanitized. They commonly emphasize elementary tasks such as binary spatial relations, coarse depth categories, or schematic navigation cues, often in synthetic or simplified settings that reduce clutter and homogenize object appearance (Du et al., 2024; Jia et al., 2025; Szymanska et al., 2024; Shiri et al., 2024; Kamath et al., 2023; Fu et al., 2024; Song et al., 2024; Yang et al., 2024; Qi et al.,

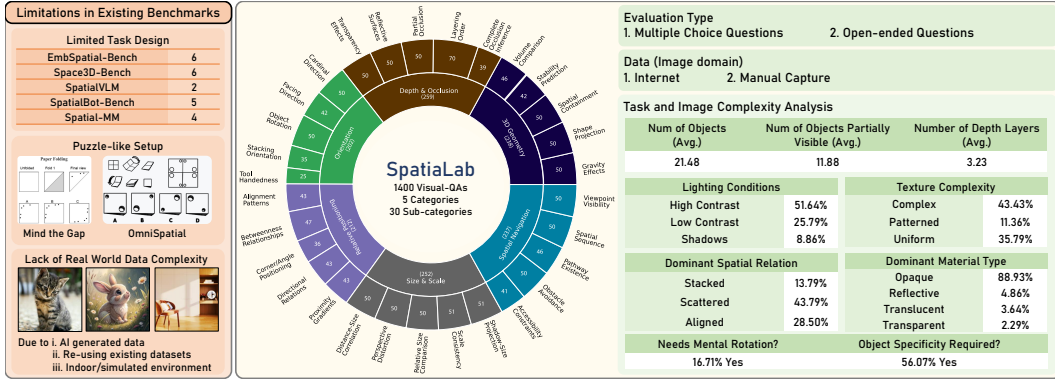


Figure 1: **Contribution overview of SPATIALLAB.** The benchmark targets limitations of prior datasets (left), contributes 1,400 visual QA pairs spanning 5 categories and 30 subcategories (center), and supports reviewer-facing evaluation through both multiple-choice and open-ended tasks. The images and questions stress diverse object counts, scene layers, lighting, textures, relations, and materials (right).

2025; Cai et al., 2024). These choices are useful for controlled diagnosis, but they can also lower the perceptual and reasoning burden, creating apparent saturation while masking distribution-shift failures. Important spatial capabilities—including occlusion inference, scale-consistent judgments across viewpoints, and planning under partial observability—remain under-sampled (Kosoy et al., 2025; Pothiraj et al., 2025; Chen et al., 2025; Cai et al., 2025). Consequently, strong benchmark scores can coexist with failures on integrated, multi-step spatial inference, which limits safe deployment in real-world settings where spatial errors have practical consequences.

A useful benchmark for this problem must make the full contribution space visible. It should cover the core axes of spatial reasoning—relative position, depth and occlusion, orientation, size and scale, navigation, and 3D geometry—because human spatial cognition integrates these dimensions under noise and ambiguity (Driess et al., 2023; Brohan et al., 2023; Zitkovich et al., 2023; Collaboration et al., 2023; Yuan et al., 2024; Qi et al., 2025). It should also evaluate both selection and generation, compare models against human performance, and use real-world scenes that reveal failures hidden by simulator or puzzle-style datasets. These principles define the methodological framing of our work: we treat benchmark construction, dual-format evaluation, human comparison, and error diagnosis as linked contributions rather than isolated components.

We introduce **SPATIALLAB**, a benchmark for assessing VLM spatial reasoning in realistic, unconstrained visual contexts, as summarized in Figure 1. The benchmark contributes a balanced visual QA suite with a broad spatial taxonomy, photographic diversity, complex relational queries, and tasks that require integrating depth, perspective, and 3D structure. It further contributes two evaluation views—multiple-choice and open-ended—so that models can be tested both for answer selection and for generative spatial reasoning. We evaluate a wide collection of state-of-the-art VLMs, including open-source, closed-source, reasoning-focused, and spatially specialized systems; compare them to human performance; and provide diagnostic error analyses. The contributions are:

- Benchmark contribution:** We introduce **SPATIALLAB**, a large-scale benchmark of 1,400 visual question-answer pairs spanning six principal categories and thirty task types in realistic, unconstrained contexts. With at least 200 questions per category and 25 per subcategory, it ensures balanced coverage and supports both multiple-choice (SPATIALLAB-MCQ) and open-ended (SPATIALLAB-OPEN) evaluation.
- Evaluation contribution:** We perform a comprehensive evaluation of 25+ state-of-the-art VLMs—open-source, closed-source, reasoning-oriented, and spatially specialized reasoning models—against human baselines, exposing a consistent and significant gap in spatial reasoning.
- Diagnostic contribution:** We provide in-depth error analyses and diagnostic studies that identify systematic failure modes in depth perception, navigation, occlusion, and 3D geome-

Table 1: Reviewer-facing comparison of spatial reasoning benchmarks, highlighting how SPATIALAB differs in real-world coverage, taxonomy, and evaluation format.

Dataset	E.	T.C.	Data				QAs			Venue/ Year	Eval			A.L.	Md.	Best Baseline	B.B.P.
			Domain	Source	A.	Size	Count	Df.	Cmpl.	Type	Depth						
ScanQA	Y	7	Indoor	E.D.	M	800	41K	H	E	CVPR'22C	Close-ended	L	L	I	-	-	
Visual Spatial	Y	7	R.W.	E.D.	T	10K	10K	E	E	TACL'23	MCQ,Bin.	L	M	I	LXMERT	70.10	
What's up	N	6	Indoor	E.D.	T	5K	5K	E	E	EMNLP'23C	MCQ	S	Hi	I	XVLM-COCO	60.40	
EmbSpatial-Bench	Y	6	Indoor	E.D.	T	2.2K	3.6K	H	E	ACL'24C	MCQ,%	M	S	I	Qwen-VL-Max	49.11	
Space3D-Bench	N	6	Indoor	E.D.	M	211	1K	Mm	E	ECCV'24W	Mix	L	S	I	RAG3D-Chat	66.80	
SpatialRGPT-Bench	N	12	Mix	E.D.	T	1.5K	1.5K	Mm	E	NeurIPS'24C	MCQ,Bin.	M	L	I	GPT-4o	57.83	
BLINK-Spatial	Y	14	Mix	E.D.	M	7.3K	3.8K	H	E	ECCV'24C	MCQ,Bin.	M	M	I	GPT-4o	59.03	
Spatial-MM	N	4	Internet	Internet	T	2.3K	2.3K	E	Mm	EMNLP'24C	Open	L	Hi	I	GPT-4 Vision	63.82	
RoboSpatial	Y	4	Indoor	E.D.	T	1M	3M	H	E	CVPR'25C	Open,Bin.	L	S	I	LEO	71.90	
SpatialVLM	N	2	Web	E.D.	T	10M	2B	H	E	CVPR'24C	Open	L	S	I	GPT-4V	68.00	
VSI-Bench	Y	8	Indoor	M,E.D.	T	288	5K	Mm	Mm	CVPR'25C	MCQ,Nm.	M	M	V	Gemini-1.5 Pro	45.40	
OmniSpatial	Y	50	Internet	Internet	M,P	1387	1.5K	H	Mm	2025	MCQ,Bin.	M	S	I	o3-2025-04-16	56.33	
Mind the Gap	N	6	Mix	AI-Gen	M,P	1.8K	1.8K	E	Mm	2025	Mix	M	S	I	InternVL2.5-26B	48.83	
VLM4D	N	4	Mix	Mix	M	1K	1.8K	E	H	ICCV'25	MCQ	Hi	M	Mix	Gemini-2.5 Pro	62.00	
SPATIALAB-MCQ	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	H	2025	MCQ	Hi	H	Mix	InternVL3.5-72B	54.93	
SPATIALAB-OPEN	Y	30	Mix	Mix,M	M	1.2K+	1.5K	H	VH	2025	Open	Hi	H	Mix	GPT-5-mini	40.93	

try, yielding concrete insights and actionable directions for robust, human-aligned spatial understanding.

- Improvement-analysis contribution:** We examine strategies for improving spatial reasoning in VLMs, including **supervised fine-tuning (SFT)**, **chain-of-thought (CoT) prompting**, **CoT with self-reflection**, and **multi-agent architectures (SPATIOXOLVER) with multi-step reasoning**, and analyze their benefits and limitations across multiple-choice and open-ended tasks.

Our results show why these contributions are needed. In the multiple-choice setting, leading models achieve roughly 50–55% accuracy, while human annotators exceed 85%. In open-ended generation, model performance drops by another 10–25 percentage points, with the best systems near 40% compared with about 65% for humans. Error analysis concentrates the gap in occlusion inference, scale consistency, and multi-step navigational planning, suggesting limits in compositional spatial representations and grounded reasoning. Across prompting, fine-tuning, and agentic interventions, our study identifies which approaches provide robust gains, which amplify existing biases, and why perceptual grounding remains critical for generalizable spatial reasoning. By packaging tasks, baselines, and diagnostics together, SPATIALAB establishes a foundation for reproducible comparison and targeted progress in spatial reasoning evaluation.

2 PRELIMINARIES: SPATIAL REASONING IN VLMs

We frame visual-spatial reasoning as the ability to perceive, represent, and manipulate spatial relations among objects—a faculty central to human cognition and to embodied agents that must navigate and act in the world (Palmer, 1999; Chabris et al., 2006). Formally, a scene can be described as a set of objects $O = \{o_1, \dots, o_n\}$ with attributes such as position, orientation, and size in a world coordinate system W . A VLM spatial reasoning task can then be viewed as learning a mapping $f : (I, Q) \mapsto R$, where I is an image (or image sequence), Q is a linguistic query, and R is a structured representation of spatial relations $R = \{r(o_i, o_j) \mid o_i, o_j \in O\}$ capturing predicates such as *left-of*, *occludes*, or *supports*. This framing highlights the capability that SPATIALAB is designed to test: models must connect noisy visual perception with compositional language under embodied constraints such as perspective, gravity, and physical interaction (Dai et al., 2017; Alayrac et al., 2022).

Prior spatial reasoning evaluations provide useful control, but they leave the contribution opportunity for SPATIALAB clear. Early VLM benchmarks often use basic spatial relations or narrow task designs (Cheng et al., 2024; Du et al., 2024; Szymanska et al., 2024), with examples in synthetic datasets such as CLEVR (Johnson et al., 2016), structured QA benchmarks such as GQA (Hudson & Manning, 2019), or puzzle-like simulated setups such as MGT (Stogiannidis et al., 2025) and OmniSpatial (Stogiannidis et al., 2025). Table 1 summarizes this landscape¹. These settings are valuable for controlled measurement, yet they often simplify spatial structure, reduce environmental noise, and underrepresent temporal change or viewpoint shifts. Our benchmark contribution responds to this gap by moving evaluation toward real-world contexts that better match embodied and interactive

¹ Here, E. → Embodied, R.W. → Real World, T.C. → Task Categories, E.D. → Existing Dataset, A. → Annotation, T → Template, M → Manual, P → Puzzles, Md. → Modality (I → Image, V → Video), Bin. → Bin., Nm. → Nm., Df. → Difficulty, Cmpl. → Complexity (E → Easy, Mm → Medium, H → Hard, S → Shallow, L → Low, Hi → Hi, VH/VHi → Very Hard/Very High), A.L. → Analysis Level, B.B.P. → Best Baseline Performance.

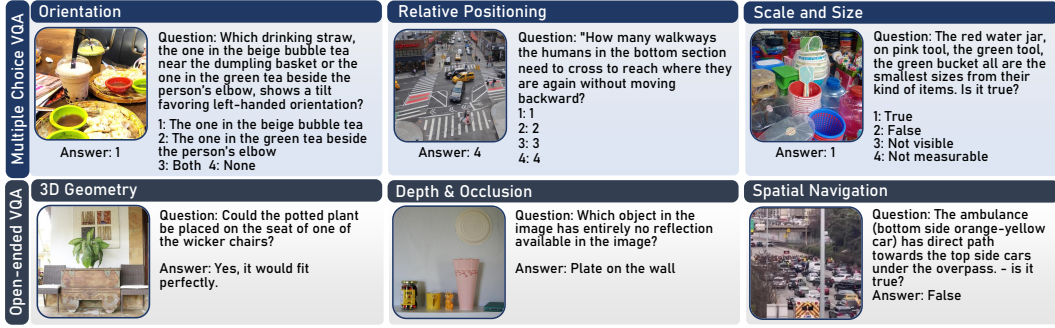


Figure 2: Representative SPATIALAB examples from the six spatial categories, showing how the benchmark contributes both open-ended and MCQ task formats.

applications such as robotics, AR/VR, and autonomous navigation (Mu et al., 2023; Collaboration et al., 2023).

The taxonomy of SPATIALAB is the paper’s organizing methodological contribution. We divide spatial reasoning into six principal categories—*Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*—each linked to cognitive faculties studied in psychology and spatial cognition (Baddeley, 1998; Previc, 1998). As shown in Figure 1 and Appendix B, each category is decomposed into five subcategories, yielding thirty task types that isolate fine-grained skills such as directional alignment, occlusion inference, and stability prediction. This decomposition makes the benchmark easier to interpret: model scores can be read not only as aggregate success or failure, but also as evidence about which spatial dimensions remain fragile or underdeveloped (Gardner, 2011; Huang et al., 2022; Gupta & Kembhavi, 2023; Yang et al., 2023).

The evaluation format is the second methodological contribution. SPATIALAB uses both multiple-choice and open-ended question answering: the multiple-choice setting probes recognition and fine discrimination among distractors, while the open-ended setting measures generative reasoning and compositional description (Driess et al., 2023; Liu et al., 2025; Zheng et al., 2022). Reporting both formats makes the benchmark reviewer-facing because it separates selection-based competence from free-form spatial reasoning and reveals different failure modes. By combining a principled taxonomy, diverse real-world imagery, and dual evaluation protocols, SPATIALAB provides a structured basis for fine-grained model comparison, human-level alignment analysis, and the study of robust spatial cognition in artificial agents.

3 SPATIALAB BENCHMARK

The benchmark contribution of this paper is **SPATIALAB**, a structured testbed for evaluating VLM spatial reasoning under naturalistic visual conditions. The design targets cases where occlusion, perspective distortion, clutter, and noisy backgrounds co-occur, because these are precisely the settings in which simplified benchmarks can hide model brittleness. **SPATIALAB** contains 1,400 visual question-answer pairs organized into six major categories: *Relative Positioning*, *Depth & Occlusion*, *Orientation*, *Size & Scale*, *Spatial Navigation*, and *3D Geometry*. Each category is divided into five subcategories, yielding 30 distinct task types; every subcategory has at least 25 QA pairs, and every top-level category has at least 200. This balance makes the contribution scannable and diagnostic: the benchmark supports aggregate evaluation while also enabling fine-grained analysis by spatial skill. It further supports both multiple-choice and open-ended evaluation, so reviewers can compare discriminative answer selection with generative spatial reasoning. Figure 2 illustrates representative examples across the six categories and both task formats.

3.1 BENCHMARK CONSTRUCTION

Image Collection. The data-construction contribution begins with a multi-source image corpus. We curated images through three complementary strategies: (i) automated large-scale web crawling, (ii) targeted online retrieval of scene-specific images, and (iii) manual snapshots from diverse indoor and outdoor environments (Figure 3). This pipeline increases variation in object categories, environmental

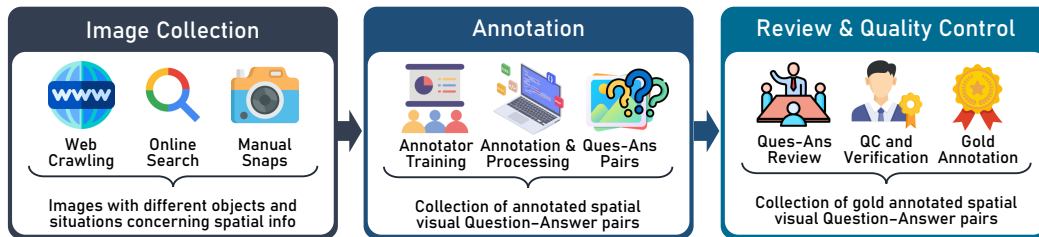


Figure 3: **Data creation pipeline for SPATIALAB.** The pipeline contributes real-world image collection, structured spatial QA annotation, and validation into a final benchmark of reviewed question-answer pairs.

conditions, and spatial configurations. To make that variation explicit, we profiled images along six meta-dimensions: *lighting condition* (high contrast, low contrast, shadows, and reflective settings), *texture complexity* (uniform, patterned, and complex), *edge complexity* (sharp and smooth boundaries), *dominant spatial relation* (stacked, scattered, aligned), *material type* (transparent, translucent, opaque, reflective), and *gravity constraints* (normal, floating, and unconstrained). Representing these dimensions in the final corpus ensures that SPATIALAB evaluates spatial reasoning in real-world, in-the-wild scenes rather than only in simplified or synthetic setups.

Corpus statistics show why this construction matters for spatial reasoning evaluation. On average, images contain 21.48 objects, 11.88 partially visible objects, and 3.23 distinct depth layers. Spatial reference chains, such as “object A is left of B, which is behind C,” contain 2.07 links on average, indicating that many questions require multi-step relational reasoning. In addition, 16.71% of the questions require mental rotation and 56.07% require object specificity, such as color, pattern, or other details. These statistics support the intended benchmark contribution: **SPATIALAB** captures cluttered, layered, and relationally rich scenes that are characteristic of real-world spatial reasoning but underrepresented in synthetic datasets.

Annotation. The annotation contribution is a structured conversion of real-world images into comparable spatial QA items. Annotation proceeded in three phases. First, annotators completed targeted training (details are available in Appendix C.2) to align interpretations of spatial categories and subcategories. Second, each image was paired with one or more spatial QA tasks covering reasoning types such as proximity gradients, alignment patterns, occlusion inference, and stability prediction. Questions were written to balance perceptual grounding with higher-order inference, including cases such as predicting stability under gravity constraints. Third, each QA item was encoded in both multiple-choice (04 options) and open-ended formats, enabling direct comparison across evaluation paradigms. This process initially produced around 1,500 spatial visual QA pairs across six categories and 30 subcategories.

Review and Quality Control. The quality-control contribution is a three-tier validation pipeline that turns the initial annotation pool into the final benchmark. In the first stage, QA items were checked for semantic validity and alignment with the assigned subcategory. In the second stage, independent annotators verified both question phrasing and answer correctness, with emphasis on removing ambiguity and enforcing task clarity. In the third stage, a gold-standard annotation round established high-confidence QA pairs for evaluation. All image metadata was removed, and the dataset was curated to avoid copyright concerns, though minor oversights may still occur. This procedure yielded the final SPATIALAB benchmark of 1,400 validated QA items.

4 EXPERIMENTS AND EVALUATION

Models. The experimental contribution is a broad, reviewer-facing comparison across model families rather than a single leaderboard. We group the evaluated models into four categories. **Proprietary models** include GPT-4o-mini (OpenAI, 2024), GPT-5-nano, Gemini-2-Flash and Gemini-2.5-Flash (Team, 2025a), and Claude 3.5 Haiku (Anthropic, 2024). **Open-source models** include Intern-VL3 (1B, 2B, 4B) (Zhu et al., 2025), Qwen-VL2.5 (3B, 7B, 32B, 72B Instruct) (Bai et al., 2025), GLM 4.5V-3.06B and GLM-4.1V-9B-Thinking (Team et al., 2025), Gemma-3 (4B, 27B)-it (Gemma Team, Google DeepMind, 2025), Llama-3.2 (11B, 90B) Vision-Instruct (Grattafiori et al., 2024), and Step-3-321B-MoE (StepFun, 2025). **Reasoning models** include o4-

Table 2: MCQ evaluation contribution: accuracy (%) (↑) on SPATIALAB-MCQ by question category.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
<i>Random Choice</i>	25.00	25.00	25.00	25.00	25.00	25.00	25.00
<i>Proprietary Models</i>							
GPT-4o-mini	47.06	39.00	47.03	47.17	49.60	49.79	46.50
GPT-5-mini	48.74	54.83	60.40	62.74	44.84	56.54	54.29
Gemini-2.0-Flash	47.06	55.21	53.96	58.02	54.37	46.84	52.50
Gemini-2.5-Flash	44.96	48.26	48.02	56.13	42.46	51.05	48.29
Gemini-2.5-Pro	47.48	50.19	49.50	58.49	43.65	52.32	50.07
Claude 3.5 Haiku	42.44	42.08	46.53	46.23	35.71	45.99	42.93
Mistral Medium 3.1	46.64	49.81	47.52	61.79	41.67	41.77	47.93
<i>Open-Source Models</i>							
InternVL3.5-1B	33.61	32.43	23.27	37.26	31.75	30.80	31.64
InternVL3.5-2B	34.03	31.66	31.68	40.57	32.54	32.49	33.71
Qwen-VL2.5-3B-Instruct	41.18	35.52	46.04	40.09	47.22	39.24	41.43
InternVL3.5-4B	42.86	42.86	42.08	54.72	36.51	42.19	43.29
Gemma-3-4B-it	43.70	34.36	46.53	45.75	37.30	37.97	40.57
Qwen-VL2.5-7B-Instruct	42.86	37.84	42.57	46.23	42.06	35.44	41.00
Llama-3.2-11B-Vision-Instruct	26.47	30.50	20.30	42.92	30.56	32.07	30.50
Gemma-3-27B-it	43.28	40.15	48.02	54.25	48.02	47.26	46.57
Qwen-VL2.5-32B-Instruct	41.18	40.15	46.53	45.28	45.24	41.77	43.21
InternVL3.5-72B	50.00	57.14	53.47	66.04	49.21	54.85	54.93
Qwen-VL2.5-72B-Instruct	47.06	48.65	51.98	54.25	43.65	48.95	48.86
Llama-3.2-90B-Vision-Instruct	46.22	52.12	50.50	58.96	46.83	48.52	50.36
<i>Reasoning Models</i>							
o3-mini	50.00	54.83	51.98	61.32	38.89	50.21	50.93
o4-mini-medium	51.26	58.30	54.95	64.15	40.87	51.48	53.21
Gemini-2-Flash-Thinking	37.82	41.31	41.58	45.75	50.40	43.04	43.36
Gemini-2.5-Flash-Thinking	45.80	53.67	52.97	56.60	55.16	53.59	52.93
Kimi-VL-A3B-Thinking-2506	42.86	41.31	40.59	51.42	39.68	41.35	42.71
<i>Spatial Reasoning Models</i>							
SpaceOm	42.44	38.61	48.02	37.74	42.86	39.24	41.36
SpaceThinker-Qwen2.5VL-3B	40.34	37.84	47.03	38.21	43.25	37.97	40.64
SpaceQwen2.5-VL-3B-Instruct	31.51	35.14	37.62	37.74	50.79	47.26	40.14
<i>Human Baseline</i>	93.70	74.13	91.58	91.51	88.89	87.76	87.57

mini (OpenAI, 2025), Gemini-2-Flash and Gemini-2.5-Flash thinking variants (Team, 2025a), and Kimi-VL-A3B-Thinking-2506 (Team, 2025b). **Spatial reasoning specialists** include SpaceOm, SpaceThinker-Qwen2.5VL-3B, and SpaceQwen2.5-VL-3B-Instruct (Chen et al., 2024a). For GLM-4.5V-106B-MoE, GLM-4.1V-9B-Thinking, and Step-3-321B-MoE, we report only open-ended evaluation because these models are not instruction-tuned and were trained primarily on open-ended tasks, leading to excessive refusals during MCQ evaluation. For readability in the results tables, we use color coding for **proprietary**, **open-source up to 7B**, **larger than 7B**, **proprietary reasoning**, **open-source reasoning**, **spatial specialists**, and **human baseline**.

Evaluation Process and Metrics. The evaluation protocol is designed to expose two complementary forms of spatial competence while keeping accuracy as the primary metric. For **MCQ evaluation**, models are directly prompted to return the option number corresponding to their selected answer, and responses are automatically checked against the gold answer; prompts and procedural details appear in Appendix D.1. For **open-ended evaluation**, models generate free-form answers that are assessed by a large language model judge, with prompts and details in Appendix D.2. To make this open-ended protocol credible, we also evaluate agreement between the LLM judge and human annotators, as detailed in Appendix D.2. Table 5 shows that the LLM judge (Gemini-2.5-Flash) achieves substantial agreement with humans: Cohen’s kappa is 0.738 against majority vote and 0.681–0.795 against individual annotators; raw accuracy against majority vote is 0.880, and Fleiss’ kappa among human annotators is 0.774. Additional experiment setup, prompts, and evaluation procedures are provided in Appendix D.

Improving Visual Reasoning Capabilities. Beyond benchmarking, we include an improvement-analysis contribution that tests whether common reasoning interventions close the spatial gap. We examine inherent reasoning in VLMs (Appendix H.1), Chain-of-Thoughts (CoT) (Appendix H.2), CoT with self-reflection (Appendix H.3), supervised fine-tuning (SFT) (Appendix H.4), and multi-agent systems (Appendix H.5). For SFT, Qwen-VL2.5-3B-Instruct was fine-tuned on 40% of the dataset with stratified sampling and evaluated on the remaining 60%, improving generalization across relational and geometric tasks and reducing errors in navigation and orientation, with additive gains

Table 3: Open-ended evaluation contribution: accuracy (%) (↑) on SPATIALAB-OPEN by question category.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	23.53	16.60	23.27	30.66	17.86	21.94	26.00
GPT-5-mini	45.38	34.75	37.13	49.53	42.46	37.13	40.93
Gemini-2.0-Flash	31.93	24.32	27.23	31.13	26.19	24.47	27.43
Gemini-2.5-Flash	34.03	26.64	31.68	38.68	26.59	29.54	30.93
Gemini-2.5-Pro	37.14	45.45	36.36	37.14	23.91	24.44	33.61
Claude 3.5 Haiku	26.05	18.92	24.75	25.94	20.24	21.10	22.64
Mistral Medium 3.1	25.21	19.31	21.78	29.25	15.08	16.88	21.00
Open-Source Models							
InternVL3.5-1B	05.88	09.65	9.90	13.68	09.13	10.13	09.64
InternVL3.5-2B	12.18	11.20	10.89	23.58	11.90	18.14	14.50
Qwen-VL2.5-3B-Instruct	15.55	8.49	15.35	10.85	18.25	9.28	12.93
InternVL3.5-4B	19.33	17.76	15.84	19.81	16.27	18.99	18.00
Gemma-3-4B-it	20.17	13.13	14.85	23.58	15.08	19.83	17.64
Qwen-VL2.5-7B-Instruct	15.13	15.83	20.30	27.83	15.87	19.83	18.86
Llama-3.2-11B-Vision-Instruct	16.81	16.99	22.28	25.00	13.49	18.57	18.57
Gemma-3-27B-it	22.69	16.22	24.75	34.43	22.62	21.94	23.43
Qwen-VL2.5-32B-Instruct	16.39	09.65	14.85	16.98	09.92	13.50	13.36
InternVL3.5-72B	22.69	20.46	20.30	31.60	19.84	26.16	23.36
Qwen-VL2.5-72B-Instruct	26.89	20.85	25.25	30.66	24.60	20.68	24.64
Llama-3.2-90B-Vision-Instruct	22.69	23.17	21.29	28.30	21.83	27.00	24.00
GLM-4.5V-106B-MoE	31.09	20.46	25.25	26.42	24.21	24.47	25.21
Reasoning Models							
o3-mini	39.08	30.12	29.70	39.15	41.27	30.38	35.00
o4-mini-medium	40.76	32.82	32.18	42.92	44.05	34.18	37.86
Gemini-2-Flash-Thinking	31.09	27.41	31.19	34.43	29.37	29.54	30.36
Gemini-2.5-Flash-Thinking	37.14	45.45	36.36	37.14	21.74	22.22	32.77
Kimi-VL-A3B-Thinking-2506	13.45	14.29	11.88	17.92	12.70	18.57	14.79
GLM-4.1V-9B-Thinking	21.43	19.31	26.73	24.53	19.84	22.78	22.21
Step-3-321B-MoE	32.35	18.53	27.72	27.36	29.76	27.85	27.14
Spatial Reasoning Models							
SpaceOm	12.61	06.95	15.84	11.79	18.65	12.24	12.93
SpaceThinker-Qwen2.5VL-3B	13.45	09.27	17.82	10.38	19.44	10.13	13.36
SpaceQwen2.5-VL-3B-Instruct	12.61	03.86	13.86	09.43	11.90	11.39	10.36
Human Baseline							
	73.53	50.19	70.30	69.81	65.48	62.87	64.93

when CoT and SFT are integrated. At the same time, agentic reasoning shows an important limitation: orientation benefits substantially, but other categories stagnate or degrade. This pattern reinforces the benchmark’s diagnostic value by showing that multi-step reasoning alone is insufficient without deeper perceptual and spatial grounding.

5 FINDINGS AND ANALYSIS

5.1 OVERALL RESULTS

SPATIALAB-MCQ. The first empirical finding is that SPATIALAB-MCQ exposes a large gap between model performance and human spatial reasoning. Table 2 reports multiple-choice accuracy across models, tasks, and model families. Overall model accuracies range roughly from 30–55% (random choice = 25%), while the human baseline reaches 87.57%, leaving substantial headroom. InternVL3.5-72B is the best-performing model with 54.93% accuracy, showing that even the strongest evaluated system remains far from human performance. Reasoning- and instruction-tuned models (e.g., o4-mini: 53.21%, Gemini-2.5-Flash-Thinking: 52.93%, GPT-5-mini: 54.29%) and some very large open-source models (e.g., InternVL3.5-72B: 54.93%, Llama-3.2-90B: 50.36%) rank near the top, but scale alone does not determine success: Llama-3.2-11B attains only 30.50% despite its substantial size.

The category-level pattern is the second MCQ contribution because it shows where current systems are and are not improving. Many strong models perform best on Orientation and 3D Geometry, with several exceeding 60% on one of these categories, whereas Spatial Navigation, Depth & Occlusion, and Size & Scale are more variable and generally harder. Spatially specialized models (SpaceOm, SpaceThinker, SpaceQwen) achieve only mid-40s to low-40s overall, indicating that specialization does not automatically transfer to higher MCQ accuracy across the diverse SPATIALAB task set. Depth & Occlusion and Size & Scale show mixed behavior: some architectures handle occlusion relatively well, while others fail, pointing to different inductive biases or training distributions. Overall, MCQ results suggest that instruction tuning and reasoning capability especially support

geometric and orientation reasoning, while multi-step grounding tasks such as navigation remain an open weakness.

SPATIALAB-OPEN. The open-ended results in Table 3 provide the clearest evidence that generative spatial reasoning is substantially harder than answer selection. Overall accuracy ranges from about 9.6% (InternVL3.5-1B) to 40.9% (GPT-5-mini), while the human baseline is 64.9%. Proprietary models generally lead on open-ended tasks (e.g., GPT-5-mini 40.93%, GPT-4o-mini 26.00%, Gemini-2.5-Flash 30.93%), and reasoning-tuned variants are particularly competitive: o4-mini (37.86%) and Gemini-2.5-Flash-Thinking (32.77%) outperform many baseline open-source models, suggesting that instruction or CoT-style tuning improves free-form spatial responses. Most small open-source models cluster near the bottom (many in the teens; InternVL3.5-1B = 9.64%, Qwen-VL2.5-3B = 12.93%), while some very large open models rise only to the low-20s (e.g., InternVL3.5-72B = 23.36%, Qwen-VL2.5-72B = 24.64%), again showing that scale alone yields limited gains.

The open-ended subtask breakdown turns the benchmark into a diagnostic tool rather than only a leaderboard. Spatial-specialist models (SpaceOm, SpaceThinker, SpaceQwen) reach only 10–13% overall, so architecture or task specialization does not automatically produce robust generative spatial reasoning. Top models perform relatively better on orientation and some relative-position items (e.g., GPT-5-mini Rel.Pos = 49.53%, Ori = 37.13%), but depth & occlusion, size & scale, and especially spatial navigation remain consistently difficult, with many models scoring <30% on these subtasks. This pattern reinforces the central empirical contribution: multi-step grounding and perceptual reasoning, rather than answer formatting alone, are the bottlenecks for open generative spatial answers.

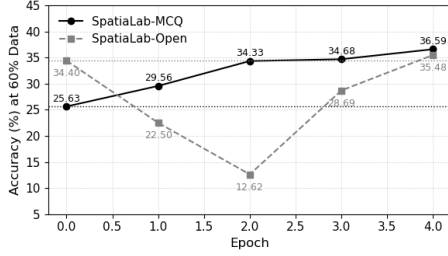
5.2 PERFORMANCE DROP IN OPEN-ENDED EVALUATION COMPARED TO MCQ

The format-gap analysis contributes a direct measurement of how much MCQ evaluation can overstate practical spatial competence. Across 25 models, the average MCQ→open-ended performance gap in SPATIALAB is 23.0% ($\sigma = 5.5\%$), with subtask means ranging from 22.89% (spatial navigation) to 24.57% (3D geometry). Specialist spatial reasoning models exhibit the largest gaps (about 27%), especially in spatial navigation (up to 36.68%) and orientation (34.44%), while reasoning-oriented models show smaller gaps (around 19%) and lower subtask variance, consistent with a stabilizing effect from instruction tuning and CoT decoding. Correlations further identify spatial navigation as the dominant contributor to overall disparity (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) also contributing. Negative or near-zero gaps, such as Llama-3.2-11B at -1.98 in depth & occlusion and o4-mini at -3.18 in relative position, show that poorly designed MCQ distractors can misrepresent true competence.

These results make the evaluation contribution actionable. We hypothesize that format sensitivity arises from MCQ’s structural advantage, specialization bias toward categorical tasks, sequential reasoning challenges in spatio-navigation, and calibration differences in generative output. Therefore, MCQ scores should be complemented with open-ended evaluation, distractors should be audited, stepwise generation or instruction tuning should be tested carefully, and per-subtask diagnostics should be reported to reduce format-dependent artifacts. Additional details appear in Appendix G and Table 18.

5.3 ERROR ANALYSIS

SPATIALAB-MCQ. The MCQ error analysis in Appendix E contributes a finer explanation for the aggregate gap: VLMs show selective peaks but not holistic spatial competence. Closed-source models reach the highest scores, such as 85.71% in *Stacking Orientation*, yet collapse to near-chance in *Relative Size Comparison*. Open-source scaling raises some ceilings, including 80.56% in *Corner/Angle Positioning*, but does not prevent catastrophic failures such as 2.0% in *Object Rotation*. Reasoning-augmented and spatially specialized models add localized gains in areas such as occlusion inference or navigation, but remain capped below 55% in many physical abstraction tasks, including *Gravity Effects* and *Stability Prediction*. Error distributions further identify weaknesses in embodied reasoning (*Tool Handedness*), recursive relational chaining (*Pathway Existence*), and non-local cues (*Reflective Surfaces*); stronger results in *Obstacle Avoidance* may reflect shortcut exploitation rather than robust planning. Together, these patterns show that current VLMs often rely on surface-level correlations and lack stable encodings of orientation, physics, and compositional



Category	MCQ			Open-ended		
	Before	After	Gain	Before	After	Gain
3D Geometry	30.07	37.76	7.69	20.28	23.78	3.50
Depth & Occlusion	23.87	32.90	9.03	43.23	45.81	2.58
Orientation	23.33	36.67	13.33	39.67	42.15	2.48
Relative Positioning	26.19	27.78	1.59	51.18	50.39	-0.79
Size & Scale	27.63	44.74	17.11	18.42	14.47	-3.95
Spatial Navigation	22.38	38.46	16.08	36.62	39.44	2.82
Grand Total	25.63	36.59	10.97	34.40	35.48	1.07

(a) SFT contribution analysis: performance trends over epochs.

(b) SFT contribution analysis: before-versus-after performance across MCQ and open-ended categories.

Figure 4: SFT training contribution: (a) learning trends over epochs, and (b) final accuracy values.

logic. Progress will likely require richer training distributions, such as embodied and physics-driven data, and architectural mechanisms for geometric and reference-frame grounding.

SPATIALAB-OPEN. The open-ended error analysis in Appendix F shows that the same contribution holds under generative evaluation, but the failures are sharper. Even top closed-source models, such as GPT-5-mini with 58.14% in *Directional Relations*, collapse on tasks such as *Proximity Gradients*, so their dominance is task-specific rather than absolute. Large open-source models can achieve strong results in *Depth & Occlusion*, such as 60.0%, but still fall to 9.3% on *Proximity*, showing that scale alone cannot ensure robustness. Smaller models show isolated strengths, such as 26.0% in *Relative Size Comparison*, but can break down completely, including 0.0% in *Betweenness*. Reasoning-tuned systems show promise—Gemini-2.5-Flash-Thinking reaches 75.0% in *Tool Handedness*—yet similar models fall to single digits, revealing fragile integration. Specialized spatial models perform worst overall, rarely exceeding 20% and frequently failing on core relational tasks. These failure patterns point to systemic weaknesses in occlusion handling, orientation, and multi-step relational chaining, while partial success in size-scale cues suggests reliance on superficial correlations. The root causes are missing representations for spatial analogies, architectural limits tied to scale and training diversity, and brittle reasoning pipelines; addressing them will require unified geometric encodings, physics-aware reasoning, and embodied data.

Qualitative Error Analysis. The qualitative analysis in Appendix I contributes an interpretable taxonomy of model failures. Errors cluster into recurring classes rather than random noise: spatial mislocalization, where models confuse referents in crowded scenes; perspective and scale mistakes, where object-size priors override image evidence; occlusion or ordering failures, especially with thin or partially hidden structures; attribute confusion, where perceptual and functional properties are mixed; and open-ended rationalizations that are fluent but visually ungrounded. These patterns appear across architectures and prompting styles, suggesting structural biases in current VLMs rather than dataset-specific noise. Failures increase when models must fuse multiple cues, such as depth ordering with relative size, and calibration is especially poor in open-ended settings. Our diagnostic protocols further show that models often ignore small but decisive visual features, rely on brittle heuristics, and fail to update under counterfactual edits. These observations motivate geometry-aware supervision, multi-scale feature retention, and verification pipelines as concrete directions for future work.

5.4 PERFORMANCE IMPROVEMENT APPROACHES

Inherent Reasoning Mechanisms. The first improvement analysis asks whether built-in reasoning mechanisms already help on SPATIALAB. Reasoning-enabled models consistently outperform their baselines in both MCQ and open-ended formats, with the largest gains in relational and orientation tasks, including +13.1% in Relative Positioning for MCQ. However, the gains are uneven across categories and can decline in Size & Scale. This result narrows the contribution claim: reasoning modules improve logical consistency and stabilize some open-ended outputs, but they do not by themselves solve grounding or scale sensitivity. Detailed per-category comparisons are provided in Appendix H.1.

Chain-of-Thought (CoT) Prompting. The CoT study shows that step-by-step prompting is not a universal spatial-reasoning remedy. Across models, CoT prompting provides little benefit and often reduces accuracy (Table 19), with orientation as the only category showing consistent gains. This indicates that explicit reasoning can help directional alignment but fails on tasks requiring robust

perceptual grounding, such as depth, scale, and spatial navigation. Unlike in many text-only reasoning settings, CoT on SPATIALAB can amplify flawed visual priors rather than correct them, underscoring that the benchmark measures perceptual understanding in addition to logical chaining (details in Appendix H.2).

Chain-of-Thought (CoT) with Self-Reflection. Self-reflection contributes a more targeted test of whether models can repair their own spatial reasoning. We prompt models to review prior reasoning step by step and correct mistakes (see Figure 8 in Appendix for prompt). This strategy yields modest benefits in multiple-choice settings, especially for geometry and depth, but it does not generalize to open-ended tasks (Appendix H.3). The result reinforces a central diagnostic claim: reflective reasoning cannot compensate for missing perceptual grounding.

Supervised Fine-Tuning (SFT). The SFT experiment clarifies what can and cannot be improved through benchmark-specific training. As shown in Figure 4, supervised fine-tuning on Qwen2.5-VL-3B-Instruct using a stratified 40% split of SPATIALAB (~560 samples) consistently improves MCQ accuracy across all spatial reasoning categories. However, it provides little, and sometimes negative, transfer to open-ended tasks. This divergence suggests overfitting to task-specific answer distributions, reinforcing concerns that SFT can *teach to the test* rather than foster generalizable reasoning (Wang et al., 2025). The sharp drop in open-ended performance points to biased internal representations tuned for categorical discrimination but unstable in generation. A likely cause is catastrophic forgetting of linguistic priors during fine-tuning (Huang et al., 2024a), compounded by objectives that favor discrete-choice alignment over spatial reasoning in language. Thus, SFT is useful for structured formats but risks becoming an algorithmic straightjacket for naturalistic spatial reasoning. More details are available in Appendix H.4.

AI Agents for Spatial Reasoning (SPATIOXOLVER). The agentic study evaluates whether modular multi-step reasoning can close gaps that prompting and SFT leave open. We employ SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025), to perform structured spatial reasoning on images. The framework decomposes visual reasoning into specialized sub-tasks managed by dedicated agents, including object segmentation, attribute extraction, spatial relation mapping, and transformation tracking. These outputs are consolidated into a unified structured representation intended to propagate low-level perceptual cues into higher-order relational inference. We implemented the pipeline using Gemini-2.5-Flash with low temperature to ensure deterministic outputs during multi-stage refinement. Results show a substantial gain in Orientation, improving by +8.00% in MCQ and +36.00% in open-ended evaluation. However, gains are not universal: Depth & Occlusion and Spatial Navigation decline sharply in open evaluation, by -24.00% and -12.00%, respectively. These drops show that when depth or occlusion priors are weak, multi-step reasoning can reinforce misconceptions rather than repair them. The contribution of this analysis is therefore diagnostic: agentic workflows help tasks reducible to sequential alignment but cannot substitute for fundamental perceptual grounding in complex 3D environments. More details are available in Appendix H.5.

6 DISCUSSION

The main takeaway is that SPATIALAB contributes more than a new dataset: it is a diagnostic instrument for separating apparent spatial competence from robust spatial reasoning. Existing benchmarks often rely on synthetic scenes, templated questions, or narrow task scopes that underrepresent real-world visual variability. In contrast, SPATIALAB provides 1,400 visual question-answer pairs across six major spatial categories and 30 subcategories, including 3D geometry, occlusion, and navigation. Its dual Multiple-Choice and Open-Ended protocols make the evaluation contribution explicit by separating option-based pattern use from generative spatial grounding. The benchmark is further supported by multi-stage review and complexity checks over lighting, texture, and spatial diversity, and its evaluation of more than 25 contemporary models against human baselines establishes a high-fidelity testbed for identifying weaknesses in current VLMs.

The empirical contribution is a consistent pattern of limitations across model families and intervention strategies. Detailed error analysis shows systematic failures with transparency, multi-step navigation, and occlusion, suggesting that models often rely on surface texture cues rather than internalized 3D geometric representations. Supervised Fine-Tuning improves structured MCQ accuracy but destabilizes open-ended reasoning, consistent with catastrophic forgetting of linguistic priors when

optimizing for discrete answers. Reasoning-based approaches such as Chain-of-Thought help sequential tasks like Orientation but do not repair fundamental perceptual errors in Depth and Scale, and can amplify hallucinations. Agentic decompositions show the same trade-off: they improve specific alignment tasks while degrading holistic scene understanding. Taken together, these findings limit the claims that can be made from scaling, prompting, or modular reasoning alone, and they point toward architectures that explicitly integrate geometric grounding and physics-aware representations.

7 CONCLUDING REMARKS

This paper contributes **SPATIALAB** as a benchmark, evaluation protocol, and diagnostic analysis for spatial reasoning in VLMs. The benchmark contains 1,400 diverse QA pairs across six spatial categories and thirty subcategories, and it supports both MCQ and open-ended formats so that spatial competence can be evaluated at both the selection and generation levels. Across more than 25 models, the results show substantial gaps between state-of-the-art VLMs and humans, with open-ended generation dropping an additional 10–25% relative to multiple-choice accuracy. Error analysis further localizes the gap to depth, occlusion, 3D geometry, and multi-step navigation, indicating reliance on superficial cues rather than grounded spatial representations.

The broader contribution is a structured testbed for future work on human-aligned spatial understanding. **SPATIALAB** makes it possible to compare model families, audit format sensitivity, diagnose subcategory-specific weaknesses, and evaluate improvement strategies such as SFT, CoT, self-reflection, and agentic reasoning. The findings suggest that future progress will require richer training signals, including embodied interaction and action-conditioned feedback, as well as architectures capable of geometric grounding, relational chaining, and consistent multi-scale reasoning.

ETHICS STATEMENT

The ethics contribution of our release process is to reduce privacy, licensing, and representational risks while preserving benchmark utility. We removed image metadata to protect privacy, and personally identifiable features such as human faces or license plates were minimized or excluded wherever possible. The dataset was curated with attention to sourcing and licensing to avoid copyright concerns, though minor oversights may occur despite best efforts. The benchmark construction process was designed for transparency and fairness, and no demographic groups were singled out or negatively represented. We adhered to the ICLR Code of Ethics and confirm that no human subjects or sensitive personal data were involved.

REPRODUCIBILITY STATEMENT

To support reproducible use of **SPATIALAB**, we document the dataset construction process, annotation pipeline, and evaluation protocol in the main text and appendix. Filtering, quality-control, and validation steps are described explicitly, with definitions of task categories and annotation criteria. Hyperparameters, model configurations, and training procedures are reported in the appendix so that future work can replicate the reported comparisons and extend the benchmark consistently.

REFERENCES

Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andrew Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning. In *NeurIPS*, 2022.

Anthropic. Model Card Addendum: Claude 3.5 Haiku and Upgraded Claude 3.5 Sonnet, October 2024. URL <https://assets.anthropic.com/m/1cd9d098ac3e6467/original/Claude-3-Model-Card-October-Addendum.pdf>. Accessed: 2025-09-11.

- Daichi Azuma, Taiki Miyanishi, Shuhei Kurita, and Motoaki Kawanabe. Scanqa: 3d question answering for spatial scene understanding. In *proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 19129–19139, 2022.
- Alan Baddeley. Working memory. *Comptes Rendus de l’Académie des Sciences-Series III-Sciences de la Vie*, 321(2-3):167–173, 1998.
- Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report, 2025. URL <https://arxiv.org/abs/2502.13923>.
- Yoav Bar-Anan, Nira Liberman, and Yaacov Trope. The association between psychological distance and construal level: evidence from an implicit association test. *Journal of experimental psychology: General*, 135(4):609, 2006.
- Anthony Brohan, Noah Brown, Justice Carbajal, Yevgen Chebotar, Joseph Dabis, Chelsea Finn, Keerthana Gopalakrishnan, Karol Hausman, Alexander Herzog, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Tomas Jackson, Sally Jesmonth, Nikhil J. Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Isabel Leal, Kuang-Huei Lee, Sergey Levine, Yao Lu, Utsav Malla, Deeksha Manjunath, Igor Mordatch, Ofir Nachum, Carolina Parada, Jodilyn Peralta, Emily Perez, Karl Pertsch, Jornell Quiambao, Kanishka Rao, Michael S. Ryoo, Grecia Salazar, Pannag R. Sanketi, Kevin Sayed, Jaspiar Singh, Sumedh Sontakke, Austin Stone, Clayton Tan, Huong T. Tran, Vincent Vanhoucke, Steve Vega, Quan Vuong, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Tianhe Yu, and Brianna Zitkovich. RT-1: robotics transformer for real-world control at scale. In *Robotics: Science and Systems XIX, Daegu, Republic of Korea, July 10-14, 2023*, 2023.
- Wenxiao Cai, Yaroslav Ponomarenko, Jianhao Yuan, Xiaoqi Li, Wankou Yang, Hao Dong, and Bo Zhao. Spatialbot: Precise spatial understanding with vision language models. *CoRR*, abs/2406.13642, 2024.
- Zhongang Cai, Yubo Wang, Qingping Sun, Ruisi Wang, Chenyang Gu, Wanqi Yin, Zhiqian Lin, Zhitao Yang, Chen Wei, Xuanke Shi, Kewang Deng, Xiaoyang Han, Zukai Chen, Jiaqi Li, Xiangyu Fan, Hanming Deng, Lewei Lu, Bo Li, Ziwei Liu, Quan Wang, Dahua Lin, and Lei Yang. Has gpt-5 achieved spatial intelligence? an empirical study, 2025. URL <https://arxiv.org/abs/2508.13142>.
- Christopher F Chabris, Thomas E Jerde, Anita W Woolley, Margaret E Gerbasi, Jonathon P Schuldt, Sean L Bennett, J Richard Hackman, and Stephen M Kosslyn. Spatial and object visualization cognitive styles: Validation studies in 3800 individuals. *Group brain technical report*, 2(1-20):2, 2006.
- Boyuan Chen, Zhuo Xu, Sean Kirmani, Brian Ichter, Danny Driess, Pete Florence, Dorsa Sadigh, Leonidas Guibas, and Fei Xia. Spatialvlm: Endowing vision-language models with spatial reasoning capabilities. *arXiv preprint arXiv:2401.12168*, 2024a. URL <https://arxiv.org/abs/2401.12168>.
- Boyuan Chen, Zhuo Xu, Sean Kirmani, Brian Ichter, Dorsa Sadigh, Leonidas J. Guibas, and Fei Xia. Spatialvlm: Endowing vision-language models with spatial reasoning capabilities. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2024, Seattle, WA, USA, June 16-22, 2024*, pp. 14455–14465. IEEE, 2024b.
- Shiqi Chen, Tongyao Zhu, Ruochen Zhou, Jinghan Zhang, Siyang Gao, Juan Carlos Niebles, Mor Geva, Junxian He, Jiajun Wu, and Manling Li. Why is spatial reasoning hard for vlms? an attention mechanism perspective on focus areas. *CoRR*, abs/2503.01773, 2025.
- An-Chieh Cheng, Hongxu Yin, Yang Fu, Qiushan Guo, Ruihan Yang, Jan Kautz, Xiaolong Wang, and Sifei Liu. Spatialrgpt: Grounded spatial reasoning in vision-language models. In Amir Globersons, Lester Mackey, Danielle Belgrave, Angela Fan, Ulrich Paquet, Jakub M. Tomczak, and Cheng Zhang (eds.), *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024, NeurIPS 2024, Vancouver, BC, Canada, December 10 - 15, 2024*, 2024.

- Open X.-Embodiment Collaboration, Abhishek Padalkar, Acorn Pooley, Ajinkya Jain, Alex Bewley, Alexander Herzog, Alex Irpan, Alexander Khazatsky, Anant Raj, Anikait Singh, Anthony Brohan, Antonin Raffin, Ayzaan Wahid, Ben Burgess-Limerick, Beomjoon Kim, Bernhard Schölkopf, Brian Ichter, Cewu Lu, Charles Xu, Chelsea Finn, Chenfeng Xu, Cheng Chi, Chenguang Huang, Christine Chan, Chuer Pan, Chuyuan Fu, Coline Devin, Danny Driess, Deepak Pathak, Dhruv Shah, Dieter Buehler, Dmitry Kalashnikov, Dorsa Sadigh, Edward Johns, Federico Ceola, Fei Xia, Freek Stulp, Gaoyue Zhou, Gaurav S. Sukhatme, Gautam Salhotra, Ge Yan, Giulio Schiavi, Gregory Kahn, Hao Su, Haoshu Fang, Haochen Shi, Heni Ben Amor, Henrik I. Christensen, Hiroki Furuta, Homer Walke, Hongjie Fang, Igor Mordatch, Ilija Radosavovic, and et al. Open x-embodiment: Robotic learning datasets and RT-X models. *CoRR*, abs/2310.08864, 2023.
- Lee J. Cronbach. Coefficient alpha and the internal structure of tests. *Psychometrika*, 16(3):297–334, September 1951. ISSN 1860-0980. doi: 10.1007/bf02310555. URL <http://dx.doi.org/10.1007/BF02310555>.
- Angela Dai, Angel X. Chang, Manolis Savva, Maciej Halber, Thomas A. Funkhouser, and Matthias Nießner. Scannet: Richly-annotated 3d reconstructions of indoor scenes. In *2017 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017, Honolulu, HI, USA, July 21-26, 2017*, pp. 2432–2443. IEEE Computer Society, 2017.
- Danny Driess, Fei Xia, Mehdi S. M. Sajjadi, Corey Lynch, Aakanksha Chowdhery, Brian Ichter, Ayzaan Wahid, Jonathan Tompson, Quan Vuong, Tianhe Yu, Wenlong Huang, Yevgen Chebotar, Pierre Sermanet, Daniel Duckworth, Sergey Levine, Vincent Vanhoucke, Karol Hausman, Marc Toussaint, Klaus Greff, Andy Zeng, Igor Mordatch, and Pete Florence. Palm-e: An embodied multimodal language model. In *ICML*, 2023.
- Chris Drummond. Replicability is not reproducibility: nor is it good science. In *Proceedings of the Evaluation Methods for Machine Learning Workshop at the 26th ICML*, volume 1, Montreal, Canada, 2009. National Research Council of Canada.
- Chris Drummond and Nathalie Japkowicz. Warning: statistical benchmarking is addictive. kicking the habit in machine learning. *Journal of Experimental and Theoretical Artificial Intelligence*, 22(1):67–80, March 2010. ISSN 1362-3079. doi: 10.1080/09528130903010295. URL <http://dx.doi.org/10.1080/09528130903010295>.
- Mengfei Du, Binhao Wu, Zejun Li, Xuanjing Huang, and Zhongyu Wei. Embspatial-bench: Benchmarking spatial understanding for embodied tasks with large vision-language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics, ACL 2024 - Short Papers, Bangkok, Thailand, August 11-16, 2024*, pp. 346–355. Association for Computational Linguistics, 2024.
- Xingyu Fu, Yushi Hu, Bangzheng Li, Yu Feng, Haoyu Wang, Xudong Lin, Dan Roth, Noah A. Smith, Wei-Chiu Ma, and Ranjay Krishna. BLINK: multimodal large language models can see but not perceive. In Ales Leonardis, Elisa Ricci, Stefan Roth, Olga Russakovsky, Torsten Sattler, and Gül Varol (eds.), *Computer Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024, Proceedings, Part XXIII*, volume 15081 of *Lecture Notes in Computer Science*, pp. 148–166. Springer, 2024.
- Howard E Gardner. *Frames of mind: The theory of multiple intelligences*. Basic books, 2011.
- Gemma Team, Google DeepMind. Gemma 3 technical report, 2025. URL <https://storage.googleapis.com/deepmind-media/gemma/Gemma3Report.pdf>. Accessed: 2025-09-11.
- Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego

702 Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova,
 703 Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel
 704 Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon,
 705 Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan
 706 Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet,
 707 Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelmer van der Linde,
 708 Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie
 709 Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua
 710 Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani, Kate Plawiak,
 711 Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley
 712 Chiu, Kunal Bhalla, Kushal Lakhotia, Lauren Rantala-Yearly, Laurens van der Maaten, Lawrence
 713 Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas
 714 Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri,
 715 Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie
 716 Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes
 717 Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang, Olivier Duchenne,
 718 Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal
 719 Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong,
 720 Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic,
 721 Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie
 722 Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana
 723 Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shao-liang Nie,
 724 Sharan Narang, Sharath Rapparthi, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon
 725 Vandenhennde, Soumya Batra, Spencer Whitman, Sten Sootla, Stephane Collot, Suchin Gururangan,
 726 Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas
 727 Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami,
 728 Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gonguet, Virginie Do, Vish Vogeti,
 729 Vitor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyin Fu, Whitney Meers, Xavier
 730 Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao
 731 Jia, Xuewei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwen Song,
 732 Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan, Zhengxing Chen, Zoe
 733 Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya
 734 Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei
 735 Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu,
 736 Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit
 737 Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury,
 738 Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer,
 739 Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paranjape, Bing Liu,
 740 Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido,
 741 Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changhan Wang, Changkyu
 742 Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer,
 743 Cynthia Gao, Damon Cvin, Dana Beatty, Daniel Kreymer, Daniel Li, David Adkins, David Xu,
 744 Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkang Wang, Duc
 745 Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily
 746 Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers,
 747 Fei Sun, Felix Kreuk, Feng Tian, Filippas Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank
 748 Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Swee,
 749 Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan,
 750 Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph,
 751 Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damlaj, Igor Molybog,
 752 Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James
 753 Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny
 754 Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings,
 755 Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai
 Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik
 Veeraraghavan, Kelly Michelena, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle
 Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng
 Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabsa, Manav Avalani, Manish
 Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim

- Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanchandani, Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojuan Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The llama 3 herd of models, 2024. URL <https://arxiv.org/abs/2407.21783>.
- Tanmay Gupta and Aniruddha Kembhavi. Visual programming: Compositional visual reasoning without training. In *CVPR*, 2023.
- Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J. Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H. van Kerkwijk, Matthew Brett, Allan Haldane, Jaime Fernández del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E. Oliphant. Array programming with NumPy. *Nature*, 585(7825):357–362, September 2020. doi: 10.1038/s41586-020-2649-2. URL <https://doi.org/10.1038/s41586-020-2649-2>.
- Md Tanzib Hosain, Salman Rahman, Md Kishor Morol, and Md Rizwan Parvez. Xolver: Multi-agent reasoning with holistic experience learning just like an olympiad team, 2025. URL <https://arxiv.org/abs/2506.14234>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. URL <https://arxiv.org/abs/2106.09685>.
- Jianheng Huang, Leyang Cui, Ante Wang, Chengyi Yang, Xinting Liao, Linfeng Song, Junfeng Yao, and Jinsong Su. Mitigating catastrophic forgetting in large language models with self-synthesized rehearsal. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1416–1428, Bangkok, Thailand, August 2024a. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.77. URL <https://aclanthology.org/2024.acl-long.77/>.
- Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Tomas Jackson, Noah Brown, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. Inner monologue: Embodied reasoning through planning with language models. In *CoRL*, 2022.
- Wenlong Huang, Chen Wang, Yunzhu Li, Ruohan Zhang, and Li Fei-Fei. Rekep: Spatio-temporal reasoning of relational keypoint constraints for robotic manipulation. In *CoRL*, 2024b.
- Drew A. Hudson and Christopher D. Manning. GQA: A new dataset for real-world visual reasoning and compositional question answering. In *IEEE Conference on Computer Vision and Pattern*

810 *Recognition, CVPR 2019, Long Beach, CA, USA, June 16-20, 2019*, pp. 6700–6709. Computer
811 Vision Foundation / IEEE, 2019.

812

813 Mengdi Jia, Zekun Qi, Shaochen Zhang, Wenyao Zhang, Xinqiang Yu, Jiawei He, He Wang, and
814 Li Yi. Omnispatial: Towards comprehensive spatial reasoning benchmark for vision language
815 models, 2025. URL <https://arxiv.org/abs/2506.03135>.

816

817 Justin Johnson, Bharath Hariharan, Laurens van der Maaten, Li Fei-Fei, C. Lawrence Zitnick, and
818 Ross Girshick. Clevr: A diagnostic dataset for compositional language and elementary visual
819 reasoning, 2016. URL <https://arxiv.org/abs/1612.06890>.

820

821 Amita Kamath, Jack Hessel, and Kai-Wei Chang. What’s ”up” with vision-language models?
822 investigating their struggle with spatial reasoning. In Houda Bouamor, Juan Pino, and Kalika Bali
823 (eds.), *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, EMNLP 2023, Singapore, December 6-10, 2023*, pp. 9161–9175. Association for Computational
824 Linguistics, 2023.

825

826 Terry K. Koo and Mae Y. Li. A guideline of selecting and reporting intraclass correlation coefficients
827 for reliability research. *Journal of Chiropractic Medicine*, 15(2):155–163, June 2016. ISSN
828 1556-3707. doi: 10.1016/j.jcm.2016.02.012. URL [http://dx.doi.org/10.1016/j.jcm.](http://dx.doi.org/10.1016/j.jcm.2016.02.012)
829 2016.02.012.

830

831 Eliza Kosoy, Annya Dahmani, Andrew K. Lampinen, Iulia M. Comsa, Soojin Jeong, Ishita Dasgupta,
832 and Kelsey Allen. Decoupling the components of geometric understanding in vision language
833 models. *CoRR*, abs/2503.03840, 2025.

834

835 Junnan Li, Dongxu Li, Silvio Savarese, and Steven C. H. Hoi. BLIP-2: bootstrapping language-image
836 pre-training with frozen image encoders and large language models. *CoRR*, abs/2301.12597, 2023.

837

838 Fangyu Liu, Guy Emerson, and Nigel Collier. Visual spatial reasoning. *Trans. Assoc. Comput.*
839 *Linguistics*, 11:635–651, 2023.

840

841 Siqi Liu, Ian Gemp, Luke Marris, Georgios Piliouras, Nicolas Heess, and Marc Lanctot. Re-
842 evaluating open-ended evaluation of large language models. In *The Thirteenth International*
843 *Conference on Learning Representations*, 2025. URL [https://openreview.net/forum?](https://openreview.net/forum?id=kbOAIXKWgx)
844 [id=kbOAIXKWgx](https://openreview.net/forum?id=kbOAIXKWgx).

845

846 Yao Mu, Qinglong Zhang, Mengkang Hu, Wenhai Wang, Mingyu Ding, Jun Jin, Bin Wang, Jifeng
847 Dai, Yu Qiao, and Ping Luo. Embodiedgpt: Vision-language pre-training via embodied chain of
848 thought. In *NeurIPS*, 2023.

849

850 OpenAI. Gpt-4v(ision) system card, 2023. URL [https://openai.com/research/](https://openai.com/research/gpt-4v-system-card)
851 [gpt-4v-system-card](https://openai.com/research/gpt-4v-system-card).

852

853 OpenAI. Gpt-4o system card, 2024. URL <https://arxiv.org/abs/2410.21276>.

854

855 OpenAI. Openai o3 and o4-mini system card, April 2025. URL [https://openai.com/index/](https://openai.com/index/o3-o4-mini-system-card/)
856 [o3-o4-mini-system-card/](https://openai.com/index/o3-o4-mini-system-card/). System Cards. Accessed: 2025-09-11.

857

858 Stephen Palmer. *Vision Science: From Photons to Phenomenology*, volume 1. MIT Press, 01 1999.

859

860 Atin Pothiraj, Elias Stengel-Eskin, Jaemin Cho, and Mohit Bansal. Capture: Evaluating spatial
861 reasoning in vision language models via occluded object counting. *CoRR*, abs/2504.15485, 2025.

862

863 Fred H Previc. The neuropsychology of 3-d space. *Psychological bulletin*, 124(2):123, 1998.

Zekun Qi, Runpei Dong, Shaochen Zhang, Haoran Geng, Chunrui Han, Zheng Ge, Li Yi, and
Kaisheng Ma. Shapellm: Universal 3d object understanding for embodied interaction. In *Computer*
Vision - ECCV 2024 - 18th European Conference, Milan, Italy, September 29-October 4, 2024,
Proceedings, Part XLIII, volume 15101 of *Lecture Notes in Computer Science*, pp. 214–238.
Springer, 2024.

864 Zekun Qi, Wenyao Zhang, Yufei Ding, Runpei Dong, Xinqiang Yu, Jingwen Li, Lingyun Xu, Baoyu
865 Li, Xialin He, Guofan Fan, Jiazhao Zhang, Jiawei He, Jiayuan Gu, Xin Jin, Kaisheng Ma, Zhizheng
866 Zhang, He Wang, and Li Yi. Sofar: Language-grounded orientation bridges spatial reasoning and
867 object manipulation. *CoRR*, abs/2502.13143, 2025.

868
869 Tiago Ramalho, Tomás Kociský, Frederic Besse, S. M. Ali Eslami, Gábor Melis, Fabio Viola, Phil
870 Blunsom, and Karl Moritz Hermann. Encoding spatial relations from natural language. *CoRR*,
871 abs/1807.01670, 2018.

872
873 Fatemeh Shiri, Xiao-Yu Guo, Mona Far, Xin Yu, Reza Haf, and Yuan-Fang Li. An empirical analysis
874 on spatial reasoning capabilities of large multimodal models. In Yaser Al-Onaizan, Mohit Bansal,
875 and Yun-Nung Chen (eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural
876 Language Processing, EMNLP 2024, Miami, FL, USA, November 12-16, 2024*, pp. 21440–21455.
877 Association for Computational Linguistics, 2024.

878
879 Patrick E. Shrout and Joseph L. Fleiss. Intraclass correlations: Uses in assessing rater reliability.
880 *Psychological Bulletin*, 86(2):420–428, 1979. ISSN 0033-2909. doi: 10.1037/0033-2909.86.2.420.
881 URL <http://dx.doi.org/10.1037/0033-2909.86.2.420>.

882
883 Chan Hee Song, Valts Blukis, Jonathan Tremblay, Stephen Tyree, Yu Su, and Stan Birchfield.
884 Robospatial: Teaching spatial understanding to 2d and 3d vision-language models for robotics.
885 *CoRR*, abs/2411.16537, 2024.

886
887 StepFun. Step-3 is large yet affordable: Model-system co-design for cost-effective decoding, 2025.
888 URL <https://arxiv.org/abs/2507.19427>.

889
890 Ilias Stogiannidis, Steven McDonagh, and Sotirios A. Tsafaris. Mind the gap: Benchmarking spatial
891 reasoning in vision-language models. *CoRR*, abs/2503.19707, 2025.

892
893 Emilia Szymanska, Mihai Dusmanu, Jan-Willem Buurlage, Mahdi Rad, and Marc Pollefeys. Space3d-
894 bench: Spatial 3d question answering benchmark. *CoRR*, abs/2408.16662, 2024.

895
896 Mohsen Tavakol and Reg Dennick. Making sense of cronbach’s alpha. *International Journal of
897 Medical Education*, 2:53–55, June 2011. ISSN 2042-6372. doi: 10.5116/ijme.4dfb.8dfd. URL
898 <http://dx.doi.org/10.5116/ijme.4dfb.8dfd>.

899
900 Gemini Team. Gemini: A family of highly capable multimodal models, 2025a. URL <https://arxiv.org/abs/2312.11805>.

901
902 Kimi Team. Kimi-vl technical report, 2025b. URL <https://arxiv.org/abs/2504.07491>.

903
904 V Team, Wenyi Hong, Wenmeng Yu, Xiaotao Gu, Guo Wang, Guobing Gan, Haomiao Tang, Jiale
905 Cheng, Ji Qi, Junhui Ji, Lihang Pan, Shuaiqi Duan, Weihang Wang, Yan Wang, Yean Cheng, Zehai
906 He, Zhe Su, Zhen Yang, Ziyang Pan, Aohan Zeng, Baoxu Wang, Bin Chen, Boyan Shi, Changyu
907 Pang, Chenhui Zhang, Da Yin, Fan Yang, Guoqing Chen, Jiazheng Xu, Jiale Zhu, Jiali Chen,
908 Jing Chen, Jinhao Chen, Jinghao Lin, Jinjiang Wang, Junjie Chen, Leqi Lei, Letian Gong, Leyi
909 Pan, Mingdao Liu, Mingde Xu, Mingzhi Zhang, Qinkai Zheng, Sheng Yang, Shi Zhong, Shiyu
910 Huang, Shuyuan Zhao, Siyan Xue, Shangqin Tu, Shengbiao Meng, Tianshu Zhang, Tianwei Luo,
911 Tianxiang Hao, Tianyu Tong, Wenkai Li, Wei Jia, Xiao Liu, Xiaohan Zhang, Xin Lyu, Xinyue Fan,
912 Xuancheng Huang, Yanling Wang, Yadong Xue, Yanfeng Wang, Yanzi Wang, Yifan An, Yifan
913 Du, Yiming Shi, Yiheng Huang, Yilin Niu, Yuan Wang, Yuanchang Yue, Yuchen Li, Yutao Zhang,
914 Yuting Wang, Yu Wang, Yuxuan Zhang, Zhao Xue, Zhenyu Hou, Zhengxiao Du, Zihan Wang, Peng
915 Zhang, Debing Liu, Bin Xu, Juanzi Li, Minlie Huang, Yuxiao Dong, and Jie Tang. Glm-4.5v and
916 glm-4.1v-thinking: Towards versatile multimodal reasoning with scalable reinforcement learning,
917 2025. URL <https://arxiv.org/abs/2507.01006>.

918
919 Yaacov Trope and Nira Liberman. Construal-level theory of psychological distance. *Psychological
920 review*, 117(2):440, 2010.

921
922 Raphael Vallat. Pingouin: statistics in python. *Journal of Open Source Software*, 3(31):1026,
923 November 2018. doi: 10.21105/joss.01026.

-
- Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, Evgeni Burovski, Pearu Peterson, Warren Weckesser, Jonathan Bright, Stéfan J. van der Walt, Matthew Brett, Joshua Wilson, K. Jarrod Millman, Nikolay Mayorov, Andrew R. J. Nelson, Eric Jones, Robert Kern, Eric Larson, C J Carey, İlhan Polat, Yu Feng, Eric W. Moore, Jake VanderPlas, Denis Laxalde, Josef Perktold, Robert Cimrman, Ian Henriksen, E. A. Quintero, Charles R. Harris, Anne M. Archibald, Antônio H. Ribeiro, Fabian Pedregosa, Paul van Mulbregt, and SciPy 1.0 Contributors. SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python. *Nature Methods*, 17:261–272, 2020. doi: 10.1038/s41592-019-0686-2.
- Yubo Wang, Xiang Yue, and Wenhui Chen. Critique fine-tuning: Learning to critique is more effective than learning to imitate, 2025. URL <https://arxiv.org/abs/2501.17703>.
- Jihan Yang, Shusheng Yang, Anjali W. Gupta, Rilyn Han, Li Fei-Fei, and Saining Xie. Thinking in space: How multimodal large language models see, remember, and recall spaces. *CoRR*, abs/2412.14171, 2024.
- Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin Lin, Ehsan Azarnasab, Faisal Ahmed, Zicheng Liu, Ce Liu, Michael Zeng, and Lijuan Wang. MM-REACT: prompting chatgpt for multimodal reasoning and action. *CoRR*, abs/2303.11381, 2023.
- Wentao Yuan, Jiafei Duan, Valts Blukis, Wilbert Pumacay, Ranjay Krishna, Adithyavairavan Murali, Arsalan Mousavian, and Dieter Fox. Robopoint: A vision-language model for spatial affordance prediction for robotics. *CoRR*, abs/2406.10721, 2024.
- Kaizhi Zheng, Xiaotong Chen, Odest Chadwicke Jenkins, and Xin Eric Wang. Vlmbench: A compositional benchmark for vision-and-language manipulation, 2022. URL <https://arxiv.org/abs/2206.08522>.
- Jinguo Zhu, Weiyun Wang, Zhe Chen, Zhaoyang Liu, Shenglong Ye, Lixin Gu, Hao Tian, Yuchen Duan, Weijie Su, Jie Shao, Zhangwei Gao, Erfei Cui, Xuehui Wang, Yue Cao, Yangzhou Liu, Xingguang Wei, Hongjie Zhang, Haomin Wang, Weiye Xu, Hao Li, Jiahao Wang, Nianchen Deng, Songze Li, Yinan He, Tan Jiang, Jiapeng Luo, Yi Wang, Conghui He, Botian Shi, Xingcheng Zhang, Wenqi Shao, Junjun He, Yingting Xiong, Wenwen Qu, Peng Sun, Penglong Jiao, Han Lv, Lijun Wu, Kaipeng Zhang, Huipeng Deng, Jiaye Ge, Kai Chen, Limin Wang, Min Dou, Lewei Lu, Xizhou Zhu, Tong Lu, Dahua Lin, Yu Qiao, Jifeng Dai, and Wenhui Wang. Internv13: Exploring advanced training and test-time recipes for open-source multimodal models, 2025. URL <https://arxiv.org/abs/2504.10479>.
- Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong T. Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R. Sanketi, Grecia Salazar, Michael S. Ryoo, Krista Reymann, Kanishka Rao, Karl Pertsch, Igor Mordatch, Henryk Michalewski, Yao Lu, Sergey Levine, Lisa Lee, Tsang-Wei Edward Lee, Isabel Leal, Yuheng Kuang, Dmitry Kalashnikov, Ryan Julian, Nikhil J. Joshi, Alex Irpan, Brian Ichter, Jasmine Hsu, Alexander Herzog, Karol Hausman, Keerthana Gopalakrishnan, Chuyuan Fu, Pete Florence, Chelsea Finn, Kumar Avinava Dubey, Danny Driess, Tianli Ding, Krzysztof Marcin Choromanski, Xi Chen, Yevgen Chebotar, Justice Carbajal, Noah Brown, Anthony Brohan, Montserrat Gonzalez Arenas, and Kehang Han. RT-2: vision-language-action models transfer web knowledge to robotic control. In *Conference on Robot Learning, CoRL 2023, 6-9 November 2023, Atlanta, GA, USA*, volume 229 of *Proceedings of Machine Learning Research*, pp. 2165–2183. PMLR, 2023.

972	A Related Work	21
973	A.1 Detailed Analysis of Limitations in Prior Benchmarks	22
974		
975		
976	B Task Design and Descriptions	22
977	B.1 Relative Positioning	23
978	B.2 Depth & Occlusion	24
979	B.3 Orientation	24
980	B.4 Size & Scale	24
981	B.5 Spatial Navigation	25
982	B.6 3D Geometry	25
983		
984		
985		
986		
987	C More Details on Benchmark Construction	25
988	C.1 Annotator Recruitment	25
989	C.2 Annotator Training and Protocol	26
990		
991		
992	D More details on Experiments and Evaluation	26
993	D.1 Multiple Choice Evaluation	26
994	D.2 Open-Ended Evaluation	26
995	D.2.1 Evaluation of LLM-as-a-Judge for Open-ended Evaluation	26
996		
997		
998		
999	E Error Analysis of SPATIALAB-MCQ	28
1000	E.1 Sub-Category-wise Quantitative Error Analysis	28
1001	E.2 Model Failure Analysis	31
1002		
1003		
1004	F Error Analysis of SPATIALAB-OPEN	32
1005	F.1 Sub-Category-wise Quantitative Error Analysis	32
1006	F.2 Model Failure Analysis	36
1007		
1008		
1009	G Analysis of MCQ vs. Open-Ended Performance Gaps	36
1010	G.1 Quantitative snapshot	36
1011	G.2 Patterns and observations	37
1012	G.3 Hypotheses on root causes	38
1013	G.4 Practical implications	39
1014		
1015		
1016		
1017	H Details on Improving Visual Reasoning Capabilities	39
1018	H.1 Inherent Reasoning Mechanisms	39
1019	H.2 Chain-of-Thought (CoT) Prompting	40
1020	H.3 Chain-of-Thought (CoT) Prompting with Self-Reflection	41
1021	H.4 Supervised Fine-Tuning (SFT)	42
1022	H.4.1 Setup and Implementation	42
1023	H.4.2 Supervised Fine-Tuning (SFT) Performance	43
1024		
1025		

1026	H.4.3 SFT Dynamics Analysis	44
1027	H.4.4 Performance Gain on External Benchmarks	45
1028		
1029	H.5 AI Agents for Spatial Reasoning	46
1030	H.5.1 Agents	46
1031		
1032	H.5.2 Technical Implementation Details	47
1033	H.5.3 Evaluation and Analysis	48
1034		
1035		
1036	I Qualitative Error Analysis and Error Patterns	49
1037	I.1 Overview	49
1038	I.2 Error taxonomy and analysis	49
1039		
1040	I.2.1 Spatial mislocalization and reference confusion.	49
1041	I.2.2 Perspective and scale mistakes.	49
1042	I.2.3 Occlusion and ordering failures.	50
1043	I.2.4 Attribute confusion and semantic swap.	50
1044	I.2.5 Open-ended rationalization without verification.	51
1045		
1046	I.3 Error patterns and cross-model signals	51
1047	I.4 Mitigation strategies	51
1048		
1049		
1050		
1051	J Representative Benchmark Samples Across All Sub-Categories	56
1052		
1053	K Statistical Robustness and Dataset Stability	86
1054	K.1 Model Robustness	86
1055	K.1.1 Multiple Run Averages and Deviations	86
1056	K.1.2 Intra-Class Correlation (ICC)	86
1057		
1058	K.2 Dataset Stability and Internal Consistency	86
1059		
1060	K.2.1 Resampling Study	87
1061	K.2.2 Item-Level Consistency	87
1062	K.2.3 Cronbach’s Alpha	87
1063		
1064	K.3 Empirical Results	88
1065	K.3.1 Model Evaluation Robustness	88
1066	K.3.2 Dataset Stability and Internal Consistency	89
1067	K.3.3 Overall Discussion	90
1068		
1069		
1070		
1071		
1072		
1073		
1074		
1075		
1076		
1077		
1078		
1079		

A RELATED WORK

Spatial reasoning in multimodal models. The related-work contribution of this section is to clarify why SPATIALAB is needed despite rapid progress in multimodal modeling. Spatial reasoning—the ability to understand geometric structures and relational layouts in visual scenes—is central to embodied intelligence. Early multimodal benchmarks such as CLEVR (Johnson et al., 2016) and GQA (Hudson & Manning, 2019) tested compositional reasoning through synthetic or curated templates, enabling controlled analysis but limiting ecological validity. Large VLMs (Li et al., 2023; Alayrac et al., 2022; OpenAI, 2023) have renewed interest in more naturalistic spatial reasoning, yet systematic weaknesses remain: models that perform well on captioning and visual QA still fail on depth ordering, occlusion inference, and reference-frame alignment. This motivates our central positioning claim that scaling alone does not guarantee structured spatial understanding.

Spatial reasoning benchmarks. Recent benchmarks probe parts of this gap, but their scope leaves room for a more comprehensive diagnostic benchmark (Table 1). SCANQA (Azuma et al., 2022) was an early embodied QA dataset for 3D indoor environments, but its limited scale (800 scenes) and shallow complexity restrict generalization. VISUAL SPATIAL (Liu et al., 2023) and WHAT’S UP (Kamath et al., 2023) expand toward real-world or household contexts, but template-driven questions reduce linguistic variability and emphasize simple binary or multiple-choice relationships. More recent datasets, including EMBSPATIAL-BENCH (Du et al., 2024) and SPACE3D-BENCH (Szymanska et al., 2024), increase task variety but remain constrained by scale (2.2K and 211 samples, respectively) and annotation style. SPATIALRGPT-BENCH and BLINK-SPATIAL add mixed domains and up to 14 categories, improving diversity while still focusing on static snapshots or limited multi-view simulation. The contribution gap is therefore not the absence of spatial benchmarks, but the lack of a balanced, real-world, fine-grained benchmark with both MCQ and open-ended evaluation.

Embodiment, difficulty, and diversity. Embodied datasets further illustrate the trade-offs that SPATIALAB addresses. ROBOSPATIAL (Song et al., 2024) scales to 1M robot-centric scenes, but template-based annotation can make questions easier to solve through heuristics. SPATIALVLM (Chen et al., 2024b) provides very large scale with billions of QA pairs, but its two-category coverage limits fine-grained diagnosis. High-quality manual datasets such as VSI-BENCH (Yang et al., 2024) improve annotation fidelity, but their small scale, including 288 scenes, limits training utility and robust cross-model comparison. In complexity terms, many benchmarks remain at “medium” (Mm in Table 1) and rarely reach “hard” or “extremely hard” cases that require multi-step causal inference. SPATIALAB is positioned to fill this evaluation gap through real-world imagery, category breadth, and challenging spatial compositions.

Evaluation protocols. Evaluation format is another dimension where our contribution is intentionally different. Most prior benchmarks rely on closed-ended QA or binary judgments, such as VISUAL SPATIAL and EMBSPATIAL-BENCH; these formats simplify scoring but constrain output expressivity. A smaller set of datasets, including ROBOSPATIAL, SPATIAL-MM, and SPATIALAB-OPEN, support open-ended evaluation, but SPATIALAB-OPEN pairs this format with high task complexity, manual puzzle-style annotation, and comprehensive error analysis. This combination enables analysis beyond aggregate accuracy, including calibration, cross-model divergence, and systematic failure modes. Without such granularity, reported gains can be misleading because models may exploit dataset-specific biases rather than develop robust spatial competence.

Positioning of SPATIALAB. SPATIALAB directly addresses the limitations identified above. Compared with prior datasets spanning 2 to 14 task categories, SPATIALAB covers 30 distinct categories across geometry, occlusion, orientation, relative positioning, size and scale, and navigation. Unlike template-based benchmarks, its questions are manually authored and puzzle-inspired, increasing linguistic richness and reducing shortcut opportunities. It is also embodied and multimodal, including both static and dynamic visual contexts, thereby bridging robotics-style datasets such as ROBOSPATIAL and web-scale corpora such as SPATIALVLM. As shown in Table 1, even strong baselines such as InternVL3.5-72B and GPT-5-mini remain below 60%, with the open-ended variant reaching only 40.93% accuracy. This performance gap makes the benchmark difficult enough to support future progress rather than merely rank saturated systems.

Toward next-generation evaluation. By combining difficulty, coverage, manual annotation, and open-ended tasks, SPATIALAB extends spatial reasoning evaluation from narrow template-driven tests toward comprehensive embodied diagnosis. Its design supports not only correctness measurement but also error-pattern analysis, causal and counterfactual probing, and calibration reliability. In this role, SPATIALAB complements datasets such as MIND THE GAP and VLM4D, which explore temporal or generative aspects while remaining narrower in coverage. The intended contribution is a high-bar diagnostic tool that encourages models to move beyond surface recognition toward grounded spatial intelligence.

A.1 DETAILED ANALYSIS OF LIMITATIONS IN PRIOR BENCHMARKS

To make the positioning explicit, we further summarize the structural limitations of representative spatial reasoning benchmarks. Earlier efforts provide important foundations, but they remain constrained by synthetic-domain bias, limited task scope, and restrictive evaluation formats. Table 4 presents these limitations in a reviewer-facing form and clarifies how SPATIALAB is designed to address them.

Synthetic and Template-Based Constraints. Benchmarks such as CLEVR and GQA rely on synthetic environments or template-generated language. These settings are useful for isolating compositional reasoning, but the visual domain lacks real-world complexity and the language space is narrow. Models can therefore exploit template regularities instead of grounded spatial interpretation. Simulated benchmarks such as ROBOSPATIAL face related constraints, which can produce a sim-to-real gap.

Domain and Scale Limitations. Datasets based on 3D reconstruction, including ScanQA and Space3D-Bench, provide depth supervision but remain restricted to indoor scenes. This prevents evaluation in outdoor or mixed environments that matter for embodied agents. High-quality manual datasets such as VSI-Bench provide richer annotations, but their small scale limits high-resolution subcategory analysis.

Taxonomic and Evaluation Narrowness. Large-scale datasets such as SpatialVLM cover broad tasks but lack detailed taxonomies, making them less suitable for diagnosing specific weaknesses in orientation, occlusion, or layout reasoning. Many recent benchmarks, including EMBSPATIAL-BENCH and WHAT’S UP, rely heavily on Binary or MCQ formats, which can inflate performance through option elimination rather than requiring generative spatial reasoning. SPATIALAB mitigates these issues by combining MCQ and Open-Ended formats across a detailed taxonomy of 30 spatial subcategories.

Table 4: Reviewer-facing summary of prior spatial benchmark limitations compared with SPATIALAB. The comparison highlights domain constraints, evaluation format, and task taxonomy as the main dimensions addressed by our benchmark contribution.

Benchmark	Domain	Eval. Format	Taxonomy	Primary Limitations Identified
CLEVR and GQA	Synthetic	Closed-Ended	Relational	Artificial visual domain; template-driven language; prone to shortcut learning.
ScanQA	Indoor 3D	Open-Ended	Navigation	Restricted to indoor point clouds; limited scale ($< 1K$ scenes).
SpatialVLM	Web Data	Open-Ended	General	Covers only a small number of high-level spatial categories.
EmbSpatial-Bench	Indoor	MCQ / Binary	6 Categories	Relies on closed-ended formats; limited complexity in reasoning chains.
RoboSpatial	Simulator	Hybrid	Embodied	Template-generated questions in simulated scenes produce a sim-to-real gap.
Space3D-Bench	Indoor	MCQ	6 Categories	Very small scale (~ 200 samples) prevents robust statistical evaluation.
OmniSpatial	Mix	MCQ / Binary	Metric	Emphasizes metric outputs; limited support for explanatory reasoning.
SpatialLab (Ours)	In-the-Wild	MCQ + Open	30 Sub-cats	Addresses identified limitations with real-world images, detailed taxonomy, and diverse evaluation formats.

B TASK DESIGN AND DESCRIPTIONS

This appendix makes the task-design contribution of SPATIALAB explicit. Spatial reasoning is not simply object recognition; it requires models to infer how objects relate, align, occlude, support, and constrain one another in physical space. SPATIALAB therefore decomposes spatial understanding into everyday but fine-grained tasks, from relative positioning (e.g., *What’s to the left of the microwave?*) to geometric inference (e.g., *Would this stack fall if another book is added?*). Each category is divided into subcategories that expose specific reasoning demands such as occlusion, orientation, navigation, and gravity. This structure helps reviewers see exactly what the benchmark measures and lets the

results diagnose whether models generalize across settings relevant to robotics, AR/VR, autonomous driving, and human-computer interaction.

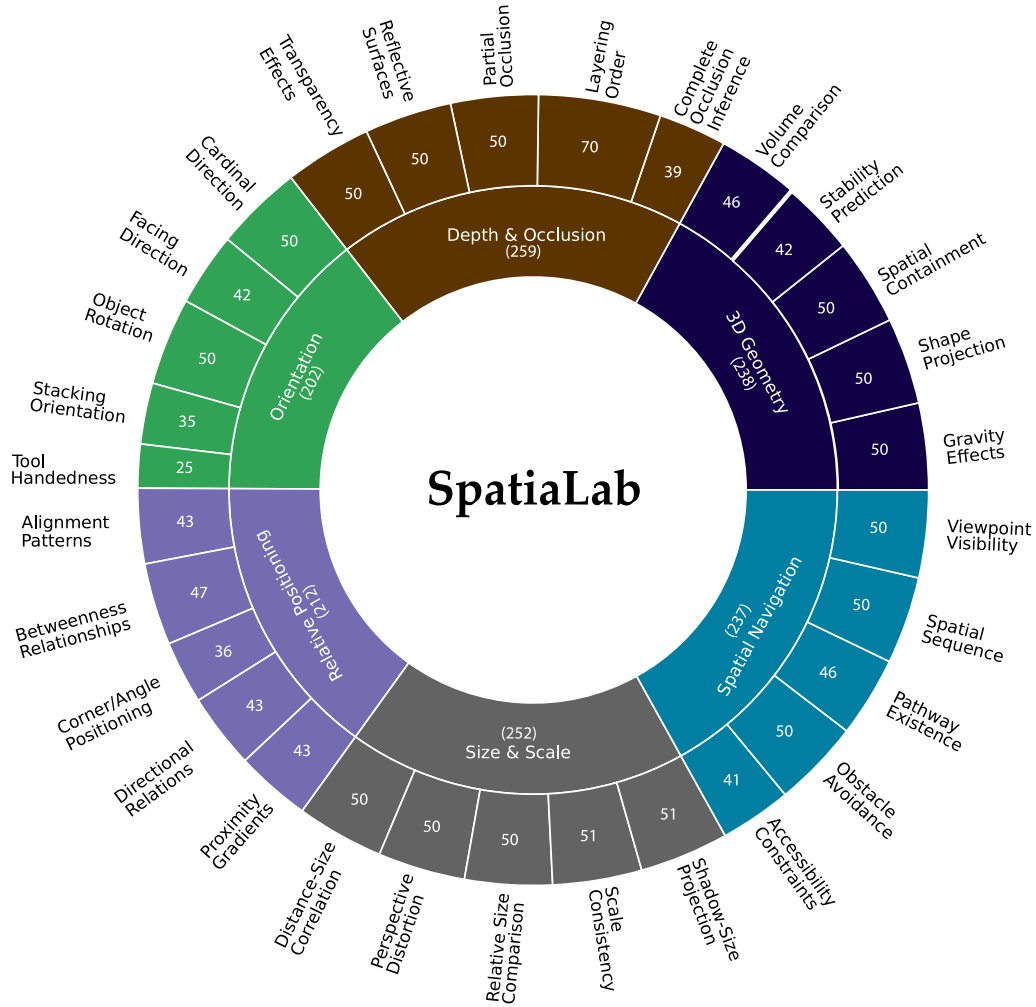


Figure 5: **Task taxonomy of SPATIALAB.** The benchmark contribution is a six-category structure, with five subcategories per category and thirty task types in total.

The following descriptions summarize the rationale and real-world utility of each task type.

B.1 RELATIVE POSITIONING

Directional Relations. This subcategory tests whether models can identify relative locations such as left, right, above, or below. The contribution of these items is to measure the most basic relational grounding needed for scene description and robot navigation, such as identifying what lies immediately to the left of an appliance.

Proximity Gradients. Proximity items ask which objects are closest or farthest from a reference point. They contribute a graded distance-reasoning test for questions such as “Which object is nearest to the doorway?” and are relevant to navigation and assistive technologies.

Alignment Patterns. Alignment items evaluate whether objects follow an orientation or order, such as vertical stacking or straight-line arrangement. This subcategory contributes a test of environmental organization in shelves, tables, and warehouses, where robotic sorting and automated inspection require stable alignment judgments.

1242 **Betweenness Relationships.** Betweenness items ask models to identify objects occupying the middle
1243 region between two references. This contributes a continuity-based relational test, such as determining
1244 what sits between a lamp and an armchair, with implications for navigation, retrieval, and situational
1245 awareness.

1246 **Corner/Angle Positioning.** Corner and angle items use room or scene geometry as reference anchors.
1247 They contribute localized spatial tests such as “What’s positioned in the back-right corner?” that
1248 matter for mapping, placement, and robotics.
1249

1250 1251 B.2 DEPTH & OCCLUSION 1252

1253 **Layering Order.** Layering items test whether models can distinguish objects in front from objects
1254 behind. This contributes direct depth-order evaluation, which is important for AR/VR applications
1255 where visual stacking affects user experience.

1256 **Partial Occlusion.** Partial-occlusion items measure whether models can reason when one object
1257 hides part of another. The contribution is to test robustness in cluttered scenes where missing visual
1258 evidence must be completed from context.

1259 **Complete Occlusion Inference.** Complete-occlusion items ask models to infer hidden objects from
1260 indirect evidence such as shadows or bulges. This contributes a test of uncertainty-aware visual
1261 inference that mirrors human reasoning in incomplete scenes.

1262 **Transparency Effects.** Transparency items evaluate whether models can reason through glass, water,
1263 or other transparent surfaces. They contribute cases where occlusion reasoning must be combined
1264 with material perception, which matters for simulation, graphics, and embodied AI.
1265

1266 **Reflective Surfaces.** Reflection items test indirect spatial mapping through mirrors, shiny metals, or
1267 glass. This contributes a difficult non-local reasoning setting relevant to autonomous-vehicle safety
1268 and robotic situational awareness.
1269

1270 1271 B.3 ORIENTATION 1272

1273 **Cardinal Direction.** Cardinal-direction items require mapping objects to directions such as north,
1274 south, east, and west. They contribute a global-orientation test needed for navigation and geospatial
1275 alignment.

1276 **Object Rotation.** Rotation items measure whether a model can estimate how an object has turned
1277 around an axis. This contributes a 3D modeling and manipulation test, because robots often need
1278 accurate object orientation before acting.

1279 **Facing Direction.** Facing-direction items ask models to identify the front-facing orientation of an
1280 object. They contribute a surface-orientation test for surveillance, robotics, and reference-frame
1281 alignment.

1282 **Stacking Orientation.** Stacking-orientation items test whether objects are arranged vertically,
1283 horizontally, or in mixed orientations. This contributes a stability-relevant layout test for warehouses,
1284 homes, and robotic stacking.

1285 **Tool Handedness.** Tool-handedness items require fine-grained shape and orientation perception for
1286 objects such as scissors or gloves. They contribute a human-centered manipulation test for assistive
1287 robotics.
1288

1289 1290 B.4 SIZE & SCALE 1291

1292 **Relative Size Comparison.** Relative-size items ask whether one object is larger, taller, or smaller
1293 than another. They contribute proportional reasoning tests relevant to packing, fitting, and recognition.

1294 **Perspective Distortion.** Perspective-distortion items test whether models can discount image-size
1295 illusions caused by camera viewpoint. This contributes robustness evaluation for cases such as a
distant car appearing smaller than a nearby bicycle.

Distance-Size Correlation. Distance-size items require joint reasoning over perceived size and object distance. They contribute a navigation and visual-grounding test where apparent size should not be confused with true size.

Scale Consistency. Scale-consistency items ask whether object sizes are plausible in context. They contribute realism checks for simulated and real scenes, such as detecting when a mouse appears larger than a chair.

Shadow-Size Projection. Shadow-size items use shadows as indirect evidence about object size and light position. This contributes a geometry-based reasoning test relevant to graphics, architecture, and physics-aware modeling.

B.5 SPATIAL NAVIGATION

Pathway Existence. Pathway items ask whether a clear route exists between two points. They contribute a navigability test for indoor planning, such as determining whether a desk can be reached without moving chairs.

Obstacle Avoidance. Obstacle-avoidance items require identifying routes that minimize collision risk. They contribute a planning-oriented test for autonomous driving, indoor mobility aids, and robot navigation.

Viewpoint Visibility. Viewpoint-visibility items ask what is visible or blocked from a specified viewpoint. They contribute perspective-simulation tests needed for AR/VR realism and robotic vision.

Spatial Sequence. Sequence items require ordering the objects encountered along a path. They contribute a navigation-memory test analogous to knowing what one passes when walking from a kitchen to a balcony.

Accessibility Constraints. Accessibility items identify objects that cannot be reached because of obstacles. They contribute a practical reasoning test for assistive AI and ergonomic design.

B.6 3D GEOMETRY

Volume Comparison. Volume-comparison items ask which object occupies more three-dimensional space. They contribute a containment and logistics test for packing, simulation, and design.

Stability Prediction. Stability items require predicting whether adding or moving objects would topple a stack. They contribute a physical-intuition test relevant to safety in robotics and construction.

Shape Projection. Shape-projection items ask what two-dimensional shape an object would cast under rotation or lighting. They contribute a geometric-abstraction test, such as distinguishing projections of spheres and cubes.

Spatial Containment. Containment items ask whether one object can fit inside another. They contribute intuitive geometry tests for packing, design, and object interaction.

Gravity Effects. Gravity items require predicting motion under physical forces. They contribute a real-world simulation test, such as whether a rolling ball stops under a table because of friction and barriers.

C MORE DETAILS ON BENCHMARK CONSTRUCTION

C.1 ANNOTATOR RECRUITMENT

This section details the human-annotation contribution behind SPATIALAB. We recruited annotators aged 24 to 30 years, all of whom were engineering graduates. They participated voluntarily and worked independently without external incentives. Before annotation began, we provided structured training so that task requirements, category definitions, and quality expectations were consistent across annotators. Ethical and quality-control standards were maintained throughout the process, supporting reliable annotations aligned with the benchmark design.

C.2 ANNOTATOR TRAINING AND PROTOCOL

The training protocol was designed to make the benchmark taxonomy operational. We first introduced annotators to the six core categories (*Relative Positioning, Depth & Occlusion, Orientation, Size & Scale, Spatial Navigation, and 3D Geometry*) and their thirty subcategories. For each subcategory, annotators reviewed curated examples of the target spatial relation, such as directional relations, occlusion inference, and stability prediction, along with counter-examples that illustrated common mistakes. This ensured that the annotation process preserved the intended contribution of fine-grained spatial diagnosis.

We also trained annotators to use six meta-dimensions of image variability: lighting, texture complexity, edge density, dominant relation, material type, and gravity constraints. These dimensions encouraged question writers to use visual diversity rather than ignore it. Training examples showed how real-world factors interact, such as occlusion under low-contrast lighting or scale consistency under perspective distortion, so that questions would stress realistic spatial reasoning rather than simplified recognition.

The protocol further covered the two evaluation formats. In multiple-choice settings, annotators designed distractors that test fine-grained discrimination rather than superficial elimination. In open-ended settings, they wrote prompts that encourage compositional spatial reasoning. Calibration sessions required annotators to generate both formats for sample images and refine them through feedback, making the dual-format evaluation contribution consistent across the dataset.

To situate task design within prior work, we reviewed benchmarks such as SpatialBot-Bench, EmbSpatial, and SpatialMM with annotators and discussed their limitations. These resources often emphasize simplified relations, synthetic or staged scenes, and narrow reasoning types, while underrepresenting shadows, reflections, and gravity-sensitive stability. This gap analysis clarified why SPATIALAB prioritizes richer, more ambiguous, and dynamically layered spatial contexts.

Finally, annotators received detailed demonstrations of the full annotation protocol. These demonstrations specified how to select images with sufficient spatial complexity, map each question unambiguously to a subcategory, and avoid trivial formulations. Annotators were instructed to prioritize multi-step reasoning, such as depth layering, relational chains, and cross-view scale inference, over binary labels. Through systematic training and iterative feedback, the final dataset balances taxonomic coverage with real-world relevance.

D MORE DETAILS ON EXPERIMENTS AND EVALUATION

D.1 MULTIPLE CHOICE EVALUATION

For MCQ evaluation, Figure 6 shows the answer-model prompt used to standardize option selection across systems. We set `temperature` to 0 and `max_tokens` to 10, while keeping all other generation parameters at their default values, so that the protocol measures discriminative spatial accuracy with minimal decoding variability.

D.2 OPEN-ENDED EVALUATION

For open-ended evaluation, Figure 6 reports both the answer-model prompt and the judge-model prompt. We set `temperature` to 0.7 and `max_tokens` to 1200, with all other parameters kept at their default values, allowing models to produce free-form spatial answers while keeping the evaluation protocol fixed.

D.2.1 EVALUATION OF LLM-AS-A-JUDGE FOR OPEN-ENDED EVALUATION

To validate the open-ended evaluation contribution, we evaluated *Gemini-2.5-Flash* as an automatic judge on a balanced set of 240 samples covering all categories. Each sample included the question, ground truth, and model output, and three human annotators independently assigned binary scores: 1 for correct and 0 for incorrect. We compared the LLM judgments with human annotations using complementary agreement metrics, computed overall and per category. Specifically, we report **Cohen’s Kappa**, **Accuracy with Majority Vote**, and **Fleiss’ Kappa**, defined as follows.



Figure 6: Evaluation prompts used to support the dual-format SPATIALAB protocol.

Cohen’s Kappa. Cohen’s kappa (κ) provides the pairwise agreement measure used to check whether judge decisions align with human scores beyond chance. For binary task, it is defined as $\kappa = \frac{p_o - p_e}{1 - p_e}$, where p_o is the observed agreement and p_e is the expected agreement by chance: $p_e = \sum_{c \in \{0,1\}} p_c^{(1)} p_c^{(2)}$. Here, $p_c^{(1)}$ and $p_c^{(2)}$ are the marginal probabilities that each rater assigns class c . A value of $\kappa = 1$: perfect agreement; $\kappa = 0$: agreement equals chance; $\kappa < 0$: systematic disagreement) We computed Cohen’s kappa between the LLM and each annotator individually ($\kappa_{\text{Human1}}, \kappa_{\text{Human2}}, \kappa_{\text{Human3}}$), as well as between the LLM and the majority vote of the annotators (κ_{majority}).

Accuracy with Majority Vote. We also report raw agreement between the LLM and the annotator majority, which makes the judge validation easy to interpret. Accuracy is defined as $\text{Accuracy} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}(y_i^{\text{LLM}} = y_i^{\text{maj}})$, where y_i^{LLM} is the LLM’s decision for sample i , and y_i^{maj} is the majority vote among annotators. Unlike Cohen’s kappa, accuracy does not correct for chance agreement; it simply reflects the proportion of identical labels.

Fleiss’ Kappa. To establish that the human reference labels are themselves reliable, we computed Fleiss’ kappa (κ_F), which generalizes Cohen’s kappa to multiple raters. Let n_{ij} denote the number of annotators assigning category j to item i , and define $P_i = \frac{1}{n(n-1)} \sum_j n_{ij}(n_{ij} - 1)$, where n is the number of annotators. The observed agreement is $\bar{P} = \frac{1}{N} \sum_{i=1}^N P_i$, and the expected agreement is $\bar{P}_e = \sum_j p_j^2$, with $p_j = \frac{1}{Nn} \sum_i n_{ij}$. Fleiss’ kappa is then computed as $\kappa_F = \frac{\bar{P} - \bar{P}_e}{1 - \bar{P}_e}$. Higher values indicate stronger inter-annotator reliability beyond chance.

Interpretation Table 5 supports the reliability of the open-ended scoring protocol. The LLM judge (Gemini-2.5-Flash) achieves a Cohen’s kappa of 0.738 with the human majority vote and 0.681–0.795 against individual annotators. Raw accuracy against the majority is 0.880, and Fleiss’ kappa among human annotators is 0.774, indicating substantial consistency in the reference judgments. Some categories, such as 3D Geometry, reach perfect agreement with $\kappa = 1.000$ and accuracy of 1.000. Depth & Occlusion also shows high agreement ($\kappa_{\text{majority}} = 0.805$, accuracy = 0.909), while

Table 5: Judge-validation results for open-ended SPATIALAB evaluation. The table reports agreement between Gemini-2.5-Flash and three human annotators using Cohen’s kappa, raw majority-vote accuracy, and Fleiss’ kappa; bold values mark the highest agreement within each category.

Category	κ Majority	κ Human1	κ Human2	κ Human3	Acc. (majority)	Fleiss κ (all)
Overall	0.738	0.738	0.681	0.795	0.880	0.774
3D Geometry	1.000	1.000	0.875	1.000	1.000	0.916
Depth & Occlusion	0.805	0.805	0.899	0.697	0.909	0.871
Orientation	0.500	0.500	0.426	0.588	0.810	0.571
Relative Positioning	0.833	0.657	0.676	0.833	0.917	0.778
Size & Scale	0.417	0.571	0.417	0.696	0.714	0.704
Spatial Navigation	0.815	0.815	0.667	1.000	0.933	0.773

Orientation ($\kappa = 0.500$, accuracy = 0.810) and Size & Scale ($\kappa = 0.417$, accuracy = 0.714) show moderate but acceptable agreement.

These results justify using LLM-as-judge as a practical component of the SPATIALAB-OPEN evaluation contribution. Gemini-2.5-Flash performs comparably to human evaluators in correctness scoring, with most Cohen’s kappa values in the substantial range (0.61–0.80) and some categories reaching almost-perfect agreement (higher than 0.81). Fleiss’ kappa of 0.774 confirms that human annotators are themselves consistent, validating the reference scores. Near-perfect agreement in clearly defined categories such as 3D Geometry ($\kappa = 1.000$, accuracy = 1.000) and Spatial Navigation ($\kappa = 0.815$, accuracy = 0.933) demonstrates that the judge captures many spatial correctness decisions accurately, while moderate agreement in Orientation and Size & Scale reflects task complexity and inherent subjectivity. Overall, the validation shows that automated judging can closely mirror human evaluation for this benchmark.

E ERROR ANALYSIS OF SPATIALAB-MCQ

E.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models. For closed-source systems, the MCQ error analysis shows high ceilings but incomplete coverage (Table 6). **GPT-4o-mini** reaches 85.71% in *Stacking Orientation*, the highest score overall. **Gemini-2.0-Flash** performs well on occlusion and projection, with 64.0% in *Partial Occlusion* and 64.71% in *Shadow-Size Projection*. **GPT-5-mini** also shows strong localized performance, scoring 68.0% in *Tool Handedness* and 55.71% in *Layering Order*. The contribution of this breakdown is to show that strong proprietary training pipelines raise peak performance, but do not produce uniform spatial competence: **Claude 3.5 Haiku** reaches only 32.0% in *Relative Size Comparison*, and within-model variation often exceeds 50 percentage points. Closed-source models outperform open-source systems by 15–25 percentage points in several categories, yet the remaining variance confirms that spatial competence is still uneven.

Open-source Large Models. Large open-source models contribute a different pattern: scaling improves selected subcategories but leaves severe fragilities (Table 7). **InternVL3.5-72B** reaches 80.56% in *Corner/Angle Positioning*, one of the strongest open-source results. Both **InternVL3.5-72B** and **llama-3.2-90b-vision-instruct** score 66.67% in *Complete Occlusion Inference*. However, **llama-3.2-11b-vision-instruct** falls to 2.0% in *Object Rotation*, and gains in *Stacking Orientation*, such as 71.43% for **llama-3.2-90b-vision-instruct**, coexist with losses elsewhere. Within-model spread often exceeds 70 percentage points, and gravity reasoning remains modest at 56.0%. Thus, the diagnostic contribution is clear: model size amplifies selective strengths but does not close foundational representation gaps.

Open-source Small Models. Small open-source models make the brittleness of compact VLMs especially visible (Table 8). **qwen-2.5-vl-7b-instruct** reaches 74.29% in *Stacking Orientation*, while **InternVL3.5-1B** collapses to 12.0% in *Cardinal Direction*. *Pathway Existence* remains difficult, with the best models tied at 34.78%, and gravity-related reasoning reaches only 50.0% for **InternVL3.5-4B** and **Gemma-3-4B-it**. Within-model spread often exceeds 60 percentage points. These results

Table 6: MCQ subcategory diagnostic accuracy on SPATIALAB-MCQ for Closed-source Models. Bold and underline mark the best score within each model category.

Category	Sub-Category	GPT-4o-mini	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	52.0	50.0	46.0	52.0	52.0	46.0
	Shape Projection	54.0	48.0	50.0	50.0	46.0	60.0
	Spatial Containment	36.0	46.0	44.0	38.0	34.0	38.0
	Stability Prediction	52.38	52.38	59.52	50.0	45.24	52.38
	Volume Comparison	41.3	47.83	39.13	34.78	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	61.54	69.23	61.54	61.54	61.54	46.15
	Layering Order	32.86	55.71	50.0	42.86	32.86	45.71
	Partial Occlusion	42.0	54.0	64.0	50.0	44.0	52.0
	Reflective Surfaces	34.0	54.0	46.0	48.0	44.0	44.0
	Transparency Effects	32.0	44.0	60.0	44.0	36.0	62.0
Orientation	Cardinal Direction	26.0	48.0	44.0	30.0	44.0	38.0
	Facing Direction	45.24	57.14	57.14	47.62	35.71	45.24
	Object Rotation	40.0	60.0	50.0	46.0	38.0	44.0
	Stacking Orientation	85.71	77.14	62.86	65.71	68.57	62.86
	Tool Handedness	52.0	68.0	64.0	64.0	56.0	56.0
Relative Positioning	Alignment Patterns	44.19	58.14	55.81	53.49	46.51	58.14
	Betweenness Relationships	51.06	63.83	70.21	63.83	44.68	61.7
	Corner/Angle Positioning	47.22	63.89	50.0	58.33	50.0	77.78
	Directional Relations	48.84	79.07	67.44	69.77	53.49	69.77
	Proximity Gradients	44.19	48.84	44.19	34.88	37.21	44.19
Size and Scale	Distance-Size Correlation	56.0	56.0	56.0	42.0	34.0	42.0
	Perspective Distortion	50.0	40.0	46.0	36.0	38.0	34.0
	Relative Size Comparison	42.0	44.0	58.0	46.0	32.0	38.0
	Scale Consistency	47.06	45.1	47.06	39.22	33.33	41.18
	Shadow-Size Projection	52.94	39.22	64.71	49.02	41.18	52.94
Spatial Navigation	Accessibility Constraints	56.1	60.98	65.85	65.85	48.78	56.1
	Obstacle Avoidance	38.0	48.0	34.0	34.0	46.0	40.0
	Pathway Existence	54.35	41.3	43.48	45.65	39.13	36.96
	Spatial Sequence	66.0	78.0	48.0	62.0	56.0	44.0
	Viewpoint Visibility	36.0	54.0	48.0	50.0	40.0	34.0

Table 7: MCQ subcategory diagnostic accuracy on SPATIALAB-MCQ for Open-source Large Models. Bold and underline mark the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-72b-instruct	InternVL3-5-72B	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct
3D Geometry	Gravity Effects	22.0	48.0	54.0	52.0	52.0	56.0
	Shape Projection	32.0	42.0	44.0	50.0	46.0	50.0
	Spatial Containment	32.0	34.0	36.0	44.0	46.0	36.0
	Stability Prediction	33.33	54.76	38.1	52.38	47.62	52.38
	Volume Comparison	13.04	39.13	32.61	52.17	43.48	36.96
Depth and Occlusion	Complete Occlusion Inference	30.77	61.54	58.97	66.67	48.72	66.67
	Layering Order	34.29	34.29	37.14	50.0	50.0	38.57
	Partial Occlusion	32.0	34.0	44.0	60.0	48.0	58.0
	Reflective Surfaces	28.0	38.0	26.0	54.0	50.0	44.0
	Transparency Effects	26.0	40.0	40.0	60.0	46.0	62.0
Orientation	Cardinal Direction	22.0	48.0	34.0	46.0	48.0	40.0
	Facing Direction	21.43	42.86	52.38	54.76	50.0	50.0
	Object Rotation	2.0	40.0	38.0	48.0	48.0	44.0
	Stacking Orientation	28.57	65.71	65.71	68.57	62.86	71.43
	Tool Handedness	40.0	48.0	52.0	56.0	56.0	56.0
Relative Positioning	Alignment Patterns	27.91	55.81	48.84	53.49	55.81	53.49
	Betweenness Relationships	42.55	55.32	55.32	65.96	61.7	61.7
	Corner/Angle Positioning	58.33	55.56	36.11	80.56	50.0	75.0
	Directional Relations	60.47	62.79	51.16	83.72	62.79	65.12
	Proximity Gradients	27.91	41.86	32.56	48.84	39.53	41.86
Size and Scale	Distance-Size Correlation	36.0	50.0	54.0	52.0	44.0	46.0
	Perspective Distortion	22.0	42.0	36.0	46.0	34.0	52.0
	Relative Size Comparison	28.0	48.0	40.0	48.0	52.0	46.0
	Scale Consistency	29.41	47.06	41.18	43.14	37.25	37.25
	Shadow-Size Projection	37.25	52.94	54.9	56.86	50.98	52.94
Spatial Navigation	Accessibility Constraints	29.27	58.54	43.9	75.61	53.66	43.9
	Obstacle Avoidance	24.0	38.0	38.0	46.0	42.0	40.0
	Pathway Existence	28.26	54.35	36.96	39.13	54.35	47.83
	Spatial Sequence	34.0	52.0	56.0	66.0	54.0	62.0
	Viewpoint Visibility	44.0	36.0	34.0	50.0	42.0	48.0

indicate that small models are cue-dependent: they perform well where visual templates are strong but fail when tasks require deeper spatial abstraction. The subcategory analysis therefore contributes evidence that compact architectures can exhibit localized competence without robustness across the full taxonomy.

Reasoning Models. Reasoning-enhanced models show that explicit reasoning helps some MCQ tasks, but not enough to provide broad spatial competence (Table 9). **o4-mini** scores 65.12% in *Alignment Patterns* and 56.0% in *Gravity Effects*. **Gemini-2.0-Flash-Thinking** reaches 62.75% in *Shadow-Size Projection* and 52.17% in *Volume Comparison*. The strongest reasoning results come from **Gemini-2.5-Flash-Thinking**, with 71.79% in *Complete Occlusion Inference* and 70.21% in *Betweenness Relationships*. Yet the same family still fails in other cases, including 34.0% in *Cardinal Direction* and 34.0% in *Shape Projection*. Within-model spreads frequently exceed 35 percentage points, supporting the paper’s diagnostic claim that reasoning augmentation adds localized capacity while leaving structural gaps in spatial understanding.

Table 8: MCQ subcategory diagnostic accuracy on SPATIALAB-MCQ for Open-source Small Models. Bold and underline mark the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	46.0	38.0	48.0	50.0	50.0	50.0
	Shape Projection	30.0	42.0	38.0	42.0	40.0	44.0
	Spatial Containment	30.0	28.0	34.0	34.0	36.0	34.0
	Stability Prediction	30.95	35.71	50.0	52.38	45.24	52.38
	Volume Comparison	30.43	26.09	36.96	36.96	47.83	34.78
Depth and Occlusion	Complete Occlusion Inference	30.77	30.77	41.03	64.1	33.33	41.03
	Layering Order	30.0	24.29	30.0	32.86	22.86	27.14
	Partial Occlusion	38.0	30.0	38.0	54.0	46.0	42.0
	Reflective Surfaces	38.0	34.0	36.0	32.0	42.0	40.0
	Transparency Effects	26.0	42.0	36.0	40.0	32.0	44.0
Orientation	Cardinal Direction	12.0	18.0	36.0	24.0	38.0	30.0
	Facing Direction	21.43	28.57	52.38	40.48	42.86	38.1
	Object Rotation	12.0	28.0	42.0	34.0	42.0	34.0
	Stacking Orientation	45.71	57.14	62.86	68.57	68.57	74.29
	Tool Handedness	40.0	36.0	40.0	60.0	48.0	48.0
Relative Positioning	Alignment Patterns	27.91	30.23	39.53	44.19	41.86	41.86
	Betweenness Relationships	38.3	38.3	46.81	65.96	46.81	53.19
	Corner/Angle Positioning	50.0	55.56	47.22	69.44	55.56	50.0
	Directional Relations	51.16	58.14	27.91	60.47	46.51	48.84
	Proximity Gradients	20.93	23.26	39.53	34.88	39.53	37.21
Size and Scale	Distance-Size Correlation	32.0	20.0	40.0	36.0	38.0	36.0
	Perspective Distortion	26.0	40.0	42.0	38.0	44.0	44.0
	Relative Size Comparison	34.0	32.0	46.0	38.0	30.0	42.0
	Scale Consistency	27.45	35.29	41.18	31.37	27.45	41.18
	Shadow-Size Projection	39.22	35.29	66.67	39.22	47.06	47.06
Spatial Navigation	Accessibility Constraints	36.59	29.27	51.22	41.46	39.02	43.9
	Obstacle Avoidance	24.0	26.0	38.0	38.0	38.0	36.0
	Pathway Existence	34.78	30.43	32.61	34.78	34.78	28.26
	Spatial Sequence	24.0	42.0	40.0	56.0	42.0	44.0
	Viewpoint Visibility	36.0	34.0	36.0	40.0	36.0	26.0

Table 9: MCQ subcategory diagnostic accuracy on SPATIALAB-MCQ for Reasoning Models. Bold and underline mark the best score within each model category.

Category	Sub-Category	o4-mini	Gemini-2.0-Flash-Thinking	Gemini-2.5-Flash-Thinking	Kimi-VL-A3B-Thinking-2506
3D Geometry	Gravity Effects	56.0	38.0	46.0	50.0
	Shape Projection	50.0	36.0	40.0	34.0
	Spatial Containment	56.0	28.0	48.0	42.0
	Stability Prediction	50.0	35.71	50.0	50.0
	Volume Comparison	43.48	52.17	45.65	39.13
Depth and Occlusion	Complete Occlusion Inference	69.23	51.28	71.79	51.28
	Layering Order	57.14	30.0	41.43	35.71
	Partial Occlusion	50.0	42.0	58.0	40.0
	Reflective Surfaces	54.0	40.0	52.0	42.0
	Transparency Effects	64.0	50.0	54.0	42.0
Orientation	Cardinal Direction	46.0	34.0	50.0	28.0
	Facing Direction	50.0	42.86	54.76	45.24
	Object Rotation	56.0	36.0	48.0	30.0
	Stacking Orientation	71.43	57.14	57.14	65.71
	Tool Handedness	56.0	44.0	60.0	44.0
Relative Positioning	Alignment Patterns	65.12	41.86	44.19	34.88
	Betweenness Relationships	65.96	55.32	70.21	63.83
	Corner/Angle Positioning	75.0	55.56	61.11	47.22
	Directional Relations	65.12	44.19	65.12	60.47
	Proximity Gradients	51.16	32.56	41.86	48.84
Size and Scale	Distance-Size Correlation	44.0	42.0	56.0	44.0
	Perspective Distortion	40.0	44.0	50.0	36.0
	Relative Size Comparison	46.0	46.0	58.0	32.0
	Scale Consistency	39.22	56.86	52.94	43.14
	Shadow-Size Projection	35.29	62.75	58.82	43.14
Spatial Navigation	Accessibility Constraints	56.1	41.46	53.66	31.71
	Obstacle Avoidance	32.0	44.0	44.0	36.0
	Pathway Existence	45.65	41.3	47.83	34.78
	Spatial Sequence	64.0	48.0	68.0	54.0
	Viewpoint Visibility	60.0	40.0	54.0	48.0

Spatial Reasoning Models. Spatially specialized models provide an important negative result: task specialization does not automatically produce generality (Table 10). **SpaceQwen2.5-VL-3B-Instruct** achieves 48.0% in *Distance-Size Correlation* and 49.02% in *Scale Consistency*, surpassing many open-source baselines on size-scale reasoning. It also leads in navigation tasks, with 53.66% in *Accessibility Constraints* and 44.0% in *Obstacle Avoidance*. **SpaceOm** and **SpaceThinker-Qwen2.5VL-3B** reach 50.0% in *Stability Prediction*. However, performance drops sharply elsewhere, including 24.0% in *Spatial Containment*, and these models rarely exceed 55.0% even in their strongest areas. The subcategory spread frequently surpasses 30 percentage points, showing that specialization creates narrow peaks rather than holistic spatial competence.

Table 10: MCQ subcategory diagnostic accuracy on SPATIALAB-MCQ for Spatial Reasoning Models. Bold and underline mark the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5-VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	48.0	46.0	32.0
	Shape Projection	38.0	36.0	36.0
	Spatial Containment	40.0	36.0	24.0
	Stability Prediction	50.0	50.0	28.57
	Volume Comparison	36.96	34.78	36.96
Depth and Occlusion	Complete Occlusion Inference	38.46	35.9	38.46
	Layering Order	30.0	32.86	24.29
	Partial Occlusion	44.0	46.0	38.0
	Reflective Surfaces	44.0	44.0	38.0
	Transparency Effects	40.0	32.0	42.0
Orientation	Cardinal Direction	40.0	38.0	30.0
	Facing Direction	59.52	54.76	50.0
	Object Rotation	40.0	42.0	28.0
	Stacking Orientation	62.86	62.86	40.0
	Tool Handedness	40.0	40.0	48.0
Relative Positioning	Alignment Patterns	39.53	44.19	39.53
	Betweenness Relationships	44.68	42.55	53.19
	Corner/Angle Positioning	41.67	41.67	38.89
	Directional Relations	25.58	25.58	20.93
	Proximity Gradients	37.21	37.21	34.88
Size and Scale	Distance-Size Correlation	40.0	44.0	48.0
	Perspective Distortion	40.0	38.0	42.0
	Relative Size Comparison	42.0	46.0	48.0
	Scale Consistency	33.33	33.33	49.02
	Shadow-Size Projection	58.82	54.9	66.67
Spatial Navigation	Accessibility Constraints	48.78	46.34	53.66
	Obstacle Avoidance	36.0	32.0	44.0
	Pathway Existence	32.61	34.78	41.3
	Spatial Sequence	48.0	44.0	50.0
	Viewpoint Visibility	32.0	34.0	48.0

Overall Discussion. Across model families, the MCQ analysis contributes a reviewer-facing map of where spatial reasoning fails. Closed-source models provide the highest ceilings but retain large within-model gaps. Large open-source models improve selected categories yet leave catastrophic lows, and small open-source models expose the limits of capacity and training diversity. Reasoning models improve certain relational and geometric tasks but remain inconsistent, while specialized spatial models show that targeted training can still fail to generalize. Overall, the evidence points to three causes: incomplete geometric representations, limited physical and reference-frame grounding, and brittle task-specific reasoning. Future progress therefore requires models that integrate geometric encodings, embodied data, and cross-task transfer rather than optimizing isolated subcategories.

E.2 MODEL FAILURE ANALYSIS

Table 11 turns the MCQ failures into item-level evidence. The distribution is skewed toward lower bins, meaning many questions are answered correctly by few models. 3D Geometry is broadly difficult, with *Shape Projection* and *Stability Prediction* producing many unsolved items, suggesting weak physical and geometric reasoning. Depth & Occlusion is especially challenging because *Layering Order* and *Transparency Effects* require non-local and partially hidden visual cues. Orientation remains brittle in *Stacking Orientation* and *Tool Handedness*, while Relative Positioning shows inconsistency in *Betweenness Relationships* and *Proximity Gradients*. Size & Scale is somewhat stronger, indicating that some models exploit coarse perspective cues, but *Shadow-Size Projection* remains difficult. Spatial Navigation is mixed: *Spatial Sequence* and *Obstacle Avoidance* sometimes succeed, whereas *Pathway Existence* and *Accessibility Constraints* expose multi-step relational weaknesses.

The root-cause contribution of this failure analysis is that errors reflect representational limits more than benchmark ambiguity. Depth and occlusion errors show reliance on surface appearance instead of structured scene layers. Orientation failures such as *Tool Handedness* reveal missing embodied affordance modeling. Relative Positioning and Spatial Navigation failures indicate weak multi-hop spatial chaining, since tasks such as *Pathway Existence* require integrating local connectivity with global layout. Partial success in Size & Scale suggests that models can use frequent perspective

Table 11: MCQ diagnostic distribution: question counts by the number of models (0–5) that correctly answer each item across SPATIALAB-MCQ categories and subcategories. “0” means no model answered correctly, while “5” means all five analyzed models answered correctly.

Category	Sub-Category	Number of Models Successfully Answered						Total
		0	1	2	3	4	5	
3D Geometry	Gravity Effects	9	7	3	2	3	1	25
	Shape Projection	4	6	0	9	5	7	31
	Spatial Containment	6	9	6	3	6	1	31
	Stability Prediction	5	6	4	2	2	3	22
	Volume Comparison	6	5	6	5	4	2	28
3D Geometry Total		30	33	19	21	20	14	137
Depth & Occlusion	Complete Occlusion Inference	0	4	0	6	3	4	17
	Layering Order	9	8	7	8	6	5	43
	Partial Occlusion	2	2	5	7	5	2	23
	Reflective Surfaces	6	5	5	7	4	1	28
	Transparency Effects	4	6	3	2	5	7	27
Depth & Occlusion Total		21	25	20	30	23	19	138
Orientation	Cardinal Direction	1	7	10	7	3	4	32
	Facing Direction	2	3	8	5	3	1	22
	Object Rotation	4	6	7	3	5	4	29
	Stacking Orientation	2	3	0	2	1	1	9
	Tool Handedness	2	1	4	3	1	1	12
Orientation Total		11	20	29	20	13	11	104
Relative Positioning	Alignment Patterns	3	4	4	4	4	1	20
	Betweenness Relationships	3	1	2	2	5	7	20
	Corner/Angle Positioning	0	1	3	2	3	3	12
	Directional Relations	1	4	1	2	3	2	13
	Proximity Gradients	4	6	6	5	2	3	26
Relative Positioning Total		11	16	16	15	17	16	91
Size & Scale	Distance-Size Correlation	6	5	6	1	1	7	26
	Perspective Distortion	8	7	3	3	6	4	31
	Relative Size Comparison	5	5	7	4	4	4	29
	Scale Consistency	7	6	7	4	5	2	31
	Shadow-Size Projection	5	7	2	1	1	4	20
Size & Scale Total		31	30	25	13	17	21	137
Spatial Navigation	Accessibility Constraints	3	2	1	5	4	3	18
	Obstacle Avoidance	2	6	9	6	10	2	35
	Pathway Existence	2	10	4	1	4	3	24
	Spatial Sequence	3	2	3	2	7	8	25
	Viewpoint Visibility	2	2	5	9	8	5	31
Spatial Navigation Total		12	22	22	23	33	21	133
Grand Total		116	146	131	122	123	102	740

regularities but collapse on rarer cases such as shadows. Addressing these weaknesses will require architectures that explicitly model reference frames, occlusion layers, and physical dynamics, along with training data enriched by embodied simulations and counterfactual augmentations.

F ERROR ANALYSIS OF SPATIALAB-OPEN

F.1 SUB-CATEGORY-WISE QUANTITATIVE ERROR ANALYSIS

Closed-source Models (Table 12). Closed-source models provide the strongest open-ended results, but the subcategory analysis shows that their advantage is uneven. GPT-5-mini is the clear leader, with the best scores in **3D Geometry** and **Relative Positioning**, including peaks of 58.14% in Directional Relations and 55.81% in Alignment Patterns. At the same time, Gemini-2.5-Flash and Claude

Table 12: Open-ended subcategory diagnostic accuracy on SPATIALAB-OPEN for Closed-source Models. Bold and underline mark the best score within each model category.

Category	Sub-Category	GPT-5-mini	Gemini-2.0-Flash	Gemini-2.5-Flash	Claude 3.5 Haiku	Mistral Medium 3.1
3D Geometry	Gravity Effects	48.0	36.0	30.0	24.0	20.0
	Shape Projection	40.0	18.0	22.0	16.0	16.0
	Spatial Containment	50.0	38.0	50.0	36.0	36.0
	Stability Prediction	50.0	26.19	35.71	30.95	26.19
	Volume Comparison	39.13	41.3	32.61	23.91	28.26
Depth and Occlusion	Complete Occlusion Inference	46.15	35.9	38.46	28.21	25.64
	Layering Order	30.0	22.86	21.43	17.14	15.71
	Partial Occlusion	38.0	20.0	28.0	14.0	14.0
	Reflective Surfaces	32.0	28.0	32.0	22.0	22.0
	Transparency Effects	32.0	18.0	18.0	16.0	22.0
Orientation	Cardinal Direction	34.0	40.0	34.0	26.0	30.0
	Facing Direction	28.57	28.57	26.19	19.05	14.29
	Object Rotation	42.0	14.0	28.0	30.0	14.0
	Stacking Orientation	37.14	20.0	31.43	22.86	28.57
	Tool Handedness	48.0	36.0	44.0	24.0	24.0
Relative Positioning	Alignment Patterns	55.81	34.88	37.21	37.21	34.88
	Betweenness Relationships	48.94	23.4	31.91	23.4	21.28
	Corner/Angle Positioning	44.44	30.56	41.67	16.67	36.11
	Directional Relations	58.14	46.51	58.14	34.88	37.21
	Proximity Gradients	39.53	20.93	25.58	16.28	18.6
Size and Scale	Distance-Size Correlation	48.0	32.0	36.0	20.0	22.0
	Perspective Distortion	48.0	24.0	22.0	20.0	12.0
	Relative Size Comparison	36.0	26.0	28.0	18.0	16.0
	Scale Consistency	43.14	21.57	23.53	19.61	9.8
	Shadow-Size Projection	37.25	27.45	23.53	23.53	15.69
Spatial Navigation	Accessibility Constraints	41.46	31.71	39.02	24.39	21.95
	Obstacle Avoidance	40.0	22.0	22.0	26.0	14.0
	Pathway Existence	23.91	15.22	21.74	10.87	19.57
	Spatial Sequence	52.0	32.0	34.0	24.0	20.0
	Viewpoint Visibility	28.0	22.0	32.0	20.0	10.0

3.5 Haiku perform poorly in Proximity Gradients (16.28%) and Perspective Distortion (12.0%), respectively. This contrast is central to the diagnostic contribution: even high-performing proprietary models remain fragile when open-ended answers require subtle continuous cues. Tasks such as Proximity Gradients are particularly challenging because they require graded analogical reasoning rather than discrete classification. GPT-5-mini’s success suggests that scaling and proprietary training provide advantages, but its remaining vulnerabilities show that closed-source dominance is relative rather than a solution to human-like spatial reasoning.

Open-source Large Models (Table 13). Large open-source models show distributed strengths rather than consistent coverage. llama-3.2-90b-vision-instruct reaches 60.0% in Depth and Occlusion, while GLM-4.5V-106B-MoE leads in Spatial Containment at 38.0%. However, qwen-2.5-vl-32b-instruct and qwen-2.5-vl-72b-instruct both fall to 9.3% on Proximity Gradients, showing that scale does not guarantee success on continuous spatial reasoning. This variation likely reflects heterogeneous training data: models exposed to richer depth and geometry corpora perform better in occlusion and containment, while others generalize poorly. The persistent weakness on Proximity Gradients indicates that these systems are not optimized for subtle spatial analogies and distance scaling. Thus, the open-source-large analysis contributes evidence that scale helps selected tasks but must be paired with targeted pre-training and fine-tuning.

Open-source Small Models (Table 14). Small open-source models lag overall, but their results are useful because they reveal how limited capacity interacts with task structure. qwen-2.5-vl-3b-instruct reaches 18.0% in Perspective Distortion and 26.0% in Relative Size Comparison, while Gemma-3-4B-it leads in Spatial Containment (24.0%) and Stability Prediction (21.43%). These local successes are offset by catastrophic failures: InternVL3.5-2B scores 0.0% in Betweenness Relationships, and InternVL3.5-1B reaches only 2.0% in Shape Projection. The contribution of this breakdown is to show that compact models can learn isolated competencies but lack robust representations of geometric and relational principles. Subcategories requiring precise orientation or hierarchy often trigger complete breakdowns, confirming a structural ceiling for small architectures under open-ended generation.

Reasoning Models (Table 15). Reasoning-oriented models expose a sharp divide between well-integrated and weakly integrated reasoning systems. Gemini-2.5-flash-thinking stands far above

Table 13: Open-ended subcategory diagnostic accuracy on SPATIALAB-OPEN for Open-source Large Models. Bold and underline mark the best score within each model category.

Category	Sub-Category	llama-3.2-11b-vision-instruct	Gemma-3-27B-it	qwen-2.5-vl-32b-instruct	qwen-2.5-vl-72b-instruct	llama-3.2-90b-vision-instruct	GLM-4.5V-106B-MoE
3D Geometry	Gravity Effects	14.0	22.0	12.0	30.0	24.0	28.0
	Shape Projection	10.0	20.0	10.0	20.0	16.0	20.0
	Spatial Containment	14.0	30.0	18.0	22.0	20.0	38.0
	Stability Prediction	23.81	30.95	16.67	21.43	26.19	38.1
	Volume Comparison	23.91	10.87	<u>26.09</u>	41.3	28.26	32.61
Depth and Occlusion	Complete Occlusion Inference	25.64	20.51	12.82	25.64	30.77	33.33
	Layering Order	8.57	15.71	10.0	14.29	15.71	10.0
	Partial Occlusion	12.0	16.0	8.0	14.0	20.0	16.0
	Reflective Surfaces	26.0	16.0	8.0	34.0	32.0	24.0
	Transparency Effects	18.0	14.0	10.0	20.0	22.0	26.0
Orientation	Cardinal Direction	32.0	34.0	20.0	38.0	28.0	30.0
	Facing Direction	11.9	26.19	14.29	21.43	11.9	16.67
	Object Rotation	26.0	16.0	10.0	18.0	28.0	20.0
	Stacking Orientation	17.14	25.71	17.14	25.71	5.71	37.14
	Tool Handedness	20.0	20.0	12.0	20.0	32.0	24.0
Relative Positioning	Alignment Patterns	23.26	46.51	20.93	46.51	30.23	30.23
	Betweenness Relationships	34.04	31.91	17.02	25.53	38.3	29.79
	Corner/Angle Positioning	27.78	36.11	11.11	27.78	27.78	16.67
	Directional Relations	30.23	39.53	25.58	34.88	25.58	32.56
	Proximity Gradients	9.3	18.6	9.3	18.6	18.6	20.93
Size and Scale	Distance-Size Correlation	20.0	26.0	12.0	26.0	30.0	26.0
	Perspective Distortion	10.0	18.0	10.0	18.0	12.0	20.0
	Relative Size Comparison	8.0	26.0	14.0	32.0	22.0	24.0
	Scale Consistency	11.76	21.57	3.92	19.61	21.57	23.53
	Shadow-Size Projection	17.65	21.57	9.8	27.45	23.53	27.45
Spatial Navigation	Accessibility Constraints	21.95	26.83	26.83	34.15	39.02	24.39
	Obstacle Avoidance	12.0	18.0	12.0	20.0	30.0	22.0
	Pathway Existence	19.57	21.74	6.52	17.39	15.22	17.39
	Spatial Sequence	22.0	28.0	12.0	22.0	28.0	34.0
	Viewpoint Visibility	18.0	16.0	12.0	12.0	24.0	24.0

Table 14: Open-ended subcategory diagnostic accuracy on SPATIALAB-OPEN for Open-source Small Models. Bold and underline mark the best score within each model category.

Category	Sub-Category	InternVL3.5-1B	InternVL3.5-2B	qwen-2.5-vl-3b-instruct	InternVL3.5-4B	Gemma-3-4B-it	qwen-2.5-vl-7b-instruct
3D Geometry	Gravity Effects	4.0	6.0	14.0	24.0	16.0	22.0
	Shape Projection	2.0	8.0	10.0	4.0	10.0	0.0
	Spatial Containment	6.0	4.0	12.0	20.0	24.0	8.0
	Stability Prediction	2.38	11.9	14.29	19.05	21.43	19.05
	Volume Comparison	15.22	32.61	28.26	30.43	30.43	28.26
Depth and Occlusion	Complete Occlusion Inference	2.56	5.13	10.26	17.95	20.51	23.08
	Layering Order	5.71	7.14	10.0	14.29	11.43	14.29
	Partial Occlusion	14.0	8.0	12.0	16.0	4.0	10.0
	Reflective Surfaces	16.0	16.0	6.0	14.0	18.0	26.0
	Transparency Effects	10.0	20.0	4.0	28.0	14.0	8.0
Orientation	Cardinal Direction	12.0	16.0	16.0	26.0	20.0	24.0
	Facing Direction	9.52	4.76	28.57	14.29	7.14	19.05
	Object Rotation	4.0	10.0	12.0	8.0	14.0	16.0
	Stacking Orientation	17.14	11.43	8.57	17.14	20.0	28.57
	Tool Handedness	8.0	12.0	8.0	12.0	12.0	12.0
Relative Positioning	Alignment Patterns	13.95	27.91	32.56	11.63	41.86	34.88
	Betweenness Relationships	12.77	17.02	0.0	19.15	17.02	25.53
	Corner/Angle Positioning	22.22	30.56	2.78	22.22	19.44	22.22
	Directional Relations	11.63	25.58	9.3	34.88	23.26	37.21
	Proximity Gradients	9.3	18.6	9.3	11.63	16.28	18.6
Size and Scale	Distance-Size Correlation	6.0	10.0	14.0	16.0	20.0	12.0
	Perspective Distortion	8.0	12.0	18.0	16.0	12.0	16.0
	Relative Size Comparison	10.0	16.0	26.0	24.0	10.0	16.0
	Scale Consistency	7.84	9.8	11.76	11.76	7.84	11.76
	Shadow-Size Projection	13.73	11.76	21.57	13.73	25.49	23.53
Spatial Navigation	Accessibility Constraints	4.88	26.83	7.32	17.07	19.51	19.51
	Obstacle Avoidance	16.0	16.0	12.0	18.0	28.0	26.0
	Pathway Existence	0.0	8.7	2.17	8.7	17.39	8.7
	Spatial Sequence	12.0	22.0	8.0	26.0	20.0	28.0
	Viewpoint Visibility	16.0	18.0	16.0	24.0	14.0	16.0

its peers, recording top scores in many subcategories, including Tool Handedness at 75.0% and Complete Occlusion Inference at 57.14%. In contrast, Kimi-VL-A3B-Thinking-2506 scores only 6.0% in Spatial Containment and lags broadly. This gap suggests that the benefit of reasoning depends on how effectively multimodal evidence is integrated into the reasoning process. Gemini-2.5-flash-thinking appears better able to form and query internal world models for object interactions and physical properties, while weaker models rely more on surface patterns. However, failures in Shape Projection and Betweenness Relationships show that even strong reasoning modules are not universally effective. The contribution is therefore nuanced: reasoning helps, but it does not yet solve open-ended spatial reasoning.

Spatial Reasoning Models (Table 16). Spatial reasoning models, despite being designed for this domain, record the lowest open-ended results overall. Several models tie at 16.0% in Gravity Effects, 8.0% in Shape Projection, and 14.29% in Stability Prediction. SpaceQwen2.5-VL-3B-Instruct stands out slightly with 23.91% in Volume Comparison and 13.73% in Scale Consistency, but these values remain far below competitive thresholds. Betweenness Relationships and Corner/Angle Positioning collapse almost entirely, with scores of 0.0% and 2.78%, respectively. These results contribute an important negative finding: narrow specialization without sufficient capacity or training diversity

Table 15: Open-ended subcategory diagnostic accuracy on SPATIALAB-OPEN for Reasoning Models. Bold and underline mark the best score within each model category.

Category	Sub-Category	gemini-2.0-flash-thinking	gemini-2.5-flash-thinking	Kimi-VL-A3B-Thinking-2506	Step3
3D Geometry	Gravity Effects	32.0	40.0	14.0	38.0
	Shape Projection	24.0	0.0	10.0	30.0
	Spatial Containment	36.0	50.0	6.0	30.0
	Stability Prediction	26.19	42.86	14.29	33.33
	Volume Comparison	36.96	44.44	23.91	30.43
Depth and Occlusion	Complete Occlusion Inference	33.33	57.14	17.95	33.33
	Layering Order	20.0	20.0	14.29	14.29
	Partial Occlusion	28.0	50.0	8.0	14.0
	Reflective Surfaces	34.0	63.64	10.0	16.0
	Transparency Effects	26.0	57.14	22.0	20.0
Orientation	Cardinal Direction	40.0	20.0	18.0	26.0
	Facing Direction	35.71	37.5	14.29	28.57
	Object Rotation	18.0	40.0	4.0	22.0
	Stacking Orientation	22.86	0.0	5.71	34.29
	Tool Handedness	44.0	75.0	20.0	32.0
Relative Positioning	Alignment Patterns	46.51	66.67	25.58	41.86
	Betweenness Relationships	27.66	20.0	12.77	25.53
	Corner/Angle Positioning	36.11	50.0	11.11	11.11
	Directional Relations	39.53	44.44	27.91	37.21
	Proximity Gradients	23.26	16.67	11.63	18.6
Size and Scale	Distance-Size Correlation	32.0	57.14	14.0	22.0
	Perspective Distortion	18.0	11.11	10.0	30.0
	Relative Size Comparison	28.0	11.11	14.0	36.0
	Scale Consistency	33.33	20.0	11.76	31.37
	Shadow-Size Projection	35.29	18.18	13.73	29.41
Spatial Navigation	Accessibility Constraints	31.71	45.45	17.07	41.46
	Obstacle Avoidance	26.0	10.0	16.0	18.0
	Pathway Existence	23.91	25.0	13.04	19.57
	Spatial Sequence	34.0	12.5	30.0	36.0
	Viewpoint Visibility	32.0	12.5	16.0	26.0

Table 16: Open-ended subcategory diagnostic accuracy on SPATIALAB-OPEN for Spatial Reasoning Models. Bold and underline mark the best score within each model category.

Category	Sub-Category	SpaceOm	SpaceThinker-Qwen2.5VL-3B	SpaceQwen2.5-VL-3B-Instruct
3D Geometry	Gravity Effects	14.0	12.0	16.0
	Shape Projection	8.0	8.0	6.0
	Spatial Containment	8.0	10.0	10.0
	Stability Prediction	14.29	14.29	7.14
	Volume Comparison	19.57	23.91	23.91
Depth and Occlusion	Complete Occlusion Inference	7.69	10.26	7.69
	Layering Order	8.57	11.43	4.29
	Partial Occlusion	10.0	12.0	4.0
	Reflective Surfaces	4.0	4.0	2.0
	Transparency Effects	4.0	8.0	2.0
Orientation	Cardinal Direction	18.0	24.0	8.0
	Facing Direction	23.81	23.81	16.67
	Object Rotation	20.0	20.0	18.0
	Stacking Orientation	8.57	8.57	17.14
	Tool Handedness	0.0	4.0	8.0
Relative Positioning	Alignment Patterns	32.56	32.56	23.26
	Betweenness Relationships	4.26	2.13	0.0
	Corner/Angle Positioning	2.78	2.78	2.78
	Directional Relations	11.63	6.98	11.63
	Proximity Gradients	6.98	6.98	9.3
Size and Scale	Distance-Size Correlation	18.0	18.0	16.0
	Perspective Distortion	16.0	20.0	2.0
	Relative Size Comparison	22.0	20.0	8.0
	Scale Consistency	9.8	11.76	13.73
	Shadow-Size Projection	27.45	27.45	19.61
Spatial Navigation	Accessibility Constraints	7.32	7.32	9.76
	Obstacle Avoidance	10.0	8.0	14.0
	Pathway Existence	6.52	6.52	8.7
	Spatial Sequence	16.0	12.0	16.0
	Viewpoint Visibility	20.0	16.0	8.0

does not yield generalizable spatial reasoning. In open-ended settings, even small representational errors can cause total failure on geometric and positional tasks.

Overall Discussion. Across open-ended categories, the diagnostic pattern is stronger than in MCQ evaluation. Closed-source models deliver the highest peaks, such as GPT-5-mini reaching 58.14% in Directional Relations, yet still fail on Proximity Gradients. Large open-source models show isolated successes, including 60.0% in Depth and Occlusion, but also collapse to 9.3% on the same Proximity task. Small models reveal localized strengths, such as 26.0% in Relative Size Comparison, but fail completely in tasks such as Betweenness Relationships. Reasoning models show that explicit reasoning can improve results, with Gemini-2.5-flash-thinking reaching 75.0% in Tool Handedness, while peers such as Kimi-VL-A3B fall to single digits. Spatial reasoning models underperform the most, often below 15% and with total failures on Betweenness Relationships. These findings point to three causes: weak representations for continuous and analogical spatial cues, structural limits tied to scale and training diversity, and brittle reasoning pipelines. No current approach—proprietary scaling, open-source expansion, reasoning integration, or specialization—delivers comprehensive coverage.

F.2 MODEL FAILURE ANALYSIS

Table 17 provides item-level evidence for the open-ended failure modes. The distribution is skewed toward lower bins (0–2 models correct), showing that many items remain difficult across model families. 3D Geometry is broadly challenging, with Shape Projection (19 items with zero correct) and Stability Prediction (10 items with zero correct) showing especially poor outcomes. Depth and Occlusion is the hardest category overall, with the highest zero-correct counts (72), especially in Layering Order (20 zero) and Transparency Effects (16 zero). Orientation remains brittle: Stacking Orientation has 11 zero-correct items, and Tool Handedness has 5 zero-correct items with no perfect-success cases. Relative Positioning is inconsistent, with Betweenness Relationships and Proximity Gradients rarely solved by more than two or three models. Size and Scale is relatively stronger, including Scale Consistency with 6 full successes, but Shadow-Size Projection still has 17 zero-correct items. Spatial Navigation remains split between occasional successes in Spatial Sequence and Obstacle Avoidance and persistent failures in Pathway Existence (16 zero) and Accessibility Constraints (10 zero). Across 356 items, zero-correct cases dominate (356), while only 74 items are solved by all five models.

The root-cause interpretation is that open-ended failures arise from representational and reasoning limitations, not merely from scoring format. Depth and Occlusion errors show that models rely on surface appearance rather than structured scene layers. Orientation failures such as Tool Handedness reveal missing embodied affordance modeling because internet-scale pretraining rarely annotates left/right handedness or stacking physics explicitly. Relative Positioning and Spatial Navigation failures point to weak multi-hop spatial chaining, since Pathway Existence requires integrating local connectivity with global layout rather than using co-occurrence priors. Partial success in Size and Scale suggests that models exploit frequent perspective or scale regularities, but fail on rarer cases such as shadows. These patterns show that models succeed when cues align with training priors but fail under counterfactual, compositional, or embodied inference. Future progress will require reference-frame modeling, occlusion-layer representations, physical dynamics, embodied simulations, and counterfactual augmentation.

G ANALYSIS OF MCQ VS. OPEN-ENDED PERFORMANCE GAPS

This appendix expands the paper’s format-sensitivity contribution: SPATIALAB is designed to show not only whether a model answers spatial questions correctly, but also whether that competence survives the shift from multiple-choice (MCQ) selection to open-ended generation. Table 18 reports the category-wise gaps used to diagnose this shift.

G.1 QUANTITATIVE SNAPSHOT

Across 25 models, the average MCQ-to-open-ended gap is **23.0%** with a standard deviation of 5.5%. Dataset-wide means per subtask are 3D geometry (24.57), depth & occlusion (23.45), orientation (23.70), relative positioning (23.11), size & scale (23.42), and spatial navigation (22.89), yielding an overall average of 23.01. The overall gap ranges from 11.93 to 31.57, with a median of $\approx$ 23.34. At the family level, spatial reasoning models have the largest average gap (27.03), while reasoning-oriented models have the smallest (19.11). Proprietary models have lower gaps on average

Table 17: Open-ended diagnostic distribution: question counts by the number of models (0–5) that correctly answer each item across SPATIALLAB-OPEN categories and subcategories. “0” means no model answered correctly, while “5” means all five analyzed models answered correctly.

Category	Sub-Category	Number of Models Successfully Answered					
		0	1	2	3	4	5
3D Geometry	Gravity Effects	13	7	3	4	1	2
	Shape Projection	19	5	6	2	5	1
	Spatial Containment	10	4	3	6	2	3
	Stability Prediction	10	3	7	3	0	2
	Volume Comparison	11	3	4	3	4	2
3D Geometry Total		63	22	23	18	12	10
Depth & Occlusion	Complete Occlusion Inference	7	5	4	2	5	2
	Layering Order	20	14	10	6	3	3
	Partial Occlusion	14	8	5	5	4	2
	Reflective Surfaces	15	4	1	5	2	5
	Transparency Effects	16	7	7	3	1	2
Depth & Occlusion Total		72	38	27	21	15	14
Orientation	Cardinal Direction	8	10	0	1	5	2
	Facing Direction	13	5	5	1	3	2
	Object Rotation	12	10	6	4	1	2
	Stacking Orientation	11	5	1	2	1	1
	Tool Handedness	5	4	1	4	2	0
Orientation Total		49	34	13	12	12	7
Relative Positioning	Alignment Patterns	7	5	3	1	1	1
	Betweenness Relationships	11	5	7	2	1	3
	Corner/Angle Positioning	7	6	2	1	3	2
	Directional Relations	5	4	4	4	2	0
	Proximity Gradients	10	10	3	3	2	4
Relative Positioning Total		40	30	19	11	9	10
Size & Scale	Distance-Size Correlation	7	5	9	4	1	5
	Perspective Distortion	12	6	6	8	5	2
	Relative Size Comparison	14	7	4	2	2	4
	Scale Consistency	14	7	5	3	3	6
	Shadow-Size Projection	17	4	4	4	3	2
Size & Scale Total		64	29	28	21	14	19
Spatial Navigation	Accessibility Constraints	10	5	4	0	1	2
	Obstacle Avoidance	15	5	8	2	2	5
	Pathway Existence	16	6	7	3	1	1
	Spatial Sequence	14	3	3	3	3	5
	Viewpoint Visibility	13	3	9	5	3	1
Spatial Navigation Total		68	22	31	13	10	14
Grand Total		356	175	141	96	72	74

(21.09) than many long open-source models (24.51), and the human baseline shows an overall gap of 22.64. The largest gaps occur for InternVL3.5-72B (31.57%), Qwen-VL2.5-32B (29.85%), and SpaceQwen2.5-VL-3B-Instruct (29.78%); the smallest occur for Llama-3.2-11B (11.93%), Gemini-2-Flash-Thinking (13.00%), and GPT-5-mini (13.36). Correlational analysis identifies spatial navigation as the dominant driver of the overall gap (Pearson $r = 0.99$), with orientation ($r = 0.83$) and 3D geometry ($r = 0.79$) also strongly aligned.

G.2 PATTERNS AND OBSERVATIONS

Three reviewer-facing patterns emerge from the gap analysis. First, specialist spatial reasoning models show the largest MCQ→open-ended gaps, suggesting that their spatial training may favor

Table 18: **Format-sensitivity contribution: performance gap between MCQ and Open-ended (%) ($\downarrow$) across spatial categories on SPATIALAB.** A lower value indicates that a model performs comparably in both setups, whereas a higher value reflects greater disparity. We bold and underline the highest gap within each model group.

Model	3D Geom. (#238)	Dep. & Occu. (#259)	Orientation (#202)	Relat. Posit. (#212)	Size & Scale (#252)	Spati. Navig. (#237)	Overall (#1400)
Proprietary Models							
GPT-4o-mini	22.40	23.76	16.51	31.74	27.85	20.50	23.53
GPT-5-mini	20.08	23.27	13.21	2.38	19.41	13.36	3.36
Gemini-2.0-Flash	30.89	26.73	26.89	28.18	22.37	25.07	15.13
Gemini-2.5-Flash	21.62	16.34	17.45	15.87	21.51	17.36	10.93
Claude 3.5 Haiku	23.16	21.78	20.29	15.47	24.89	20.29	16.39
Mistral Medium 3.1	30.50	25.74	32.54	26.59	24.89	26.93	21.43
Open-Source Models (Small)							
InternVL3.5-1B	22.78	13.37	23.58	22.62	20.67	22.00	27.73
InternVL3.5-2B	20.46	20.79	16.99	20.64	14.35	19.21	21.85
Qwen-VL2.5-3B-Instruct	27.03	30.69	29.24	28.97	29.96	28.50	25.63
InternVL3.5-4B	25.10	26.24	34.91	20.24	23.20	25.29	23.53
Gemma-3-4B-it	21.23	31.68	22.17	22.22	18.14	22.93	23.53
Qwen-VL2.5-7B-Instruct	22.01	22.27	18.40	26.19	15.61	22.14	27.73
Llama-3.2-11B-Vision-Instruct	13.51	-1.98	17.92	17.07	13.50	11.93	9.66
Gemma-3-27B-it	23.93	23.27	19.82	25.40	25.32	23.14	20.59
Qwen-VL2.5-32B-Instruct	30.50	31.68	28.30	35.32	28.27	29.85	24.79
InternVL3.5-72B	36.68	33.17	34.44	29.37	28.69	31.57	27.31
Qwen-VL2.5-72B-Instruct	27.80	26.73	23.59	19.05	28.27	24.22	20.17
Llama-3.2-90B-Vision-Instruct	28.95	29.21	30.66	25.00	21.52	26.36	23.53
Reasoning Models							
o4-mini	25.48	22.77	21.23	-3.18	17.30	15.35	10.50
Gemini-2-Flash-Thinking	13.90	10.39	11.32	21.03	13.50	13.00	6.73
Gemini-2.5-Flash-Thinking	8.22	16.61	19.46	33.42	31.37	20.16	8.66
Kimi-VL-A3B-Thinking-2506	27.02	28.71	33.50	26.98	22.78	27.92	29.41
Spatial Reasoning Models							
SpaceOm	31.66	32.18	25.95	24.21	27.00	28.43	29.83
SpaceThinker-Qwen2.5VL-3B	28.57	29.21	27.83	23.81	27.84	27.28	26.89
SpaceQwen2.5-VL-3B-Instruct	31.28	23.76	28.31	38.89	35.87	29.78	18.90

classification-style supervision over free-form generation. Reasoning-focused models show smaller gaps, consistent with the stabilizing effect of chain-of-thought-style training on generative outputs. Second, spatial navigation is the strongest predictor of overall disparity: when models fail to express navigation reasoning in open-ended form, the aggregate gap grows. Orientation and 3D geometry also contribute substantially, while relative position and size/scale vary more idiosyncratically. Third, scale alone does not explain robustness: smaller models such as Llama-3.2-11B show among the lowest gaps, whereas very large models such as InternVL3.5-72B show the highest. Negative or near-zero gaps, such as depth & occlusion for Llama-3.2-11B at -1.98 and relative position for o4-mini at -3.18 , further show that open-ended evaluation can sometimes be more reliable than MCQ when distractors are misleading.

G.3 HYPOTHESES ON ROOT CAUSES

We identify five interacting causes behind the MCQ $\rightarrow$ open-ended gaps in SPATIALAB, each tied to the statistics above:

- MCQ structural advantage.** MCQ constrains the output space, allowing models to exploit surface cues or eliminate distractors. Across models, MCQ scores are systematically higher than open-ended scores, with an average gap of 23.0% and $\sigma = 5.5\%$. Negative gaps for Llama-3.2-11B in depth & occlusion (-1.98) and o4-mini in relative position (-3.18) show that open-ended generation can occasionally outperform MCQ when distractors are misleading, confirming that MCQ can inflate or distort measured performance.
Recommendation: Report open-ended tasks and subtask-level diagnostics alongside MCQ scores to reduce overestimation from option-based cues.
- Specialization bias in spatial reasoning models.** Spatial reasoning models have the largest average gaps (27.03%) despite being optimized for spatial tasks, suggesting that their training may emphasize categorical or synthetic selection rather than free-form generation. InternVL3.5-72B, a 72B parameter spatial reasoning model, has the highest overall gap

(31.57%), especially in spatial navigation (36.68%) and orientation (34.44%). Correlations between subtask gaps and overall gap (SpNav: $r = 0.99$, Orientation: $r = 0.83$) show that these format-dependent failures concentrate in sequential and geometric reasoning tasks.

Recommendation: Use open-ended spatial generation during fine-tuning or instruction-tuning to reduce reliance on categorical supervision.

3. **Instruction-tuning and stepwise decoding.** Reasoning-oriented models such as Gemini-2-Flash-Thinking and o4-mini have smaller average gaps (around 19.11%) and lower variance across subtasks. Gemini-2-Flash-Thinking has an overall gap of 13.0% and modest subtask gaps, including Difference in 3D Geometry at 13.9% and Difference Relative Positioning at 21.03%. This suggests that chain-of-thought or stepwise decoding helps stabilize open-ended responses and mitigates format sensitivity.

Recommendation: Combine stepwise decoding with instruction-tuning for open-ended spatial tasks to improve generative consistency.

4. **Spatio-navigation stresses sequential grounding.** Spatial navigation gaps are the strongest predictor of MCQ-to-open-ended disparity (Pearson $r = 0.99$). Models with high spatial-navigation gaps, such as SpaceQwen2.5-VL-3B-Instruct (Spatial Navigation gap = 35.87%), also have large overall gaps (29.78%). This indicates that open-ended formats disproportionately stress multi-step referential reasoning, while MCQ options can simplify the reasoning chain and hide sequential-grounding weaknesses.

Recommendation: Include dedicated open-ended spatio-navigation tasks and targeted multi-step supervision to measure and improve sequential grounding.

5. **MCQ distractor artifacts.** Negative or near-zero gaps show that distractor quality can suppress MCQ performance, as with o4-mini’s relative position gap (−3.18%). Conversely, large gaps for models such as Qwen-VL2.5-32B-Instruct (overall = 29.85%) show that MCQ distractors can also exaggerate perceived competence.

Recommendation: Audit and refine MCQ distractors, and pair MCQ with open-ended evaluation to obtain less biased measures of spatial reasoning.

G.4 PRACTICAL IMPLICATIONS

The practical contribution of this analysis is a reporting recommendation for future spatial-reasoning benchmarks. MCQ and open-ended results should be reported together because MCQ-only evaluation can inflate estimates of practical spatial competence by around 23%. Subtask-level breakdowns are necessary because spatial navigation disproportionately drives the observed differences. Evaluations should explicitly distinguish classification from generative reasoning, use instruction-tuning and stepwise decoding carefully, and audit MCQ distractors to support fair cross-format comparison.

H DETAILS ON IMPROVING VISUAL REASONING CAPABILITIES

This appendix expands the improvement-analysis contribution of the paper. We test whether common intervention strategies—built-in reasoning, CoT prompting, self-reflection, SFT, transfer evaluation, and agentic decomposition—actually improve spatial reasoning on the diverse tasks in SPATIALAB.

H.1 INHERENT REASONING MECHANISMS

We first isolate the contribution of built-in reasoning mechanisms by comparing standard models with their reasoning-enabled counterparts (Table 2, Table 3). Across MCQ evaluations, reasoning usually improves performance, but the magnitude depends strongly on category. For instance, Gemini-2.5-Flash-Thinking improves over its non-reasoning variant by +4.64% overall (52.93% vs. 48.29%), with particularly strong gains in Relative Positioning (+13.1%) and Orientation (+8.6%). A similar trend holds for o4-mini-medium, which reaches the highest overall MCQ accuracy (53.21%), suggesting that explicit reasoning scaffolds are especially effective in structured, discrete-answer settings.

Open-ended evaluation makes the contribution more nuanced. Reasoning-equipped models again outperform their baselines, but the gains are uneven and do not remove format sensitivity. Gemini-2.5-Flash-Thinking improves slightly over Gemini-2.5-Flash (32.77% vs. 30.93%), but the largest leap is observed in o4-mini-medium (37.86%), which significantly surpasses all non-reasoning

2106
2107
2108
2109
2110
2111
2112
2113
2114
2115
2116
2117
2118
2119
2120
2121
2122
2123
2124
2125
2126
2127
2128
2129
2130
2131
2132
2133
2134
2135
2136
2137
2138
2139
2140
2141
2142
2143
2144
2145
2146
2147
2148
2149
2150
2151
2152
2153
2154
2155
2156
2157
2158
2159

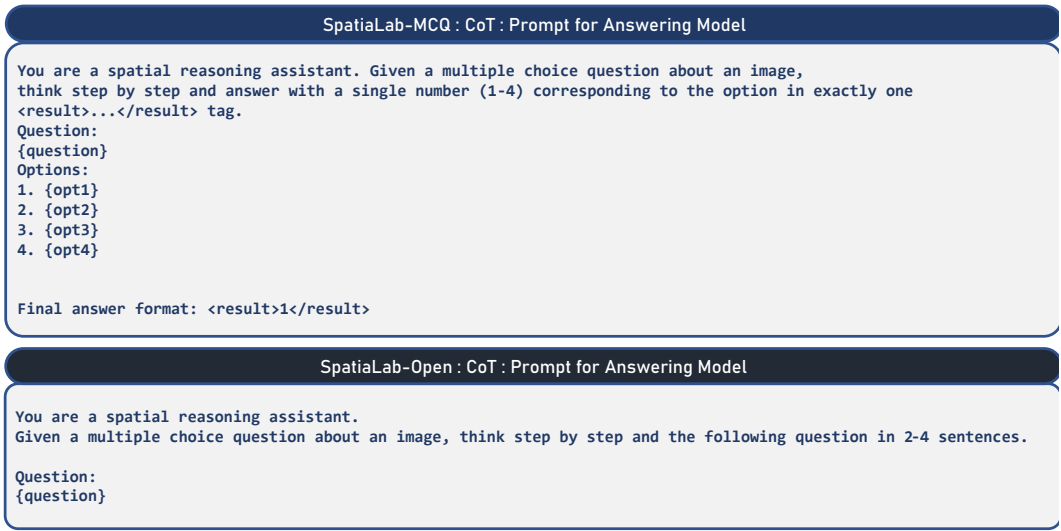


Figure 7: Chain-of-thought (CoT) prompts used to test whether explicit reasoning scaffolds improve SPATIALAB performance.

models. These results indicate that reasoning mechanisms help mitigate the instability seen in free-form generation, especially in complex categories such as 3D Geometry and Spatial Navigation. However, gains are not universal, reasoning models sometimes show regressions, e.g., Gemini-2.5-Flash-Thinking drops sharply in Size & Scale (21.74% vs. 26.59%), echoing our earlier finding that generative settings amplify brittleness.

Overall, this comparison contributes a bounded conclusion: explicit reasoning modules improve structured MCQ performance and can stabilize open-ended responses, but their benefits remain category-specific. The results suggest that reasoning mechanisms strengthen logical consistency and relational chaining, yet they do not fully resolve the challenges of scale sensitivity, occlusion, and free-form linguistic grounding.

H.2 CHAIN-OF-THOUGHT (CoT) PROMPTING

To test the contribution of structured prompting, we evaluated models under Chain-of-Thought (CoT) prompting. For each of the 30 sub-categories in SPATIALAB, we sampled 5 items (seed fixed to 42) to construct a balanced test set covering all major spatial phenomena. This design preserves balanced coverage while keeping the analysis computationally tractable. Example CoT prompts are shown in Figure 7, which illustrate the step-by-step reasoning scaffolds provided to the models. The setup allows us to isolate the contribution of CoT reasoning from model scale or training.

Table 19: Contribution analysis for CoT prompting across categories for InternVL-3-78b and Gemini-2.5-Flash. Values are accuracies (%); gains are with-CoT minus without-CoT.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain	w/o CoT	with CoT	Gain
3D Geometry	52.00	52.00	0.00	28.00	28.00	0.00	48.00	20.00	-28.00	60.00	28.00	-32.00
Depth & Occlusion	60.00	60.00	0.00	16.00	12.00	-4.00	48.00	28.00	-20.00	40.00	16.00	-24.00
Orientation	60.00	64.00	4.00	24.00	28.00	4.00	56.00	24.00	-32.00	0.00	32.00	32.00
Relative Positioning	80.00	72.00	-8.00	44.00	56.00	12.00	60.00	36.00	-24.00	50.00	32.00	-18.00
Size & Scale	56.00	44.00	-12.00	28.00	20.00	-8.00	48.00	60.00	12.00	25.00	8.00	-17.00
Spatial Navigation	64.00	64.00	0.00	48.00	40.00	-8.00	68.00	52.00	-16.00	40.00	28.00	-12.00
Overall	62.00	59.33	-2.67	31.33	30.67	-0.66	54.67	36.67	-18.00	38.24	24.00	-14.24

Our analysis (Table 19) shows that orientation is the only category where CoT prompting reliably improves performance, while most other categories stagnate or decline. This suggests that reasoning helps when the task is reducible to logical alignment of directions, but fails when more complex perceptual priors (e.g., depth, scale, relative positioning) are required. We anticipate this happens because models lack inherent knowledge about 3D perception and geospatial relationships; they are



Figure 8: Chain-of-thought (CoT) with Self-Reflection Prompts.

not fundamentally “aware” of these concepts in the way humans are. As a result, when CoT prompting is applied, the step-by-step reasoning does not correct misconceptions but instead reinforces them, propagating and amplifying errors across reasoning steps.

This finding highlights the benchmark contribution: SPATIALLAB cannot be solved merely by adding multi-step reasoning to existing models. Unlike textual reasoning tasks where CoT reliably improves outcomes, our benchmark exposes deeper architectural limitations in perceptual and spatial grounding. In categories like depth and size, incorrect priors mean that every additional reasoning step moves the model further from the correct solution. The sharp performance drops, particularly for Gemini in MCQ settings, demonstrate how fragile reasoning becomes when it is not supported by robust perceptual knowledge.

Thus, the task design contributes a stress test that goes beyond surface-level reasoning and probes whether models understand geospatial and 3D relational concepts. Solving it requires more than chaining logic; it demands embeddings and training that capture core aspects of perception and spatial cognition. In this way, our benchmark challenges models at a fundamental level and highlights an important gap in current architectures.

H.3 CHAIN-OF-THOUGHT (CoT) PROMPTING WITH SELF-REFLECTION

We further extend the CoT setup by adding a self-reflection stage, where the model is prompted to review and revise its initial reasoning before finalizing an answer (see Figure 8). The evaluation protocol remains identical to the standard CoT setting, using 5 balanced samples per sub-category

(seed 42). This design isolates the effect of self-reflection while keeping the comparison directly aligned with baseline CoT performance.

Table 20: Contribution analysis for CoT with Self-Reflection (CoT-SR) versus CoT across categories for InternVL-3-78b and Gemini-2.5-Flash. Values are accuracies (%); gains are CoT-SR minus CoT.

Category	InternVL-3-78b (MCQ)			InternVL-3-78b (Open)			Gemini-2.5-Flash (MCQ)			Gemini-2.5-Flash (Open)		
	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain	CoT	CoT-SR	Gain
3D Geometry	52.00	56.00	4.00	28.00	20.00	-8.00	20.00	52.00	32.00	28.00	24.00	-4.00
Depth & Occlusion	60.00	44.00	-16.00	12.00	16.00	4.00	28.00	52.00	24.00	16.00	16.00	0.00
Orientation	64.00	64.00	0.00	28.00	32.00	4.00	24.00	48.00	24.00	32.00	36.00	4.00
Relative Positioning	72.00	64.00	-8.00	56.00	28.00	-28.00	36.00	72.00	36.00	32.00	32.00	0.00
Size & Scale	44.00	56.00	12.00	20.00	12.00	-8.00	60.00	56.00	-4.00	8.00	12.00	4.00
Spatial Navigation	64.00	56.00	-8.00	40.00	24.00	-16.00	52.00	64.00	12.00	28.00	28.00	0.00
Overall	59.33	56.67	-2.66	30.67	22.00	-8.67	36.67	57.33	20.66	24.00	24.67	0.67

Adding self-reflection to Chain-of-Thought (CoT) prompting produced mixed and highly model-dependent outcomes. For InternVL-3-78b, overall accuracy declined slightly (-2.66% MCQ, -8.67% open-ended), with sharp drops in categories like depth, relative positioning, and navigation, suggesting that reflection often reinforced errors instead of correcting them. Some localized gains were observed, such as $+12\%$ in Size & Scale (MCQ), but these were inconsistent and offset by large degradations (-28% in relative positioning open-ended). In contrast, Gemini-2.5-Flash showed strong MCQ gains ($+20.66\%$), with particularly large improvements in 3D Geometry ($+32\%$) and Depth & Occlusion ($+24\%$), though open-ended improvements remained negligible ($+0.67\%$). These results highlight a key asymmetry: reflection helps in multiple-choice contexts where models can prune unlikely options, but does not transfer to free-form generation where uncertainty propagates into ungrounded rationales. We also find that the categories most improved by reflection are those where geometric consistency is central (orientation, depth, size), while tasks requiring relational integration across multiple objects (relative positioning, navigation) degrade. Overall, reflection functions as a stabilizer in structured choice settings but fails to reliably enhance naturalistic reasoning.

This observed inconsistency mirrors our earlier findings with plain CoT prompting: reasoning enhancements only succeed when the model has stable perceptual anchors to ground its steps. Self-reflection adds an explicit verification pass, but this mechanism is only as strong as the model’s internal priors. When depth cues, occlusion ordering, or coordinate frames are poorly encoded, the second pass amplifies misconceptions rather than repairing them. This explains why Gemini, which exhibits stronger perceptual grounding, benefits under MCQ conditions, while InternVL-3-78b suffers more frequent degradations. The broader root cause lies in a pretraining mismatch: current VLMs import reasoning templates from language modeling but lack the embedded 3D and geospatial representations that humans leverage. Without these anchors, reflection cannot perform true error correction; instead, it functions as a linguistic filter that polishes reasoning syntax without improving perceptual grounding. As a result, reflective reasoning loops provide modest utility in discrete answer formats but collapse under open-ended demands where factual verification is absent.

These findings confirm that our benchmark shows a fundamental gap in current architectures: surface-level reasoning layers alone cannot compensate for missing spatial cognition, and meaningful progress will require deeper integration of geometric and perceptual structure. Moreover, because VLLMs are commonly trained on web data, the concepts that are well-represented online are those where models tend to perform better, whereas poorly represented spatial and 3D concepts suffer, reinforcing these performance disparities.

H.4 SUPERVISED FINE-TUNING (SFT)

H.4.1 SETUP AND IMPLEMENTATION

We fine-tuned the Qwen 2.5 VL 3B instruction-following model (unsloth/Qwen2.5-VL-3B-Instruct) using a parameter-efficient approach with LoRA (Hu et al., 2021). To accommodate limited GPU resources, we loaded the model in 4-bit precision (load_in_4bit=True) and inserted LoRA adapters into attention and feed-forward network projection layers. The LoRA hyperparameters, including rank, scaling factor, dropout, and target

modules, are summarized in Table 21. Gradient checkpointing was enabled to reduce memory usage, and a fixed random seed ensured reproducibility.

For tokenization, we followed the Qwen-2.5 chat template to maintain consistent formatting between training and inference. Each training example was converted into a single text sequence, and dynamic padding was applied with `DataCollatorForSeq2Seq`. Training used an effective batch size of 4 (per-device batch size 1 with gradient accumulation of 4). Trainer and optimization hyperparameters, including learning rate, optimizer, weight decay, and scheduler, are listed in Table 22. The training loop consisted of four passes, each spanning a single epoch, yielding four saved checkpoints.

Overall, this setup allowed us to efficiently fine-tune a large vision-language model on limited GPU resources while maintaining high adaptation capacity.

Table 21: LoRA Hyperparameters for SFT

Hyperparameter	Value	Notes
LoRA rank (r)	16	Low-rank update capacity
LoRA alpha	16	Scaling factor
LoRA dropout	0	No dropout applied
Gradient checkpointing	Enabled	Library-specific flag
LoRA random seed	3407	Adapter initialization

Table 22: Trainer and Optimization Hyperparameters for SFT

Hyperparameter	Value
Base model	unsloth/Qwen2.5-VL-3B-Instruct
Max sequence length	2048
Load in 4-bit	True
Per-device train batch size	1
Gradient accumulation steps	4
Effective batch size	4
Learning rate	2e-4
Optimizer	paged_adamw_8bit
Weight decay	0.01
LR scheduler	linear
Logging steps	1
Seed	42
Tokenizer template	Qwen-2.5

H.4.2 SUPERVISED FINE-TUNING (SFT) PERFORMANCE

The learning trends in Figure 4a reveal notable differences between the MCQ and Open-ended evaluation modes over training epochs. We observe that MCQ accuracy begins at a lower baseline of 25.63% but rises consistently, converging near 36.59% by the fourth epoch. This steady improvement suggests that the model benefits strongly from additional supervised fine-tuning, particularly in tasks that constrain answers within a limited candidate space. By contrast, Open-ended evaluation starts higher at 34.40% but declines sharply by the second epoch, dropping to as low as 12.62%. Only in later epochs does it partially recover, reaching 35.48% at the end. This instability points to a sensitivity in open-ended generation that is not present in the MCQ setting, where the structure of multiple-choice questions provides more stable optimization signals. We interpret the fluctuations in open-ended performance as evidence that fine-tuning may overfit to MCQ-like reasoning patterns at the expense of free-form language generalization. The dotted baselines in the plot further highlight how the MCQ mode achieves gains beyond its initial standing, whereas Open-ended remains approximately stagnant. Overall, the figure illustrates that SFT predominantly favors MCQ tasks in terms of stability and consistent accuracy gains.

The tabulated results in Table 4b complement the figure by providing a detailed before-and-after breakdown across six spatial reasoning categories. Across MCQ tasks, we find that performance

improved in every category, with gains ranging from a modest 1.59% in Relative Positioning to a substantial 17.11% in Size and Scale. The Grand Total gain for MCQ stands at 10.97%, confirming that the fine-tuning procedure effectively enhances structured-answer performance. In contrast, Open-ended results are far more mixed, with both positive and negative changes. Categories such as 3D Geometry and Spatial Navigation show small but positive gains (3.50% and 2.82%, respectively), yet Size and Scale exhibits a significant drop of -3.95% . Relative Positioning also decreases by -0.79% , reflecting that fine-tuning can inadvertently harm certain reasoning modes in free-form contexts. The overall Open-ended gain is only 1.07%, which is statistically negligible compared to MCQ improvements. These results confirm that the effectiveness of SFT is highly sensitive to the evaluation paradigm. By isolating the categories, we underscore where training helps, where it fails, and how performance benefits differ between constrained and unconstrained outputs.

A deeper inspection reveals striking disparities between categories and formats. For MCQ, Size and Scale and Spatial Navigation yield the highest improvements, suggesting that models trained with multiple-choice prompts are especially effective at learning tasks requiring scale discrimination and navigational reasoning. However, in Open-ended form, Size and Scale actually declines, showing that the same learned representations do not transfer well to generative contexts. Similarly, Relative Positioning barely improves in MCQ and worsens in Open-ended, highlighting a consistent weakness in relational reasoning regardless of format. Conversely, Orientation shows strong MCQ gains ($+13.33\%$) and small Open-ended improvements, suggesting that some categories generalize more robustly. We hypothesize that the divergence between the two formats may stem from the structural scaffolding MCQs provide, which reduces the need for the model to generate precise linguistic formulations. Open-ended tasks, by contrast, require simultaneous mastery of reasoning and natural language fluency, which amplifies weaknesses. The trends indicate that while fine-tuning strongly enhances pattern recognition in discrete-choice settings, it introduces volatility and even degradation when free expression is required. This duality makes clear that training interventions optimized for MCQs may not generalize seamlessly to unconstrained reasoning.

The implications of these findings are twofold. First, we show that SFT as currently applied provides strong benefits for structured tasks but limited or even negative transfer to open-ended reasoning. This limitation is not only an artifact of fine-tuning, but also symptomatic of how contemporary VLMs are pretrained. Existing pretraining pipelines heavily emphasize broad image-text alignment, object recognition, and captioning, yet they provide little explicit supervision on compositional spatial structures, occlusion layers, or geometric transformations. As a result, models learn to excel at surface-level associations while failing to acquire robust representations of depth, orientation, or scale. The instability we observe in open-ended spatial reasoning suggests that the model’s learned priors are fragile: without multiple-choice scaffolding, they revert to shallow pattern-matching instead of grounded inference. Categories such as Size and Scale and Relative Positioning, where declines are most pronounced, point to missing inductive biases in pretraining, biases that humans develop through embodied interaction with 3D environments. Another contributing factor is the reliance on synthetic or instruction-following datasets that flatten spatial diversity into puzzle-like abstractions, preventing models from generalizing to real-world clutter, noise, and complexity. These findings underscore the need to rethink both pretraining and fine-tuning for spatial reasoning, incorporating datasets and objectives that reflect naturalistic visual-spatial challenges. Ultimately, we argue that without targeted reforms, current VLMs risk reinforcing an algorithmic straightjacket: they appear competent under constrained evaluations yet remain brittle and unaligned with the embodied, flexible reasoning that humans bring to real-world spatial understanding.

H.4.3 SFT DYNAMICS ANALYSIS

To investigate the stability of Supervised Fine-Tuning (SFT) on spatial reasoning tasks, we conducted a multi-seed validation study ($N = 5$, seeds A–E) using the Qwen-2.5-VL-3B-Instruct model. The training trajectory, summarized in Table 23, reveals a distinct divergence in performance dynamics between Multiple-Choice Question (MCQ) and Open-Ended evaluation formats.

Divergent Learning Trajectories. As illustrated in Table 23 and Figure 9, performance on SPATIALAB-MCQ exhibits a monotonic improvement, rising from a baseline of 25.63% to an average of 35.74% at Epoch 4. This indicates that the model successfully adapts to the discriminative nature of the task, learning to map visual features to constrained output options with increasing reliability.

In sharp contrast, SPATIALAB-OPEN displays a pronounced *U-shaped* trajectory across all seeds. Rather than steady improvement, the model suffers an immediate and significant performance collapse in the early epochs (dropping from 34.40% to an average of 19.26% at Epoch 1). While performance recovers in later epochs (converging to $\approx 34.82\%$ at Epoch 4), the final gain over the zero-shot baseline is negligible ($+0.42\%$).

Table 23: Multi-seed validation of SFT dynamics ($N = 5$). While MCQ performance improves steadily, Open-Ended performance exhibits a systematic “U-shaped” collapse and recovery, indicating representational instability in generative spatial tasks.

Epoch	Type	Seed A	Seed B	Seed C	Seed D	Seed E
0 (Base)	MCQ	25.63%	25.63%	25.63%	25.63%	25.63%
	Open	34.40%	34.40%	34.40%	34.40%	34.40%
1	MCQ	29.56%	35.83%	24.13%	27.63%	26.53%
	Open	22.50%	23.20%	12.50%	14.00%	24.10%
2	MCQ	34.33%	37.76%	27.33%	31.83%	34.43%
	Open	12.62%	25.30%	28.80%	23.80%	16.65%
3	MCQ	34.68%	37.33%	32.86%	32.23%	34.73%
	Open	28.69%	22.22%	30.34%	32.90%	28.30%
4	MCQ	36.59%	38.48%	34.43%	34.36%	34.83%
	Open	35.48%	34.59%	33.92%	34.70%	35.40%

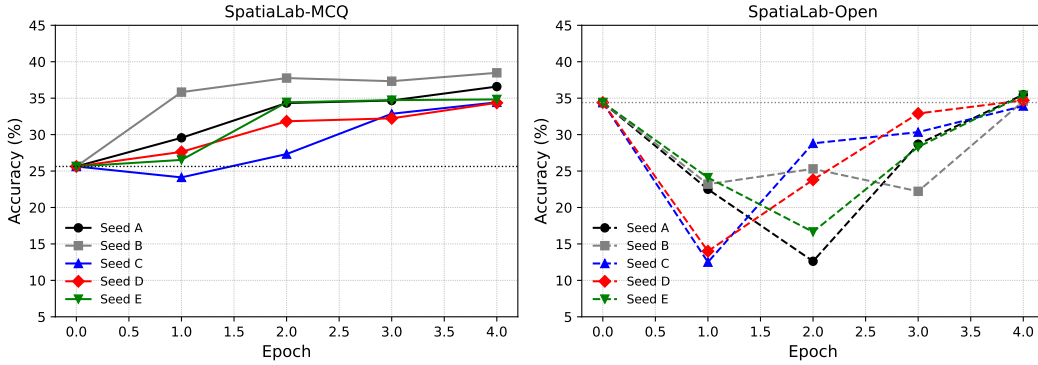


Figure 9: SFT dynamics analysis showing the contrast between structured MCQ gains and open-ended instability.

Representational Brittleness and Alignment Tax. We interpret this dynamic as evidence that spatial representations in current VLMs remain fragile under fine-tuning. The initial collapse in open-ended accuracy suggests *catastrophic forgetting* of the linguistic priors required for fluent, grounded generation. As the model optimizes for the specific instruction-following format of the training data, it overfits to the structural logic of the tasks (benefiting MCQ) at the expense of the flexible, generative spatial grounding required for open-ended descriptions.

The subsequent recovery phase suggests that the model relearns to articulate spatial concepts within the new distribution, yet it does not significantly surpass the initial baseline. This dynamic underscores a critical finding: SFT is effective for teaching models to *select* correct spatial answers, but it does not fundamentally enhance the underlying *generative spatial representation* needed for robust open-ended reasoning.

H.4.4 PERFORMANCE GAIN ON EXTERNAL BENCHMARKS

To test whether the improvement contribution transfers beyond the benchmark format, we evaluated the fine-tuned model on three external spatial reasoning datasets: OMNISPATIAL, SPACE (Multi-

modal), and MIND THE GAP. We employed the Qwen2.5-VL-3B-Instruct model, fine-tuned on SPATIALAB via the SFT protocol detailed in Appendix H.4.1, and evaluated it on these external benchmarks in a zero-shot setting.

Table 24: Transfer learning performance of Qwen2.5-VL-3B-Instruct on external benchmarks after fine-tuning on SPATIALAB. The consistent gains indicate that the model learns generalizable spatial representations rather than merely overfitting to the source dataset’s format.

Benchmark	Base Accuracy	After SFT	Improvement
OmniSpatial	40.30%	47.35%	+7.05%
SPACE (Multimodal)	23.43%	28.67%	+5.24%
Mind The Gap	35.86%	47.56%	+11.70%

As shown in Table 24, fine-tuning on SPATIALAB yields consistent improvements across all evaluated domains, ranging from +5.24% to +11.70%. The most significant gain was observed on MIND THE GAP (+11.70%), suggesting that our benchmark’s emphasis on occlusion and spatial continuity effectively addresses the reasoning gaps targeted by that dataset. Similarly, improvements on OMNISPIATIAL (+7.05%) and SPACE (+5.24%) confirm that the geometric and relational concepts encoded in our taxonomy align with broader definitions of spatial intelligence in the literature. These results support the quality and transfer value of the SPATIALAB data. While the primary purpose of this work is to serve as a rigorous diagnostic benchmark, the strong transferability demonstrates that the dataset does not merely encourage overfitting to a specific question template. Instead, the diversity of the 30 sub-categories forces the model to acquire robust, transferable spatial representations that generalize to unseen distributions and task formats.

H.5 AI AGENTS FOR SPATIAL REASONING

As an agentic improvement-analysis contribution, we developed SPATIOXOLVER, a multi-agent system adapted and extended from Xolver (Hosain et al., 2025), to perform structured spatial reasoning on images within the SPATIALAB evaluation. The system decomposes complex visual reasoning into specialized sub-tasks, each managed by a dedicated agent, allowing for focused processing of objects, attributes, relations, symmetries, and transformations. This modular design not only improves accuracy and interpretability but also supports multi-step reasoning, enabling the system to propagate low-level perceptual cues through higher-order relational and transformational inferences. The architecture, agent implementation, and their coordinated interaction are detailed in the following sections, with the full set of prompts provided in Figures 10, 11, and 12.

H.5.1 AGENTS

Base Visual Analysis Agent. Base Visual Language Model (VLM) agent initiates the pipeline by generating a detailed textual description of the input image, capturing object types—including geometric shapes, natural elements, and artificial icons—alongside attributes such as size, orientation, color, and shading. Relative positions like leftmost, center, and top are encoded explicitly, and the prompt emphasizes exhaustive detail without summarization. Images are converted to data URIs and sent with the prompt to a Gemini-2.5-based model, whose output is parsed into a structured bullet-list using regex-based extraction. This structured description serves as the foundation for downstream agents, providing a comprehensive perceptual anchor that encodes both explicit and implicit spatial cues. The approach ensures that even subtle visual features, such as partial occlusions or minor shape variations, are retained for later relational and transformational reasoning.

Object Segmentation Agent. Object Segmentation agent converts the verbose textual description into a discrete list of uniquely identified objects, differentiating between visually similar instances. Each object receives a stable identifier (e.g., Obj1, Obj2) to maintain consistent references across subsequent reasoning steps. The agent uses carefully formatted prompts and calls the same Gemini-2.5 model to structure the description, while post-processing splits lines by object ID to enforce consistency. This structured representation reduces ambiguity in downstream attribute and relation extraction, enabling precise mapping of complex scenes. By explicitly isolating objects, the agent

also facilitates detection of overlaps, repetitions, and hierarchical groupings that may not be evident in raw descriptions.

Attribute Extraction Agent. Attribute Extraction agent translates each object into a structured JSON capturing core properties such as shape, size, color, shading, orientation, and positional descriptors. Prompts enforce strict JSON output, and responses are parsed using `json.loads()`, with fallback defaults to ensure robustness against model inconsistencies. This structured encoding allows quantitative and qualitative comparisons of objects, supporting later stages of spatial reasoning. By preserving both absolute and relative measures, the agent maintains fidelity to the original perceptual content, while also normalizing descriptors for computational consistency. The JSON structure enables programmatic downstream operations, such as calculating symmetries or detecting transformations.

Spatial Relation Agent. The Spatial Relation agent encodes object interactions as triples (ObjectA,Relation,ObjectB), capturing directional relations (e.g., left_of, above, inside) and higher-order alignments like rows, grids, or repetition patterns. Consistent object IDs are enforced to prevent relational ambiguity, and parsing splits output lines starting with parentheses, followed by automated consistency checks. This step transforms static object descriptions into a relational graph that encodes spatial topology and hierarchy. The resulting structure allows reasoning over both pairwise interactions and more complex scene configurations, which is critical for tasks like symmetry detection, grouping, and multi-step navigation of spatial layouts.

Grouping and Symmetry Agent. Grouping and Symmetry agent abstracts higher-order patterns from object lists and spatial relations, identifying clusters, rows, grids, and symmetries—including vertical, horizontal, rotational, and translational. Outputs are requested in structured JSON with keys `groups` and `symmetries`, with empty structures as fallbacks to maintain downstream compatibility. This agent captures emergent scene regularities that are not apparent at the object or pairwise relation level, providing critical context for pattern recognition and reasoning. By formalizing structural redundancies and symmetries, the agent supports both perceptual inference and anticipatory reasoning for dynamic transformations.

Transformation Tracking Agent. For multi-frame sequences, the Transformation Tracking agent logs object-level changes including translations, scaling, rotations, color/shading shifts, and appearance or disappearance events. Structured logs are extracted line-by-line using regex patterns that capture directional arrows, producing a temporal representation of the scene. This encoding allows the system to track not just static spatial configurations but also dynamic evolution, supporting reasoning about motion, causality, and object interactions over time. The agent ensures that transformations are linked to consistent object IDs, enabling coherent cross-frame reasoning and predictive modeling.

Representation Standardization Agent. Finally, the Representation Standardization agent consolidates all outputs into a unified JSON format containing objects, relations, groups, symmetries, and transformations. Prompts enforce consistent object IDs and structured formatting, while JSON parsing failures revert to empty dictionaries to ensure pipeline robustness. By centralizing all perceptual, relational, and temporal information, this agent creates a comprehensive scene representation that is both human-interpretable and machine-readable. The resulting unified structure serves as a versatile foundation for downstream tasks, including reasoning evaluation, spatial pattern analysis, and multi-frame prediction.

Open-Ended Spatial Reasoning Agent. The standardized representation feeds into the reasoning agent, which answers multiple-choice or pattern recognition questions. Inputs include objects, attributes, relations, groups, symmetries, and transformations. The model considers geometric patterns, relational alignment, and dynamic transformations to generate or select the correct answer.

H.5.2 TECHNICAL IMPLEMENTATION DETAILS

For the technical implementation, we use Gemini-2.5-Flash with a low temperature (0.1) to ensure deterministic outputs and perform multiple iterations for refinement. API calls are handled via OpenRouter, supporting both text and image inputs, while parsing of model responses relies on

regex and JSON-based extraction to generate structured outputs. All interactions are logged in JSON format for reproducibility, and the final output is a unified JSON per image containing comprehensive information on objects, relations, symmetries, and transformations. This multi-agent decomposition allows each specialized model to focus on a constrained sub-task, enhancing overall accuracy, interpretability, and robustness in structured spatial reasoning.

H.5.3 EVALUATION AND ANALYSIS

The agent evaluation uses the same balanced protocol as the CoT and CoT-SR setting: 5 samples per subcategory with seed 42, for a total of 150 samples.

Table 25: Agent performance analysis for SPATIOXOLVER, showing where multi-agent reasoning helps and where it degrades spatial reasoning.

Category	MCQ		Gain	Open Evaluation		Gain
	Normal	With Agents		Normal	Agent Results	
3D Geometry	48.00%	44.00%	-4.00%	60.00%	48.00%	-12.00%
Depth & Occlusion	48.00%	44.00%	-4.00%	40.00%	16.00%	-24.00%
Orientation	56.00%	64.00%	+8.00%	0.00%	36.00%	+36.00%
Relative Positioning	60.00%	60.00%	0.00%	50.00%	36.00%	-14.00%
Size & Scale	48.00%	52.00%	+4.00%	25.00%	24.00%	-1.00%
Spatial Navigation	68.00%	64.00%	-4.00%	40.00%	28.00%	-12.00%
Overall	54.67%	57.33%	+2.66%	35.83%	31.33%	-4.50%

The evaluation of Gemini-2.5-Flash with agents contributes a clear boundary on agentic reasoning for perceptual–geometric tasks. Orientation is the only category where agents provide consistent and substantial gains, improving by +8.00% in MCQ and an impressive +36.00% in open evaluation. This pattern suggests that agent-based multi-step reasoning is effective when the task can be decomposed into sequential alignment of directional cues, where structured deliberation offers a clear advantage over single-step outputs.

In contrast, most other categories have negative or neutral gains, showing that agentic reasoning does not universally improve performance. For example, depth & occlusion suffers a severe -24.00% drop in open evaluation, while spatial navigation decreases by -12.00%. These tasks require deeper perceptual grounding in three-dimensional space, and the absence of such priors means that reasoning steps tend to reinforce misconceptions rather than correct them. Similarly, relative positioning shows stagnation in MCQ and a -14.00% decline in open evaluation, indicating that agent loops fail to add robustness when relational understanding is inherently weak.

Some categories show marginal improvements, such as size & scale (+4.00% in MCQ), but the gains are unstable and do not transfer to open evaluation. Even in 3D geometry, agents fail to provide a boost, with -4.00% and -12.00% drops in MCQ and open evaluation, respectively. These results underscore that our benchmark is complex and not solvable simply by layering multi-step agentic reasoning on top of existing models. Instead, it requires a deeper integration of geospatial understanding and perceptual representations, which current architectures lack.

Overall, the mixed results make the improvement-analysis contribution explicit: agentic workflows alone are insufficient for this domain. While orientation tasks benefit from structured reasoning, most categories expose the models’ lack of inherent awareness of perceptual concepts. This makes errors propagate through multi-step reasoning, amplifying performance drops. Consequently, our benchmark serves as a strong stress test, highlighting that future progress will require embedding genuine perceptual and geospatial priors rather than relying solely on agent-based reasoning.

I QUALITATIVE ERROR ANALYSIS AND ERROR PATTERNS

This section makes the qualitative diagnostic contribution explicit. We analyze model failures in SPATIALAB, define each error class, connect examples to root causes, identify recurring cross-model patterns, and describe verification protocols. The goal is to make the benchmark actionable for future architecture and evaluation design.

I.1 OVERVIEW

Across the sample set, models show strong coarse perception: they can often recognize object categories, colors, and simple binary relations, but they break down when multiple spatial cues must be integrated. The contribution of this section is that failures are not treated as random noise; they cluster into a small set of interpretable classes: spatial mislocalization, perspective and scale mistakes, occlusion and ordering errors, attribute confusion, and ungrounded open-ended rationalization. These classes recur across architectures and prompting strategies, which suggests that the underlying causes are common inductive biases in current vision-language pipelines rather than isolated training artifacts. Importantly, the same error often appears in both large-scale proprietary systems and small open-source baselines, indicating structural weaknesses shared across the model family. The following taxonomy unpacks each class with representative examples and root-cause hypotheses.

I.2 ERROR TAXONOMY AND ANALYSIS

I.2.1 SPATIAL MISLOCALIZATION AND REFERENCE CONFUSION.

■ **Observation.** This error class measures referential grounding: when queries require distinguishing one object from visually similar candidates, models frequently misidentify the target. For example, if asked to point out “the leftmost red cube among three stacked cubes,” the model may incorrectly output the middle cube, influenced by lighting highlights or central positioning. This problem becomes particularly acute in cluttered environments such as kitchen counters or crowded table scenes, where multiple overlapping objects share the same color and shape. In such cases, the model’s errors are not random: it tends to privilege objects that appear most salient due to size, brightness, or centrality, even when these cues contradict the explicit spatial instructions in the query. These consistent patterns highlight that the failures are systematic, not accidental.

★ **Analysis.** The recurring mistakes suggest that models lack a robust object-centric representation, or what cognitive science would call an “index” that persists across reasoning steps. Instead of maintaining a stable pointer to the correct candidate throughout the reasoning chain, models appear to drift, often reassigning attention to whichever object seems most visually striking at intermediate layers. This leads to reasoning errors such as starting with the leftmost cube but ending the chain with the brightest or largest cube. The inability to anchor spatial reference consistently makes tasks involving relative comparisons, like “the ball behind the second chair”, particularly fragile. In essence, the model operates more like a salience detector than a referential reasoner.

⊗ **Root cause.** At the architectural level, feature pooling operations, whether via attention or convolution, tend to mix contextual information from neighboring regions. While this is beneficial for global context, it erases the fine-grained, object-specific identity needed to distinguish between similar candidates. Once pooled, features of two adjacent red cubes may be nearly indistinguishable in the latent space. Without an explicit mechanism, such as slot-based representations or pointer tokens, the network collapses distinct entities into blended embeddings. This structural limitation prevents reliable referent tracking across reasoning steps and explains why models consistently confuse dense, overlapping objects.

I.2.2 PERSPECTIVE AND SCALE MISTAKES.

■ **Observation.** Tasks involving metric reasoning about relative size, fit, or scale are inconsistently handled. On simple comparisons like “is a car larger than a bicycle,” models typically succeed because such relationships align with strong statistical priors. However, when faced with near-threshold decisions such as “will the small sphere fit inside the slightly larger cylinder,” errors become frequent and unpredictable. A common pattern is reliance on canonical object sizes: for example, predicting that a door must be taller than a truck even in a rendering where the truck is scaled down. Another failure occurs in unusual visualizations, such as toy miniatures or architectural models, where familiar

proportions are deliberately inverted. In these cases, models cling to their priors and ignore the actual geometric evidence visible in the image.

★ **Analysis.** This error class shows that models rely heavily on distributional statistics of object categories rather than performing true geometric inference. Instead of reasoning about projection geometry, relative bounding box sizes, or depth cues, they substitute their world knowledge of typical scales. This shortcut works well in everyday contexts but fails catastrophically in edge cases that deviate from the training distribution. For instance, when an unusually small airplane is rendered next to a person, the model still predicts the airplane as larger. Such errors highlight the lack of grounding in pixel-level metric cues and the absence of consistent use of perspective or vanishing points.

⊗ **Root cause.** Current architectures do not include explicit geometric reasoning modules that preserve projection-aware relationships. Supervision for scale and fit tasks is limited, so models learn correlations between object category labels and size but not the underlying geometry. Furthermore, training objectives prioritize overall accuracy rather than fine-grained metric consistency, meaning that scale-sensitive errors are under-penalized. Without inductive biases for 3D reasoning or targeted supervision, the model defaults to memorized priors about canonical sizes rather than interpreting the actual perspective geometry in the image.

I.2.3 OCCLUSION AND ORDERING FAILURES.

■ **Observation.** Determining which object is in front, behind, or partially hidden remains a core weakness. Models often misreport the foreground object when two animals overlap, or when semi-transparent barriers like glass or fences are involved. For instance, in an image where a cat sits partially behind a curtain, models may incorrectly describe the curtain as being occluded by the cat. Thin or fragile structures, such as wires, tree branches, or railings, are particularly error-prone: they are frequently omitted altogether, leading to wrong answers in questions like “what object lies behind the fence.” These mistakes occur consistently, especially when occlusion cues are subtle or when the depth difference is small.

★ **Analysis.** Such failures indicate that models treat depth and occlusion as weak, secondary signals. While the pixel data contains clear occlusion cues (e.g., edge boundaries, T-junctions), the intermediate features learned by the model do not retain them with sufficient resolution. Aggregation layers blur boundaries and downweight thin structures, effectively removing the very cues necessary to resolve ordering. This is why models succeed in gross depth distinctions (foreground vs. background) but collapse in fine-grained cases where partial occlusion must be inferred. In practice, they guess rather than infer, with systematic biases toward treating the visually larger or brighter object as foreground.

⊗ **Root cause.** The architecture lacks explicit mechanisms for multi-scale feature retention and has no dedicated supervision for depth or occlusion reasoning. Thin or partially occluded signals are easily lost when features are pooled across large receptive fields. Furthermore, training data often underrepresents subtle occlusion cases, especially those involving transparency or fine boundaries. As a result, the learned representations flatten layered scenes into 2D appearance maps, preventing robust construction of depth-aware relational reasoning.

I.2.4 ATTRIBUTE CONFUSION AND SEMANTIC SWAP.

■ **Observation.** Models sometimes conflate perceptual attributes (such as color, texture, or material) with functional or semantic roles. A classic example is misjudging whether a crosswalk is safe: models may base the answer solely on the presence of painted stripes while ignoring the state of the traffic light or approaching vehicles. Similarly, a surface with a metallic texture may be misclassified as “steel” even if it is painted plastic. In tool-use contexts, the presence of handles or familiar shapes may lead the model to infer functionality that the object does not possess. These confusions are frequent in tasks requiring fine-grained visual grounding of attributes.

★ **Analysis.** These errors reveal that language priors strongly bias the model’s predictions when visual input is ambiguous. The decoder favors plausible narratives based on co-occurrence in the training data: “crosswalks are safe when stripes are visible,” “shiny textures correspond to metal,” and so on. This shortcut allows models to produce semantically coherent answers even when they do not inspect the relevant visual evidence. The result is superficially correct reasoning in familiar contexts but systematic errors in atypical or adversarial scenarios. In practice, this manifests as overfitting to textual stereotypes rather than grounding in perceptual details.

⊗ **Root cause.** The underlying training objective, maximizing likelihood of fluent and plausible

text, does not enforce visual grounding. As long as the generated answer is semantically reasonable, the loss is minimized, even if the reasoning is incorrect. Consequently, the model is incentivized to substitute common-sense priors for genuine perceptual inference. Without auxiliary losses or verification steps to tie textual claims to image regions, models will continue to swap perceptual and semantic attributes.

I.2.5 OPEN-ENDED RATIONALIZATION WITHOUT VERIFICATION.

■ **Observation.** For explanatory or why-style questions, models often generate coherent but unsupported narratives. For example, when asked “why is the object leaning,” a model might answer “because the surface is slippery” even though slipperiness cannot be visually inferred. In another case, a tilted shelf might be explained as “due to an earthquake,” a claim entirely outside the evidence in the image. These errors highlight a tendency to fill gaps with imaginative but ungrounded rationales, producing fluent explanations that mislead users about the true evidence.

★ **Analysis.** The issue stems from a decoupling between text generation and visual grounding. The model generates high-quality language that reads convincingly, but there is no mechanism ensuring that these rationales are tied to actual visual features. Confidence calibration compounds the problem: models often assign high certainty to fabricated explanations, giving the false impression of reliability. As a result, open-ended outputs may look authoritative while being factually baseless. This mode of failure is particularly concerning in safety-critical domains where plausible but false rationales could misguide decision-making.

⊗ **Root cause.** Training is dominated by language-modeling objectives that reward fluency and coherence but not factual grounding. Decoders are optimized to produce the most probable continuation of text given the prompt, without constraints linking the explanation back to specific image evidence. The lack of an explicit verification layer or grounding requirement means that models default to linguistic plausibility over perceptual truth. Without additional architectural or training constraints, open-ended rationalizations will continue to prioritize surface coherence at the expense of evidential accuracy.

I.3 ERROR PATTERNS AND CROSS-MODEL SIGNALS

When comparing failures across different models, several consistent patterns appear, revealing insights into the underlying mechanisms of errors. One clear trend is that disagreements between models are systematic rather than random: some models consistently rely on geometric cues such as relative size, depth ordering, and occlusion relationships, while others fall back on linguistic priors or common-sense assumptions derived from training text. This leads to predictable divergences, for example when two models give different answers about which object is “in front” despite seeing the same scene. Errors also spike when multiple cues must be integrated simultaneously. Tasks that require reasoning about occlusion, relative scale, and perspective together are particularly challenging; for instance, understanding whether a larger object partially occluding a smaller one is in front or behind often triggers consistent mispredictions. Calibration is another noticeable issue: high-confidence predictions frequently correspond to incorrect answers, especially in open-ended explanations or complex spatial queries. These patterns indicate that errors arise from how models balance perceptual grounding and linguistic plausibility. Different architectures exhibit distinct tendencies, some over-rely on visual geometry, others overfit to language correlations, so failures are not just isolated mistakes but reflect deeper trade-offs in reasoning strategies. This also implies that any attempt to improve performance should account for both sources of bias simultaneously rather than treating errors as independent events.

I.4 MITIGATION STRATEGIES

Given these error patterns, several targeted strategies can be pursued to reduce failures without requiring complete architectural redesign. For instance, geometry-aware supervision can be incorporated by adding auxiliary tasks like depth prediction, occlusion ordering, or overlap estimation. This encourages models to respect spatial constraints and improves consistency across predictions. Multi-scale feature retention is another approach: by using edge-aware pooling or attention mechanisms, models can maintain information from thin or partially occluded structures that would otherwise vanish in standard downsampling. Coordinate anchoring can help resolve confusion between viewer-centric and object-centric frames by explicitly representing reference points for relative positioning. Coun-

terfactual augmentation can also be applied, where distractor objects are systematically swapped or perturbed, forcing the model to rely on the true referent cues rather than heuristics. Lightweight verification modules can be added on top of existing outputs, cross-checking claims against visual or geometric constraints to filter implausible predictions. Each of these strategies can be implemented incrementally, providing clear paths to reduce systematic failures while keeping the core model largely unchanged.

In addition to these architectural and training augmentations, supervised fine-tuning (SFT) offers a practical pathway to instill more reliable spatial reasoning. By curating datasets that explicitly highlight failure cases, such as near-threshold scale comparisons, occlusion ambiguities, or reference resolution traps, SFT can directly expose models to the edge scenarios where they otherwise default to heuristics. Carefully designed instruction-style fine-tuning further encourages step-by-step reasoning, prompting the model to verbalize intermediate spatial relations rather than skipping to a guess. This not only strengthens grounding but also improves interpretability, as errors become traceable to specific reasoning steps. Moreover, SFT can be paired with counterfactual or adversarial augmentations, ensuring that the model does not simply memorize but instead learns invariances to distractors and context shifts. Importantly, these interventions require no wholesale changes to the architecture: they adapt existing models by reshaping their inductive biases through supervision. In this sense, SFT serves as a lightweight yet powerful complement to geometry-aware and verification-based strategies.

In this work, we additionally performed SFT using 40% of the training data and observed clear performance gains, particularly on multiple-choice questions. This suggests that targeted supervision helps models better internalize spatial cues and resolve distractor confusions when answer options are well-defined. However, improvements on open-ended responses were more limited. The model continues to produce fluent but weakly grounded explanations, highlighting that SFT alone cannot fully address rationalization without verification. These findings reinforce the need to combine SFT with geometry-aware supervision, counterfactual augmentations, and lightweight verifiers to achieve balanced progress across both structured and free-form tasks.

Our analysis shows that failures are not arbitrary but structured and interpretable, emerging from recurring patterns rather than random noise. This indicates that VLM errors follow systematic trajectories tied to the inductive biases of their architectures and training regimes. For example, geometric reasoning often collapses when occlusion cues are weak, and in such cases language priors take over, producing answers that sound plausible but lack visual grounding. Similarly, scale inference drifts when canonical object size priors dominate over pixel-level metric cues, leading to consistent misjudgments in near-threshold scenarios. These findings highlight that the gap is not simply one of capacity but of representation: current models lack explicit mechanisms to anchor visual features to grounded geometric relations.

By adopting the proposed verification protocols and targeted mitigation strategies, researchers can address these weaknesses directly. Geometry-aware supervision and counterfactual augmentation would help models resist shortcuts, while multi-scale feature retention ensures thin structures and partial occlusions are not discarded. SFT, when carefully applied, can further strengthen structured reasoning by aligning model predictions with task constraints, though it must be combined with grounding objectives to avoid overfitting to discrete-choice formats. Importantly, verification modules provide a lightweight way to enforce consistency between claimed relations and image evidence, counteracting the tendency to rationalize without checking. Taken together, these steps chart a practical pathway toward systems that reason more like humans: robust to clutter, interpretable in their failures, and capable of explaining not only what they predict but why.

2808
2809
2810
2811
2812
2813
2814
2815
2816
2817
2818
2819
2820
2821
2822
2823
2824
2825
2826
2827
2828
2829
2830
2831
2832
2833
2834
2835
2836
2837
2838
2839
2840
2841
2842
2843
2844
2845
2846
2847
2848
2849
2850
2851
2852
2853
2854
2855
2856
2857
2858
2859
2860
2861

Base VLM Agent Prompt

You are a Visual Scene Describer. Produce a raw, exhaustive text description of the given image.

Input image:
{image_input}

Your focus:

- Object Identification:
 - Shapes (circle, square, triangle, polygon, etc.)
 - Natural elements (tree, sun, water, etc.)
 - Artificial objects (arrows, patterns, icons)
- Attribute Noticing:
 - Size (small, medium, large, relative comparison)
 - Orientation (upright, rotated, tilted)
 - Shading or texture (solid, striped, shaded, transparent)
 - Color (if available)
- Positioning:
 - Relative location (leftmost, rightmost, top, bottom, center)
 - Proximity to other objects
- Level of Detail:
 - Do not summarize
 - Be verbose and explicit about every visible detail

Object Segmentation Agent Prompt

You are an Object Segmentation Specialist. Convert a verbose image description into a structured list of distinct objects.

Input description:
{visual_description}

Your focus:

- Object Isolation:
 - Treat each unique instance as a separate object
 - If multiple objects are similar, assign separate IDs
- Labeling:
 - Assign IDs: Obj1, Obj2, Obj3...
 - Keep descriptions short but complete
- Example Output:
Obj1: Small shaded circle (leftmost)
Obj2: Large unshaded square (center)
Obj3: Triangle rotated 45° (rightmost)

Attribute Extraction Agent Prompt

You are an Attribute Extraction Specialist. For each object, extract its properties into structured JSON.

Your focus:

- Attributes to Capture:
 - Shape
 - Size (relative if necessary)
 - Color or shading
 - Orientation or rotation
 - Position (leftmost, rightmost, center, top, bottom, etc.)
- Output Format:
JSON array, e.g.:
[
 {{ "ID": "Obj1", "Shape": "Circle", "Size": "Small", "Shading": "Yes", "Rotation": "0°", "Position":
"Leftmost"}},
 {{ "ID": "Obj2", "Shape": "Square", "Size": "Large", "Shading": "No", "Rotation": "45°", "Position":
"Center"}}
]

Figure 10: Prompts for SPATIOXOLVER (Part 1).

2862
2863
2864
2865
2866
2867
2868
2869
2870
2871
2872
2873
2874
2875
2876
2877
2878
2879
2880
2881
2882
2883
2884
2885
2886
2887
2888
2889
2890
2891
2892
2893
2894
2895
2896
2897
2898
2899
2900
2901
2902
2903
2904
2905
2906
2907
2908
2909
2910
2911
2912
2913
2914
2915

Spatial Relation Agent Prompt

You are a Spatial Relation Specialist. Extract spatial relationships among objects.

Your focus:

- Core Relations:
 - left_of, right_of, above, below
 - inside, surrounding
 - touching, overlapping
 - aligned, centered
- Precision:
 - Always specify directionality (ObjA left_of ObjB is not the same as ObjB left_of ObjA)
 - Use consistent object IDs
- Structural Patterns:
 - Detect repeated relations (e.g., "each circle is above a square")
 - Note spacing and alignment
- Output Format:

Triples in form: (ObjectA, Relation, ObjectB)

Example:

(Obj1, left_of, Obj2)

(Obj3, inside, Obj4)

Grouping Symmetry Agent Prompt

You are a Grouping and Symmetry Specialist. Identify higher-order structures from objects and relations.

Relations:
{relation_list}

Your focus:

- Grouping:
 - Clusters of objects (rows, columns, triangles, grids)
 - Repeated patterns
- Symmetry:
 - Vertical, horizontal, rotational
 - Mirror or reflective symmetry
 - Translational repetition
- Output Format:

JSON with two fields: "groups" and "symmetries"

Example:

```
{{  
  "groups": ["Obj1, Obj2, Obj3 form a horizontal row"],  
  "symmetries": ["Vertical symmetry across Obj2"]  
}}
```

Transformation Tracking Prompt

You are a Transformation Tracker. Analyze changes across multiple frames of an image sequence.

Your focus:

- Types of Transformations:
 - Movement (left, right, up, down)
 - Scaling (larger, smaller)
 - Rotation (degrees)
 - Shading or color change
 - Shape change (square → circle)
 - Appearance/disappearance
- Output Format:

Per-object transformation logs:

Obj1: Frame1 → Frame2 (rotated 90° clockwise)

Obj2: Frame2 → Frame3 (moved right)

Figure 11: Prompts for SPATIOXOLVER (Part 2).

2916
2917
2918
2919
2920
2921
2922
2923
2924
2925
2926
2927
2928
2929
2930
2931
2932
2933
2934
2935
2936
2937
2938
2939
2940
2941
2942
2943
2944
2945
2946
2947
2948
2949
2950
2951
2952
2953
2954
2955
2956
2957
2958
2959
2960
2961
2962
2963
2964
2965
2966
2967
2968
2969

Representation Standardization Agent Prompt

You are a Symbolic Abstraction Specialist. Consolidate all perception results into a standardized structured format.

Inputs:

- Objects: {object_list}
- Attributes: {attributes_json}
- Relations: {relation_list}
- Groups & Symmetries: {grouping_symmetry}
- Transformations: {transformations}

Your focus:

1. Create a unified JSON representation containing:
 - Objects with attributes
 - Relations as triples
 - Groups and symmetries
 - Transformations across frames
2. Ensure consistency in IDs and formatting
3. Example Output:

```
{
  "objects": [...],
  "relations": [...],
  "groups": [...],
  "symmetries": [...],
  "transformations": [...]}

```

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and select the correct option for the multiple-choice question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Options: {options_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.
- Select only the correct option Only (1, 2, 3, 4).
- Do NOT provide explanations, reasoning, or extra text. Only the option letter.

Output format: "<option_number>"

MCQ Spatial Option Prompt

You are a Spatial Reasoning Expert. Analyze the image data and give correct answer for the question.

Image Analysis Data:

- Objects and Attributes: {attributes_json}
- Spatial Relations: {relations_list}
- Groups & Symmetries: {grouping_symmetry_json}
- Transformations Across Frames: {transformations_json}

MCQ:

Question: {question_text}

Instructions:

- Consider shapes, positions, shading, rotations, symmetries, and transformations.
- Identify the pattern or rule in the sequence of images.

Figure 12: Prompts for SPATIOXOLVER (Part 3).

J REPRESENTATIVE BENCHMARK SAMPLES ACROSS ALL SUB-CATEGORIES

We present representative benchmark samples for all 30 sub-categories in SPATIALAB, illustrating the diverse spatial reasoning challenges covered. These samples highlight the variety of visual phenomena, including 3D geometry, occlusion, orientation, relational positioning, size and scale, and navigation tasks. Together, they provide a concrete view of the types of reasoning each model must perform to succeed across the benchmark.

3D Geometry: Gravity Effects



Question: From which direction would the wind cause the liquid in the glass which is in the front to spill onto the ground first?

Options

1. From the left side
2. From the right side
3. From the front
4. From the back

Gold MCQ Answer: 1

Gold Open ended Answer: From the left side

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** A gust blowing from the back of the patio (the top of the image, toward the viewer) would push the liquid forward over the near edge and make it spill first. Winds from the sides would tend to slosh it sideways rather than directly over the front rim.
- **Gemini-2.5-Flash:** The glass in the front is the one on the right side of the coffee table. Given its position, a wind blowing from the left (from the direction of the wooden cabinet and tall plant) would likely cause the liquid to spill first. This is because the glass is positioned with its opening facing somewhat towards the left, making it more susceptible to a gust from that direction.
- **InternVL3.5-72B:** The wind would need to come from the upper-right direction to cause the liquid in the glass on the front right of the table to spill first. This is due to the positioning of the glass towards the front edge of the table, closest to the right side of the image.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about the direction the wind would have to come from to spill the liquid in the front glass. Looking at the position of the glass: it's on the left front edge of the coffee table. The table seems to have a flat surface. If the wind blows from the left side (from the viewer's left, which would be the front of the image if the viewer.

Figure 13: Some examples from the benchmark (3D Geometry: Gravity Effects) and evaluation results.

3024
3025
3026
3027
3028
3029
3030
3031
3032
3033
3034
3035
3036
3037
3038
3039
3040
3041
3042
3043
3044
3045
3046
3047
3048
3049
3050
3051
3052
3053
3054
3055
3056
3057
3058
3059
3060
3061
3062
3063
3064
3065
3066
3067
3068
3069
3070
3071
3072
3073
3074
3075
3076
3077

3D Geometry: Shape Projection



Question: If the hanging light casts a shadow on the wall, which outline might appear?

Options

1. A rectangle narrowing at the bottom From the right side
2. A vertical cylinder
3. A circle with fuzzy edges
4. A triangle pointing downward

Gold MCQ Answer: 2

Gold Open ended Answer: A vertical cylinder

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would see the pendant's dome-shaped profile: a rounded semicircle or oval shadow with a thin vertical line or narrow neck above where the cord hangs. The shadow would fall on the wall to the right of the window, roughly over the nightstand.
- Gemini-2.5-Flash: The hanging light is a simple, cylindrical pendant. If it casts a shadow on the wall, the outline would likely be a circular or elliptical shape, depending on the angle of the light source relative to the wall. Given its position near the window, the shadow might also be influenced by ambient light or other objects in the room.
- InternVL3.5-72B: The hanging light fixture is likely to cast a shadow resembling its own shape onto the wall. Given its cylindrical form, the shadow would appear as a soft-edged, elongated circle or oval, depending on the angle of the light source relative to the wall.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking which outline a hanging light would cast on the wall if it casts a shadow. The wall near the window is light-colored, so the shadow would be visible. Since the shade is cylindrical, the shadow might be a cylinder. But maybe the outline is of the lampshade's perimeter. Alternatively, if the light is hanging from a chain, the shadow could be more elongated. But based on typical light fixtures, a cylindrical shade would cast a circular or oval shadow. The answer should be the outline of the hanging light's shade.

Figure 14: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

3078
3079
3080
3081
3082
3083
3084
3085
3086
3087
3088
3089
3090
3091
3092
3093
3094
3095
3096
3097
3098
3099
3100
3101
3102
3103
3104
3105
3106
3107
3108
3109
3110
3111
3112
3113
3114
3115
3116
3117
3118
3119
3120
3121
3122
3123
3124
3125
3126
3127
3128
3129
3130
3131

3D Geometry: Spatial Containment



Question: Look at the tall pot in the background with the wide mouth (top left). could the small jug from the center of the photo fit completely inside it?

Options

1. Yes, easily with room to spare
2. No, the small jug hand will stuck
3. No, the small jug is too tall.
4. Yes, but it would be a perfect fit with no room left

Gold MCQ Answer: 2

Gold Open ended Answer: No, the small jug hand will stuck

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** No — the tall pot's mouth appears too narrow compared with the small jug's widest part and its handle. The jug wouldn't pass through the opening to sit completely inside.
- **Gemini-2.5-Flash:** No, the small jug from the center of the photo could not fit completely inside the tall pot in the background. While the tall pot has a wide mouth, the jug's overall height and width appear to be too large to be fully contained within the pot's interior. The jug's handle would also likely prevent it from fitting snugly.
- **InternVL3.5-72B:** The tall pot in the background with the wide mouth appears large enough to potentially accommodate the small jug from the center of the photo. Considering the proportions and the wide opening of the tall pot, it is likely that the small jug could fit completely inside it.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about whether the small jug in the center of the photo could fit completely inside the tall pot in the background with the wide mouth (top left). However, without explicit scale references, this conclusion relies on typical proportional relationships between similar pottery forms. Based on typical proportions, the small jug's height is likely shorter than the tall pot's, and its width seems narrow enough to fit.

Figure 15: Some examples from the benchmark (3D Geometry: Spatial Containment) and evaluation results.

3132
3133
3134
3135
3136
3137
3138
3139
3140
3141
3142
3143
3144
3145
3146
3147
3148
3149
3150
3151
3152
3153
3154
3155
3156
3157
3158
3159
3160
3161
3162
3163
3164
3165
3166
3167
3168
3169
3170
3171
3172
3173
3174
3175
3176
3177
3178
3179
3180
3181
3182
3183
3184
3185

3D Geometry: Volume Comparison



Question: Compare the internal cooking volume of the microwave with the internal cooking volume of the larger oven below the stove.

Options

1. They both have the same volume A vertical cylinder
2. The microwave has slightly more volume
3. The oven below the stove has much more volume
4. The oven has more width, but the microwave has more height, so they are equal

Gold MCQ Answer: 3

Gold Open ended Answer: The oven below the stove has much more volume

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 3
- **InternVL3.5-72B:** 3
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** The microwave’s cooking cavity is much smaller than the larger oven below the cooktop — it’s only a small fraction of the oven’s internal volume. The countertop microwave is sized for single plates or small dishes, whereas the built-in oven can hold full baking trays and multiple racks for larger meals.
- **Gemini-2.5-Flash:** The internal cooking volume of the microwave appears to be significantly smaller than that of the larger oven below the stove. The microwave is a compact countertop model, suitable for reheating or cooking smaller items..
- **InternVL3.5-72B:** The internal cooking volume of the microwave on the right is likely smaller than that of the larger oven located below the stove. This is because microwaves typically have more compact dimensions to fit on countertops, while ovens have a greater internal capacity to accommodate larger dishes and multiple items. The size and design visible in the image suggest the oven has a deeper and wider cooking area compared to the microwave.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants me to compare the internal cooking volumes of the microwave and the larger oven below the stove in the provided image. So, the internal cooking volume of the oven below the stove is larger than that of the microwave. Based on their relative sizes in the image, the internal cooking volume of the oven below the stove is substantially larger than that of the microwave

Figure 16: Some examples from the benchmark (3D Geometry: Volume Comparison) and evaluation results.

3186
3187
3188
3189
3190
3191
3192
3193
3194
3195
3196
3197
3198
3199
3200
3201
3202
3203
3204
3205
3206
3207
3208
3209
3210
3211
3212
3213
3214
3215
3216
3217
3218
3219
3220
3221
3222
3223
3224
3225
3226
3227
3228
3229
3230
3231
3232
3233
3234
3235
3236
3237
3238
3239

3D Geometry: Shape Projection



Question: If you look at the seating arrangement from above, what shape does it form?

Options

1. A perfect square
2. A round shape
3. A rectangle
4. An irregular shape

Gold MCQ Answer: 4

Gold Open ended Answer: An irregular shape

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Viewed from above, the seating is arranged in a U-shape (a horseshoe) around the central fire pit. The L-shaped sectional plus the two facing chairs create that U configuration.
- Gemini-2.5-Flash: The seating arrangement, when viewed from above, primarily forms an L-shape with the large sectional sofa. This L-shape is complemented by two individual chairs placed opposite the open ends of the sectional. Together, the furniture creates a U-shaped configuration around the central fire pit, designed for communal gathering.
- InternVL3.5-72B: From above, the seating arrangement forms a U-shape around the central fire pit. The sectional sofa curves around the pit, leaving an open side opposite the tall fence, creating a semi-enclosed space.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user wants to know the shape formed by the seating arrangement when viewed from above. When viewed from above, the configuration clearly forms a right-angled L-shape, combining two parallel sections at a 90-degree angle. The symmetry of the layout and the placement of furniture around the fire pit reinforce this geometric form. The seating arrangement forms a right-angled L-shape when viewed from above.

Figure 17: Some examples from the benchmark (3D Geometry: Shape Projection) and evaluation results.

3240
3241
3242
3243
3244
3245
3246
3247
3248
3249
3250
3251
3252
3253
3254
3255
3256
3257
3258
3259
3260
3261
3262
3263
3264
3265
3266
3267
3268
3269
3270
3271
3272
3273
3274
3275
3276
3277
3278
3279
3280
3281
3282
3283
3284
3285
3286
3287
3288
3289
3290
3291
3292
3293

Spatial Navigation: Accessibility Constraints



Question: The notebook at the leftmost corner of the table is inaccessible because of which object?

Options

1. Chair
2. Vase
3. Table
4. Mug with pens and pencil in it

Gold MCQ Answer: 4

Gold Open ended Answer: Mug with pens and pencil in it

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** The leftmost corner notebook is blocked by a wooden chair. The chair sits directly in front of the desk, preventing easy access to the notebook.
- **Gemini-2.5-Flash:** The notebook at the leftmost corner of the table is inaccessible because of the chair. The chair is positioned directly in front of the notebook, blocking direct access to it. To reach the notebook, one would first need to move the chair out of the way.
- **InternVL3.5-72B:** The notebook at the leftmost corner of the table is inaccessible because of the chair, which is positioned in front of it. The chair's backrest and seat block direct access to the notebook.
- **Kimi-VL-A3B-Thinking-2506:** Alright, let's tackle this spatial reasoning question. So, the question is asking why the notebook at the leftmost corner of the table is inaccessible. The chair is positioned near the leftmost corner of the table, and given its placement, it's reasonable to conclude that the chair is blocking access to the notebook. The chair's presence in that space would make it difficult to reach the notebook from the left side without moving the chair first. The chair is the object that's making the notebook at the leftmost corner inaccessible.

Figure 18: Some examples from the benchmark (Spatial Navigation: Accessibility Constraints) and evaluation results.

3294
3295
3296
3297
3298
3299
3300
3301
3302
3303
3304
3305
3306
3307
3308
3309
3310
3311
3312
3313
3314
3315
3316
3317
3318
3319
3320
3321
3322
3323
3324
3325
3326
3327
3328
3329
3330
3331
3332
3333
3334
3335
3336
3337
3338
3339
3340
3341
3342
3343
3344
3345
3346
3347

Spatial Navigation: Obstacle Avoidance



Question: There is a clear direct path to the end of the hallway without colliding with any object.

Options

1. TRUE
2. FALSE
3. Maybe
4. Unable to determine

Gold MCQ Answer: 1

Gold Open ended Answer: TRUE

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

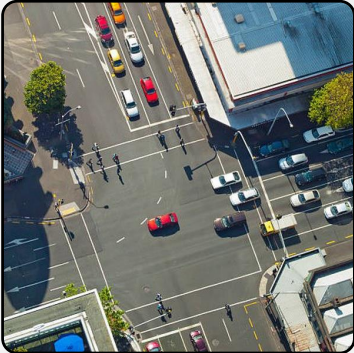
Open-ended Answers:

- **GPT-5-mini:** Yes. The central aisle is unobstructed from the foreground to the far end, with shelving stacked along both sides and no items blocking the floor. You can walk straight down the hallway without colliding with any objects.
- **Gemini-2.5-Flash:** Yes, there is a clear direct path to the end of the hallway without colliding with any object. The image shows a wide, unobstructed aisle between two rows of tall shelving units filled with boxes. The floor is clear, indicating a safe passage for movement.
- **InternVL3.5-72B:** The image shows a large warehouse with high shelving units full of boxes on both sides of a central aisle. The aisle appears to be clear and unobstructed, leading directly to the end of the hallway. There are no objects or obstacles in the pathway, ensuring a direct and clear route to the end of the aisle.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The user says there's a clear direct path to the end of the hallway without colliding with any object. A clear direct path exists because the shelving units are positioned symmetrically, leaving a central aisle free of objects. There are no visible hazards like hanging items, loose debris, or protruding elements in the visible portion of the hallway, allowing safe, uninterrupted movement to the end. The path is clear because the shelving units are symmetrically arranged, leaving the central walkway obstacle-free, with no visible hazards in the visible hallway.

Figure 19: Some examples from the benchmark (Spatial Navigation: Obstacle Avoidance) and evaluation results.

3348
3349
3350
3351
3352
3353
3354
3355
3356
3357
3358
3359
3360
3361
3362
3363
3364
3365
3366
3367
3368
3369
3370
3371
3372
3373
3374
3375
3376
3377
3378
3379
3380
3381
3382
3383
3384
3385
3386
3387
3388
3389
3390
3391
3392
3393
3394
3395
3396
3397
3398
3399
3400
3401

Spatial Navigation: Pathway Existence



Question: Which walkways are the safest to cross according to the picture?

Options

1. Top and bottom
2. Top and left
3. Left and right
4. Bottom and right

Gold MCQ Answer: 1

Gold Open ended Answer: Top and bottom

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

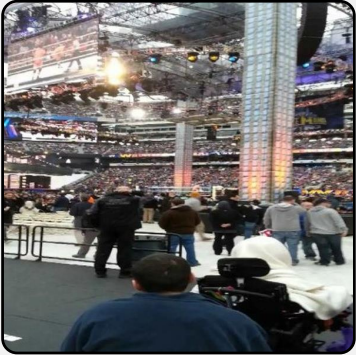
Open-ended Answers:

- **GPT-5-mini:** The marked white crosswalks at each corner of the intersection (the painted lines across the road) are the safest to use. They are signalized and where pedestrians are waiting, unlike cutting across mid-block or diagonally through the intersection.
- **Gemini-2.5-Flash:** The walkways marked with white lines, commonly known as crosswalks, are the safest to cross. These designated areas indicate where pedestrians have the right-of-way or are expected to cross the street. In the image, several people are visible using these marked crosswalks, suggesting they are the intended and therefore safest routes for pedestrians.
- **InternVL3.5-72B:** The crosswalks aligned with the pedestrian signals at the corners are the safest places to cross, as they are equipped with traffic signals ensuring vehicles are stopped. The designated pedestrian areas with painted lines across the streets are clearly marked for safety. Additionally, sidewalks provide a safe space for walking along the roads.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user wants to know which walkways are the safest to cross based on the image provided. The safest walkways to cross, based on the image, are the marked crosswalks equipped with pedestrian signals. Specifically, the crosswalk at the bottom-left corner of the intersection is safest because it features a visible pedestrian signal indicating walk, reducing ambiguity with traffic flow.

Figure 20: Some examples from the benchmark (Spatial Navigation: Pathway Existence) and evaluation results.

3402
3403
3404
3405
3406
3407
3408
3409
3410
3411
3412
3413
3414
3415
3416
3417
3418
3419
3420
3421
3422
3423
3424
3425
3426
3427
3428
3429
3430
3431
3432
3433
3434
3435
3436
3437
3438
3439
3440
3441
3442
3443
3444
3445
3446
3447
3448
3449
3450
3451
3452
3453
3454
3455

Spatial Navigation: Viewpoint Visibility



Question: The human standing the closest (blue jacket) has clear view of the stadium, how?

Options

1. Seeing directly
2. Watching on the big screen on the left side
3. There is no stadium
4. Unable to determine

Gold MCQ Answer: 2

Gold Open ended Answer: Watching on the big screen on the left side

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- Gemini-2.5-Flash: The person in the blue jacket is seated in a wheelchair, which places them at a lower vantage point than the standing crowd. However, they have a clear view of the stadium because they are positioned at the very front of a section, likely an accessible seating area. This front-row placement ensures an unobstructed line of sight over the heads of any standing individuals further back.
- InternVL3.5-72B: The person in the blue jacket is standing near the front of the arena, which allows them a clear, unobstructed view of the stadium. Their position is close to the barrier separating the audience from the performance area, providing a direct line of sight towards the stage and the large screens overhead. This proximity minimizes visual obstructions from other spectators or structures.
- Kimi-VL-A3B-Thinking-2506: Okay, let's tackle this question. The user is asking how the person in the blue jacket (who is standing the closest) has a clear view of the stadium. The person's elevated seating, minimized obstructions from nearby spectators, and potential alignment with stadium features (e.g., screens or lighting) enable their clear view despite being relatively close to the ground.

Figure 21: Some examples from the benchmark (Spatial Navigation: Viewpoint Visibility) and evaluation results.

3456
3457
3458
3459
3460
3461
3462
3463
3464
3465
3466
3467
3468
3469
3470
3471
3472
3473
3474
3475
3476
3477
3478
3479
3480
3481
3482
3483
3484
3485
3486
3487
3488
3489
3490
3491
3492
3493
3494
3495
3496
3497
3498
3499
3500
3501
3502
3503
3504
3505
3506
3507
3508
3509

Spatial Navigation: Spatial Sequence



Question: Which path is the easiest for the human sitting in the bench(reading paper) to follow to get on the train?

Options

1. Go straight to the first red colored door
2. To make a turn and go to the second floor
3. Go through the door at the end of the train
4. No viable option

Gold MCQ Answer: 1

Gold Open ended Answer: Go straight to the first red colored door

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Because he's standing on the raised walkway right behind the front-row barrier, his eye level is higher than the people immediately in front of him. That elevation (and the clear space by the railing) gives him an unobstructed line of sight over the crowd into the stadium.
- **Gemini-2.5-Flash:** The simplest way for the person to board the train is to rise from the bench, turn right, and walk a short distance along the platform. They would then cross the yellow safety line and step through one of the train doors. This route is straightforward and involves minimal effort.
- **InternVL3.5-72B:** The person sitting on the bench can follow a straightforward path to board the train by moving forward along the platform, parallel to the train tracks, while keeping to the right of the yellow safety line. This path will lead them directly to the nearest open train door. They should avoid crossing the yellow line until they are ready to enter the train to ensure safety.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is about determining the easiest path for the person sitting on the bench to get on the train. First, I need to look at the image provided. The person is sitting on the bench which is alongside the platform walls. The train is on the opposite side. To get to the train, the person would..

Figure 22: Some examples from the benchmark (Spatial Navigation: Spatial Sequence) and evaluation results.

3510
3511
3512
3513
3514
3515
3516
3517
3518
3519
3520
3521
3522
3523
3524
3525
3526
3527
3528
3529
3530
3531
3532
3533
3534
3535
3536
3537
3538
3539
3540
3541
3542
3543
3544
3545
3546
3547
3548
3549
3550
3551
3552
3553
3554
3555
3556
3557
3558
3559
3560
3561
3562
3563

Size and Scale: Scale Consistency



Question: With the electric pole's capacity can all this wires be redesigned to carry properly in aligned structured design?

Options

1. The size of wires are too big
2. The wires need to be move into another pole
3. Wires can be set there following a layout
4. No, the bulk size of the wires are too much here for one pole

Gold MCQ Answer: 4

Gold Open ended Answer: No, the bulk size of the wires are too much here for one pole

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 4
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** Not safely in its current state — the pole is clearly overloaded and the wires are tangled, so simply re-bundling them on the same pole would exceed mechanical, clearance and safety limits. A proper redesign is possible but requires removing redundant lines, separating circuits (high-voltage vs. telecom) onto organized crossarms, trays or underground ducts, and likely installing stronger or additional poles and anchors.
- **Gemini-2.5-Flash:** While it's technically possible to redesign the wiring for better alignment, the sheer volume and entanglement of wires in the image suggest that the current electric pole's capacity is severely overtaxed. Simply reorganizing the existing wires on the same pole would not address the underlying capacity issues or the safety hazards presented by such a dense and disorganized network.
- **InternVL3.5-72B:** Redesigning the extensive web of wires around the electric pole into a properly aligned and structured design is feasible but would likely require significant effort and resources. Proper planning and coordination with multiple service providers would be essential. Implementing such changes would enhance safety and aesthetics, but it involves logistical challenges and coordination to minimize service disruptions.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking if all the wires on the electric pole can be redesigned to be properly aligned in a structured way. A redesigned system is feasible but dependent on evaluating regulatory, technical, and safety constraints. Prioritizing organized routing for essential cables while removing redundant ones could create.

Figure 23: Some examples from the benchmark (Size and Scale: Scale Consistency) and evaluation results.

3564
3565
3566
3567
3568
3569
3570
3571
3572
3573
3574
3575
3576
3577
3578
3579
3580
3581
3582
3583
3584
3585
3586
3587
3588
3589
3590
3591
3592
3593
3594
3595
3596
3597
3598
3599
3600
3601
3602
3603
3604
3605
3606
3607
3608
3609
3610
3611
3612
3613
3614
3615
3616
3617

Size and Scale: Shadow-Size Projection



Question: If one light is rotated so that they are perpendicular to each other, how will the shadow look over the wall?

Options

1. It will cover almost all of the wall
2. The shadow will only take the size of the light frame as it's actual size
3. There won't be any shadow
4. Shadow will smaller than the actual light body

Gold MCQ Answer: 1

Gold Open ended Answer: It will cover almost all of the wall

MCQ Answers:

- **GPT-5-mini:** 4
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** If one lamp is rotated 90° so the two beams are perpendicular, you'll get two elongated oval patches of light on the wall at right angles to each other. Where the ovals overlap it will be brightest; outside them the wall will be darker, producing an L-shaped darker region and two perpendicular shadow lobes cast by the fixture.
- **Gemini-2.5-Flash:** If one light is rotated to be perpendicular to the other, the shadow cast on the wall would change significantly. The current shadows are likely elongated and somewhat parallel due to the lights' current orientation. With one light rotated, the shadows would intersect at a more acute angle, creating a more complex and possibly darker overlapping shadow where the light beams cross.
- **InternVL3.5-72B:** If one light is rotated so that the two lights are perpendicular to each other, the shadows on the wall will intersect at a right angle, creating intersecting patterns. The light beams will cast shadows in perpendicular directions, resulting in a cross-like effect where the shadows overlap. The intensity and shape of the shadows will depend on the orientation and angle of the lights relative to the wall.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The user is asking about rotating a light so that they're perpendicular to each other and how the shadow on the wall would look. If they're further away, the shadow would be smaller.

Figure 24: Some examples from the benchmark (Size and Scale: Shadow-Size Projection) and evaluation results.

3618
3619
3620
3621
3622
3623
3624
3625
3626
3627
3628
3629
3630
3631
3632
3633
3634
3635
3636
3637
3638
3639
3640
3641
3642
3643
3644
3645
3646
3647
3648
3649
3650
3651
3652
3653
3654
3655
3656
3657
3658
3659
3660
3661
3662
3663
3664
3665
3666
3667
3668
3669
3670
3671

Size and Scale: Perspective Distortion



Question: If similar shaped pipes are stacked together separately will it take up less space?

Options

1. Yes
2. No
3. Take more
4. Similar

Gold MCQ Answer: 1

Gold Open ended Answer: Yes

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 1
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 1

Open-ended Answers:

- **GPT-5-mini:** Yes - when like-shaped pipes are grouped you can pack them in their most efficient pattern ,which reduces wasted voids and the total footprint. The exact savings depend on orientation and dimensions; in some cases clever mixing or nesting of different shapes can beat separate stacks, but as a general rule same-shape grouping is more space-efficient.
- **Gemini-2.5-Flash:** Yes, if similarly shaped pipes are stacked together separately, it would likely take up less space. This is because organizing them by shape allows for more efficient packing and reduces wasted space between different forms. For example, square pipes can be stacked flush against each other, and circular pipes can be nested or arranged in a hexagonal pattern for optimal density.
- **InternVL3.5-72B:** Yes, stacking similar shaped pipes together separately will generally take up less space. Organizing them by shape allows for more efficient packing, reducing gaps and empty spaces between different sizes and types. This method optimizes the use of available storage area and improves accessibility.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's see. The question is asking if stacking similar shaped pipes separately would take up less space. Stacking similar-shaped pipes separately can potentially optimize space usage. When pipes of the same diameter and shape (e.g., all round or rectangular) are grouped together, they can be arranged more tightly, minimizing gaps between

Figure 25: Some examples from the benchmark (Size and Scale: Perspective Distortion) and evaluation results.

3672
3673
3674
3675
3676
3677
3678
3679
3680
3681
3682
3683
3684
3685
3686
3687
3688
3689
3690
3691
3692
3693
3694
3695
3696
3697
3698
3699
3700
3701
3702
3703
3704
3705
3706
3707
3708
3709
3710
3711
3712
3713
3714
3715
3716
3717
3718
3719
3720
3721
3722
3723
3724
3725

Size and Scale: Relative Size Comparison



Question: The ash-colored, thicker coiled spring will be surpassed by which coiled springs if all are kept in the standing position?

Options

1. Right bottom corner
2. Right top corner
3. Left top corner
4. Bottom mid spring

Gold MCQ Answer: 1

Gold Open ended Answer: Right bottom corner

MCQ Answers:

- **GPT-5-mini:** 1
- **Gemini-2.5-Flash:** 4
- **InternVL3.5-72B:** 1
- **Kimi-VL-A3B-Thinking-2506:** 2

Open-ended Answers:

- **GPT-5-mini:** The ash-colored thicker spring in the center will be shorter than the thin, taller silver spring just behind it (top-center) and the long, dark-gray spring at the right edge when all are stood upright. Those two coils extend above the ash-colored spring.
- **Gemini-2.5-Flash:** The ash-colored, thicker coiled spring, located near the center-right of the image, would be surpassed by the longer, thinner coiled spring positioned in the bottom right corner. Additionally, the two smaller, conical springs in the upper left quadrant would also be taller if fully extended and standing. These springs vary in height and coil density, indicating different potential applications.
- **InternVL3.5-72B:** The ash-colored, thicker coiled spring in the image will be surpassed in height by the two taller springs on its right. One of these taller springs is also ash-colored, while the other has a slightly darker hue. Both of these springs have more coils and are visibly longer when standing upright.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's try to figure out this question. The image shows a bunch of different coiled springs. The question is asking which coiled springs would be surpassed by the ash-colored, thicker one if all are kept in the standing position The thinner, tightly coiled springs (smaller in diameter or fewer coils) would be surpassed in height by the ash-colored, thicker spring when all are kept upright.

Figure 26: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

3726
3727
3728
3729
3730
3731
3732
3733
3734
3735
3736
3737
3738
3739
3740
3741
3742
3743
3744
3745
3746
3747
3748
3749
3750
3751
3752
3753
3754
3755
3756
3757
3758
3759
3760
3761
3762
3763
3764
3765
3766
3767
3768
3769
3770
3771
3772
3773
3774
3775
3776
3777
3778
3779

Size and Scale: Relative Size Comparison



Question: If the red bow is taken out of it's place and kept far away in the behind how will it look and what will be it's capacity comparing with it's other similar clips at front?

Options

1. Large, same
2. Same, less
3. Small, same
4. Small, large

Gold MCQ Answer: 3

Gold Open ended Answer: Small, same

MCQ Answers:

- **GPT-5-mini:** 3
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 3

Open-ended Answers:

- **GPT-5-mini:** If you move the red bow to the back it will appear smaller and more obscured by the bows in front because of perspective and overlap, and its red color will look less vivid from that angle. Physically nothing about the bow changes — its size, shape and clasp strength remain the same as the other similar clips in front, so its holding capacity is unchanged.
- **Gemini-2.5-Flash:** If the red bow were taken out and placed far behind, it would appear smaller and less distinct due to perspective. Its capacity, referring to its visual prominence or ability to hold hair, would remain the same as the other similar clips. However, its perceived capacity might diminish as it would no longer be in the foreground, drawing less attention.
- **InternVL3.5-72B:** If the red bow is moved to the back, it will be partially obscured by the other bows in the foreground, making it less prominent. Its size and capacity to hold hair will remain the same as the others, but visually, it won't stand out as much as it does currently when placed in the front.
- **Kimi-VL-A3B-Thinking-2506:** Okay, let's tackle this question. The user is asking about removing the red bow and placing it behind, then comparing its appearance and capacity to the other similar clips in front. Removing the red bow would reduce its central visual impact, making the remaining front-facing bows appear denser but less balanced. Its capacity (physical or visual space) in the front area would decrease, potentially limiting the number of similar bows that can be displayed

Figure 27: Some examples from the benchmark (Size and Scale: Relative Size Comparison) and evaluation results.

3780
3781
3782
3783
3784
3785
3786
3787
3788
3789
3790
3791
3792
3793
3794
3795
3796
3797
3798
3799
3800
3801
3802
3803
3804
3805
3806
3807
3808
3809
3810
3811
3812
3813
3814
3815
3816
3817
3818
3819
3820
3821
3822
3823
3824
3825
3826
3827
3828
3829
3830
3831
3832
3833

Depth & Occlusion : Complete Occlusion Inference



Question: What is creating the striped shadow on the wall?

Options

1. The sunflower
2. Blinds
3. Window
4. Not applicable

Gold MCQ Answer: 2

Gold Open ended Answer: Blinds

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** Sunlight streaming through window blinds (slatted shades) is casting the diagonal striped shadow on the wall. The slats break the light into parallel bands, producing the striped pattern.
- **Gemini-2.5-Flash:** The striped shadow on the wall is created by sunlight passing through a window with horizontal blinds. The blinds block portions of the light, casting distinct parallel lines onto the wall. This effect is common when direct sunlight enters a room through slatted window coverings.
- **InternVL3.5-72B:** The striped shadow on the wall is created by blinds or window slats. The sunlight is passing through the gaps between the horizontal blinds, casting parallel lines of light and shadow. This pattern is often seen when light is filtered through window coverings with narrow openings. The alignment and spacing of the lines suggest that the blinds are partially open, allowing the sun to project the shadow pattern onto the wall.
- **Kimi-VL-A3B-Thinking-2506:** The striped shadow on the wall is created by sunlight passing through a set of window blinds or shutters with narrow, evenly spaced slats. The parallel lines of the shadow result from the angular interaction between the light source (likely the sun) and the rigid, horizontal slats of the blinds, casting consistent geometric patterns onto the wall. The uniformity and sharpness of the shadows suggest the light is direct and possibly filtered through the gaps between the slats.

Figure 28: Some examples from the benchmark (Depth and Occlusion: Complete Occlusion Inference) and evaluation results.

3834
3835
3836
3837
3838
3839
3840
3841
3842
3843
3844
3845
3846
3847
3848
3849
3850
3851
3852
3853
3854
3855
3856
3857
3858
3859
3860
3861
3862
3863
3864
3865
3866
3867
3868
3869
3870
3871
3872
3873
3874
3875
3876
3877
3878
3879
3880
3881
3882
3883
3884
3885
3886
3887

Depth & Occlusion : Layering Order



Question: What is the frontmost object in the picture?

Options

1. Trolley
2. Box
3. Man
4. Woman

Gold MCQ Answer: 1

Gold Open ended Answer: Trolley

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 29: Some examples from the benchmark (Depth and Occlusion: Layering Order) and evaluation results.

3888
3889
3890
3891
3892
3893
3894
3895
3896
3897
3898
3899
3900
3901
3902
3903
3904
3905
3906
3907
3908
3909
3910
3911
3912
3913
3914
3915
3916
3917
3918
3919
3920
3921
3922
3923
3924
3925
3926
3927
3928
3929
3930
3931
3932
3933
3934
3935
3936
3937
3938
3939
3940
3941

Depth & Occlusion : Partial Occlusion



Question: What is the color of the ball behind the white colored square shaped device?

Options

1. Cyan
2. Yellow
3. Red
4. Not applicable

Gold MCQ Answer: 1

Gold Open ended Answer: Cyan

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The ball directly behind the white, square-shaped device is blue (a cyan/teal colored ball). It sits slightly to the left and behind the device.
- Gemini-2.5-Flash: The ball directly behind the white-colored square-shaped device (GameBoy) is teal. There is also a yellow-green ball further to the right of the GameBoy.
- InternVL3.5-72B: The ball behind the white, square shaped device is blue.
- Kimi-VL-A3B-Thinking-2506: The color of the ball behind the white square-shaped device is blue.

Figure 30: Some examples from the benchmark (Depth and Occlusion: Partial Occlusion) and evaluation results.

3942
3943
3944
3945
3946
3947
3948
3949
3950
3951
3952
3953
3954
3955
3956
3957
3958
3959
3960
3961
3962
3963
3964
3965
3966
3967
3968
3969
3970
3971
3972
3973
3974
3975
3976
3977
3978
3979
3980
3981
3982
3983
3984
3985
3986
3987
3988
3989
3990
3991
3992
3993
3994
3995

Depth & Occlusion : Reflective Surfaces



Question: Which object is the shiniest among all the objects in the image?

Options

1. Kettle
2. Tea glass
3. Table clock
4. Lamp chimney

Gold MCQ Answer: 4

Gold Open ended Answer: Lamp chimney

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: The frontmost object is a small orange street-cleaning cart with a blue plastic bin mounted on it. A broom is attached to the cart and it sits on the pavement in the foreground.
- Gemini-2.5-Flash: The frontmost object in the picture is the yellow cart with a blue crate on top, which a man in a yellow and black suit is sitting on. Its wheels are closest to the viewer, indicating its position in the foreground. The people sitting on the bench behind the cart are further back in the scene.
- InternVL3.5-72B: The frontmost object in the image is a cleaning cart with wheels. It has a blue container attached to it and a broom standing upright. There's also a noticeable yellow brush or shovel handle next to the container. The cart appears to be positioned on a sidewalk in an urban environment.
- Kimi-VL-A3B-Thinking-2506: The perspective is from the ground level, looking along the sidewalk. The cart is positioned in the foreground, closest to the viewer. The frontmost object is a high-visibility work cart.

Figure 31: Some examples from the benchmark (Depth and Occlusion: Reflective Surfaces) and evaluation results.

3996
3997
3998
3999
4000
4001
4002
4003
4004
4005
4006
4007
4008
4009
4010
4011
4012
4013
4014
4015
4016
4017
4018
4019
4020
4021
4022
4023
4024
4025
4026
4027
4028
4029
4030
4031
4032
4033
4034
4035
4036
4037
4038
4039
4040
4041
4042
4043
4044
4045
4046
4047
4048
4049

Depth & Occlusion : Transparency Effects



Question: How many objects has filaments inside them in the image?

Options

- 7
- 6
- 5
- 4

Gold MCQ Answer: 4

Gold Open ended Answer: 4

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: Three. The glowing bulb being held clearly shows a filament, and two more filament bulbs are visible inside the round box on the left.
- Gemini-2.5-Flash: There are three objects with filaments inside them. One is the illuminated light bulb being held, and the other two are unlit light bulbs inside the "Sunny Day Kit" box. All three are incandescent light bulbs, which contain a filament that glows when heated.
- InternVL3.5-72B: In the image, there are four objects with visible filaments. These include the lit light bulb being held, and three unlit bulbs inside the container labeled "Sunny Day Kit."
- Kimi-VL-A3B-Thinking-2506: There are 3 objects with filaments in the image - one held in hand and two in the "Sunny Day Kit" container.

Figure 32: Some examples from the benchmark (Depth and Occlusion: Transparency Effects) and evaluation results.

4050
4051
4052
4053
4054
4055
4056
4057
4058
4059
4060
4061
4062
4063
4064
4065
4066
4067
4068
4069
4070
4071
4072
4073
4074
4075
4076
4077
4078
4079
4080
4081
4082
4083
4084
4085
4086
4087
4088
4089
4090
4091
4092
4093
4094
4095
4096
4097
4098
4099
4100
4101
4102
4103

Orientation : Cardinal Direction



Question: If you go straight following route 6, which direction are you traveling?

Options

1. North
2. South
3. East
4. West

Gold MCQ Answer: 3

Gold Open ended Answer: East

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: You would be traveling east. The Route 6 shield is labeled "EAST" and the straight-ahead arrow beneath it shows that going straight follows eastbound Route 6.
- Gemini-2.5-Flash: If you go straight following route 6, you will be traveling East. The sign for Route 6 is directly below the "EAST" sign, and there is an arrow pointing straight up beneath the Route 6 shield, indicating that going straight leads to Route 6 East
- InternVL3.5-72B: If you go straight following Route 6, you are traveling east. This is indicated by the sign with the number "6" and the arrow pointing forward.
- Kimi-VL-A3B-Thinking-2506: Standard highway signs usually have the path of the route. So if it's route 11 East, the arrow would indicate the direction. But in the image, the left arrow is under the NORTH 11 EAST sign. Maybe that's indicating that route 11 East goes left? But the sign says East. Alternatively, maybe the arrows are indicating the direction of the route numbers. Like, route 11 North is going straight, route 11 East is turning left, etc. But then the route 6 sign has an upward arrow. So if you take route 6 straight, which direction is that? The arrow on the route 6 shield is pointing straight up. But the Question: is about direction—like North, South, East, West. Wait, the signs here are for routes, not cardinal directions. So maybe the answer is that going straight on route 6 would be in the direction indicated by the upward arrow. But the upward arrow might be part of the route 6 sign.

Figure 33: Some examples from the benchmark (Orientation: Cardinal Direction) and evaluation results.

4104
4105
4106
4107
4108
4109
4110
4111
4112
4113
4114
4115
4116
4117
4118
4119
4120
4121
4122
4123
4124
4125
4126
4127
4128
4129
4130
4131
4132
4133
4134
4135
4136
4137
4138
4139
4140
4141
4142
4143
4144
4145
4146
4147
4148
4149
4150
4151
4152
4153
4154
4155
4156
4157

Orientation : Facing Direction



Question: Suppose, Rabindranath Tagore's portrait is hung on the left wall and it faces directly into the room, what is the direction of the portrait with reference to the wall?

Options

- Up
- Down
- Away
- Toward

Gold MCQ Answer: 3

Gold Open ended Answer: Away

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 4

Open-ended Answers:

- GPT-5-mini: If the portrait is on the left wall and faces directly into the room, it is oriented perpendicular to that wall and pointing inward. In other words, it faces away from the wall toward the center of the room.
- Gemini-2.5-Flash: The portrait of Rabindranath Tagore is located on the wall to the left, as seen in the image. If it faces directly into the room, its direction with reference to the wall would be perpendicular to the wall. This means the portrait is facing straight out from the wall into the space of the room.
- InternVL3.5-72B: If Rabindranath Tagore's portrait is hung on the left wall and faces directly into the room, it would be positioned perpendicular to the wall. This means the portrait is oriented so that viewers inside the room can face it directly when looking toward the wall on which it is hung.
- Kimi-VL-A3B-Thinking-2506: The portrait is hung on the left wall and faces directly into the room, meaning it is oriented inward toward the interior space.

Figure 34: Some examples from the benchmark (Orientation: Facing Direction) and evaluation results.

4158
4159
4160
4161
4162
4163
4164
4165
4166
4167
4168
4169
4170
4171
4172
4173
4174
4175
4176
4177
4178
4179
4180
4181
4182
4183
4184
4185
4186
4187
4188
4189
4190
4191
4192
4193
4194
4195
4196
4197
4198
4199
4200
4201
4202
4203
4204
4205
4206
4207
4208
4209
4210
4211

Orientation : Object Rotation



Question: What is the approximate angle of the cases of the pocket watch?

Options

- 90
- 45
- 180
- 120

Gold MCQ Answer: 3

Gold Open ended Answer: 180

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: About 180°. The two covers are opened in opposite directions and lie nearly flat, forming an almost straight line.
- Gemini-2.5-Flash: The cases of the pocket watch are open, revealing the intricate mechanism inside. The front cover is open to approximately a 120-degree angle relative to the watch face. The back case is also open, but its angle is less clear, appearing to be around 90 degrees or slightly more.
- InternVL3.5-72B: The pocket watch in the image appears to be partially open, with the angle between its two cases estimated to be around 60 degrees. The case opens hinge-wise, and the visible gap suggests that it is made to display the internal mechanical components of the watch. This angle provides a good view of the watch face and some of its intricate gears..
- Kimi-VL-A3B-Thinking-2506: When the watch is open, the cases form an angle. If the covers are both open to 180 degrees relative to each other, that would be a straight line. But in this image, the back cover isn't fully open. It looks like it's at around 90 degrees from the front cover. So the angle between the cases would be approximately 90 degrees.

Figure 35: Some examples from the benchmark (Orientation: Object Rotation) and evaluation results.

4212
4213
4214
4215
4216
4217
4218
4219
4220
4221
4222
4223
4224
4225
4226
4227
4228
4229
4230
4231
4232
4233
4234
4235
4236
4237
4238
4239
4240
4241
4242
4243
4244
4245
4246
4247
4248
4249
4250
4251
4252
4253
4254
4255
4256
4257
4258
4259
4260
4261
4262
4263
4264
4265

Orientation : Stacking Orientation



Question: What is the relationship between the road and the other objects in the image?

Options

1. Behind the building and plants
2. In front of the plants and leads towards the building
3. The road is beside the building, with no clear direction
4. The road is stacked above the building and plants

Gold MCQ Answer: 2

Gold Open ended Answer: In front of the plants and leads towards the building

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- GPT-5-mini: The road runs in front of the large white building, curving around a central planted island and seating area. It separates and encircles the entrance and decorative planters, forming a driveway/roundabout in the courtyard.
- Gemini-2.5-Flash: The road in the image serves as an access path, leading towards the grand white building. It curves around a central island feature, which contains a decorative fountain and some potted plants. This suggests the road is a driveway or entrance loop designed to facilitate vehicle access to the building.
- InternVL3.5-72B: The road in the image forms a semi-circular driveway in front of a large white building with columns. It encircles a landscaped garden area with plants, a stone sculpture, and a fountain at the center, leading up to the building's entrance. The road is bordered by a curb that separates it from the garden and pedestrian space. Lamp posts are situated alongside the road, enhancing the building's stately appearance.
- Kimi-VL-A3B-Thinking-2506: The road curves around the front of the grand white building, acting as a transitional pathway that guides movement and attention toward the entrance while dividing space between the main structure and other elements like the pergola. It establishes scale and perspective while integrating with the surrounding landscape.

Figure 36: Some examples from the benchmark (Orientation: Stacking Orientation) and evaluation results.

4266
4267
4268
4269
4270
4271
4272
4273
4274
4275
4276
4277
4278
4279
4280
4281
4282
4283
4284
4285
4286
4287
4288
4289
4290
4291
4292
4293
4294
4295
4296
4297
4298
4299
4300
4301
4302
4303
4304
4305
4306
4307
4308
4309
4310
4311
4312
4313
4314
4315
4316
4317
4318
4319

Orientation : Tool Handedness



Question: Considering the likely process of creating the mural, which tool's handedness would be most evident in the smooth, curved lines of the bengali script?

Options

1. Paint roller
2. Paint brush
3. Tape measure
4. Stencil

Gold MCQ Answer: 2

Gold Open ended Answer: Paint brush

MCQ Answers:

- **GPT-5-mini:** 2
- **Gemini-2.5-Flash:** 2
- **InternVL3.5-72B:** 2
- **Kimi-VL-A3B-Thinking-2506:** 2

Open-ended Answers:

- **GPT-5-mini:** Those flowing, tapered curves are most characteristic of a paintbrush (or broad paint marker) rather than a spray can. The handedness of the brush user — most likely right-handed — would be most evident, since right-handed strokes tend to pull and taper in predictable left-to-right sweeps visible in the Bengali script.
- **Gemini-2.5-Flash:** The smooth, curved lines of the Bengali script suggest the use of a brush, likely a fine-tipped one, to achieve such fluidity. The artist's handedness would be most evident in the consistent direction and pressure applied to the brushstrokes.
- **InternVL3.5-72B:** In creating the mural, the smooth, curved lines of the Bengali script would most likely show the artist's handedness, particularly if the lines consistently curve more naturally in one direction. If the curves are more fluid and precise on one side, it could suggest that the artist is either right-handed or left-handed, depending on the direction of the dominant curves.
- **Kimi-VL-A3B-Thinking-2506:** If the artist is left-handed, they might use the brush in their left hand, leading to strokes that curve more naturally. Alternatively, the tool itself (like a specific brush tip) might be designed for left or right handed use.

Figure 37: Some examples from the benchmark (Orientation: Tool Handedness) and evaluation results.

4320
4321
4322
4323
4324
4325
4326
4327
4328
4329
4330
4331
4332
4333
4334
4335
4336
4337
4338
4339
4340
4341
4342
4343
4344
4345
4346
4347
4348
4349
4350
4351
4352
4353
4354
4355
4356
4357
4358
4359
4360
4361
4362
4363
4364
4365
4366
4367
4368
4369
4370
4371
4372
4373

Relative Positioning : Alignment Patterns



Question: Which of the objects is the closest to the centerline of the image?

Options

1. Rice
2. Nut
3. Avocado
4. Berries

Gold MCQ Answer: 4

Gold Open ended Answer: Berries

MCQ Answers:

- GPT-5-mini: 4
- Gemini-2.5-Flash: 4
- InternVL3.5-72B: 4
- Kimi-VL-A3B-Thinking-2506: 3

Open-ended Answers:

- GPT-5-mini: The red bowl of berries (raspberries and blueberries) is closest to the image's centerline. It sits almost directly in the middle of the composition, slightly left of the exact center..
- Gemini-2.5-Flash: The red bowl containing raspberries and blueberries is the closest object to the centerline of the image. Its position is almost perfectly centered horizontally, and it is located in the upper-middle section of the image vertically. While other objects like the avocado halves are also quite central, the red bowl's overall placement makes it the most aligned with the image's central axis.
- InternVL3.5-72B: In the image, the avocado halves are the closest to the centerline. One avocado is placed near the top center, and the other is slightly below it to the right, straddling the vertical centerline of the image.
- Kimi-VL-A3B-Thinking-2506: The bowl of mixed berries is closest to the centerline.

Figure 38: Some examples from the benchmark (Relative Positioning: Alignment Patterns) and evaluation results.

4374
4375
4376
4377
4378
4379
4380
4381
4382
4383
4384
4385
4386
4387
4388
4389
4390
4391
4392
4393
4394
4395
4396
4397
4398
4399
4400
4401
4402
4403
4404
4405
4406
4407
4408
4409
4410
4411
4412
4413
4414
4415
4416
4417
4418
4419
4420
4421
4422
4423
4424
4425
4426
4427

Relative Positioning : Betweenness Relationships



Question: What is the object between the building and the street lamp?

Options

1. A monument of various statues
2. Statue
3. Tree
4. Humans

Gold MCQ Answer: 1

Gold Open ended Answer: A monument of various statues

MCQ Answers:

- GPT-5-mini: 1
- Gemini-2.5-Flash: 1
- InternVL3.5-72B: 1
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- **GPT-5-mini:** Between the building and the street lamp stands a stone statue of a person. It appears to be one of the sculpted figures that are part of the fountain in the foreground.
- **Gemini-2.5-Flash:** The object located between the building and the street lamp is a large, ornate monument. This monument features multiple levels adorned with various statues and intricate carvings. At its very top, there is a prominent statue of a person.
- **InternVL3.5-72B:** The object between the building and the street lamp is a detailed stone monument or fountain. It features several statues, including a prominent figure standing on top and additional figures around its structure. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.
- **Kimi-VL-A3B-Thinking-2506:** The object between the building and the street lamp is a detailed stone monument or fountain. The monument is ornate, with decorative elements and possibly bas-reliefs on its sides, and it appears to be situated in an open, possibly public area.

Figure 39: Some examples from the benchmark (Relative Positioning: Betweenness Relationships) and evaluation results.

4428
4429
4430
4431
4432
4433
4434
4435
4436
4437
4438
4439
4440
4441
4442
4443
4444
4445
4446
4447
4448
4449
4450
4451
4452
4453
4454
4455
4456
4457
4458
4459
4460
4461
4462
4463
4464
4465
4466
4467
4468
4469
4470
4471
4472
4473
4474
4475
4476
4477
4478
4479
4480
4481

Relative Positioning : Corner/Angle Positioning



Question: What is in the top left corner of the plant?

Options

1. A lobster
2. A crab
3. An octopus
4. Nothing

Gold MCQ Answer: 2

Gold Open ended Answer: A crab

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: B

Open-ended Answers:

- GPT-5-mini: A crab is in the top left corner of the plant, perched on the rim of the fishbowl. It's hanging over the water with its claws visible.
- Gemini-2.5-Flash: The plant in the image is a type of seaweed or coral, and it is contained within a glass bowl. In the top left corner of this plant, there is a branch of the plant extending upwards and slightly to the left. This branch is a dark, greenish-brown color.
- InternVL3.5-72B: In the top left corner of the plant, there is a small crab that is perched on edge of the bowl, outside the water. It is gripping the rim with its legs, and its body is partially inside the bowl. The background shows the open water, providing contrast to the scene.
- Kimi-VL-A3B-Thinking-2506: The answer is the crab

Figure 40: Some examples from the benchmark (Relative Positioning: Corner/Angle Positioning) and evaluation results.

4482
4483
4484
4485
4486
4487
4488
4489
4490
4491
4492
4493
4494
4495
4496
4497
4498
4499
4500
4501
4502
4503
4504
4505
4506
4507
4508
4509
4510
4511
4512
4513
4514
4515
4516
4517
4518
4519
4520
4521
4522
4523
4524
4525
4526
4527
4528
4529
4530
4531
4532
4533
4534
4535

Relative Positioning : Directional Relations



Question: What is located directly behind the cheetah?

Options

1. Person
2. Tree
3. Safari vehicle
4. Cheetah

Gold MCQ Answer: 3

Gold Open ended Answer: Saffari vehicle

MCQ Answers:

- GPT-5-mini: 3
- Gemini-2.5-Flash: 3
- InternVL3.5-72B: 3
- Kimi-VL-A3B-Thinking-2506: 1

Open-ended Answers:

- GPT-5-mini: Directly behind the cheetah is a white open-sided tour vehicle or tram. Several people are seated inside the tram, watching and taking photos.
- Gemini-2.5-Flash: Directly behind the cheetah, there is a white open-sided vehicle, likely a safari or tour cart. Inside this vehicle, several people are seated, observing the cheetah. Some of these individuals are holding up cameras or phones, seemingly taking pictures or videos.
- InternVL3.5-72B: Directly behind the cheetah, there is a group of people sitting and standing around, appearing to watch or interact with the cheetah. They are positioned near a small covered vehicle, which looks like a safari cart or shuttle. The area is a grassy patch, and there are trees and a hill in the further background.
- Kimi-VL-A3B-Thinking-2506: People seated on benches under a canopy are located directly behind the cheetah.

Figure 41: Some examples from the benchmark (Relative Positioning: Directional Relations) and evaluation results.

4536
4537
4538
4539
4540
4541
4542
4543
4544
4545
4546
4547
4548
4549
4550
4551
4552
4553
4554
4555
4556
4557
4558
4559
4560
4561
4562
4563
4564
4565
4566
4567
4568
4569
4570
4571
4572
4573
4574
4575
4576
4577
4578
4579
4580
4581
4582
4583
4584
4585
4586
4587
4588
4589

Relative Positioning : Proximity Gradients

Question: Which book is closer to the book "american marriage" - "atonement" or "cloud atlas"?

Options

1. Cloud Atlas
2. Atonement
3. Crossed
4. nan

Gold MCQ Answer: 2

Gold Open ended Answer: Atonement

MCQ Answers:

- GPT-5-mini: 2
- Gemini-2.5-Flash: 2
- InternVL3.5-72B: 2
- Kimi-VL-A3B-Thinking-2506: 2

Open-ended Answers:

- **GPT-5-mini:** "Atonement" is closer to "american marriage" than "Cloud Atlas." "Atonement" sits just to the right with only one small book between them, while "Cloud Atlas" is further to the right on the shelf.
- **Gemini-2.5-Flash:** "Atonement" is closer to "An American Marriage" than "Cloud Atlas." "Atonement" is directly adjacent to "An American Marriage" on its right side. "Cloud Atlas" is further to the right, with "Americanah" separating it from "Atonement."
- **InternVL3.5-72B:** The book "atonement" is closer to "american marriage" than "cloud atlas." In the image, "atonement" is directly next to "american marriage," while "cloud atlas" is located further to the left..
- **Kimi-VL-A3B-Thinking-2506:** Cloud Atlas is closer to "American Marriage" than "Atonement."

Figure 42: Some examples from the benchmark (Relative Positioning: Proximity Gradients) and evaluation results.

K STATISTICAL ROBUSTNESS AND DATASET STABILITY

This appendix supports the benchmark contribution by testing whether reported results are stable rather than artifacts of random initialization or dataset noise. We evaluate model robustness across runs and dataset stability across repeated trials, because small- to medium-scale benchmarks can otherwise produce misleading fine-grained conclusions (Drummond, 2009; Drummond & Japkowicz, 2010).

Let N denote the number of items *questions* in the benchmark and $R = 3$ the number of independent runs of the same model. Each run uses a distinct random seed while preserving all other hyperparameters.

K.1 MODEL ROBUSTNESS

To validate the evaluation protocol, we perform several model-robustness studies.

K.1.1 MULTIPLE RUN AVERAGES AND DEVIATIONS

Let $a_{i,r} \in 0, 1$ denote the correctness of item $i \in 1, \dots, N$ in run $r \in 1, 2, 3$. The per-item average accuracy is:

$$\bar{a} * i = \frac{1}{R} \sum_{r=1}^R a_{i,r} \quad (1)$$

and its corresponding standard deviation is:

$$\sigma_i = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (a_{i,r} - \bar{a}_i)^2} \quad (2)$$

The overall mean accuracy per run is:

$$A_r = \frac{1}{N} \sum_{i=1}^N a_{i,r} \quad (3)$$

and its standard deviation across runs is:

$$\sigma_A = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_r - \bar{A})^2} \quad (4)$$

where $\bar{A} = \frac{1}{R} \sum_{r=1}^R A_r$ is the overall mean accuracy across all runs.

K.1.2 INTRA-CLASS CORRELATION (ICC)

To quantify run-to-run agreement, we compute the Intra-Class Correlation coefficient (Shrout & Fleiss, 1979), using a two-way mixed-effects model:

$$\text{ICC}(3, k) = \frac{\sigma_{\text{between}}^2}{\sigma_{\text{between}}^2 + \frac{\sigma_{\text{within}}^2}{k}} \quad (5)$$

where $\sigma_{\text{between}}^2$ is the variance between items and σ_{within}^2 is the residual variance across runs. ICC values above 0.75 indicate good reliability, while values above 0.9 imply excellent consistency (Koo & Li, 2016).

K.2 DATASET STABILITY AND INTERNAL CONSISTENCY

We also analyze dataset stability, asking whether question difficulty remains consistent across runs.

K.2.1 RESAMPLING STUDY

To further support the robustness claim, we perform a resampling analysis. From the total of $C = 30$ subcategories, we randomly sample subsets of size $S \in 20, 25$ for each sub-category from each of the $R = 3$ runs, across both models and evaluation types (MCQ and open-ended).

For each run r and subset size S , the accuracy is computed as:

$$A_{r,S} = \frac{1}{S} \sum_{i=1}^S a_{i,r,S}, \quad (6)$$

where $a_{i,r,S}$ denotes the accuracy for subcategory i in run r under subset size S .

The mean accuracy across runs for each subset size is:

$$\bar{A} * S = \frac{1}{R} \sum_{r=1}^R A_{r,S}, \quad (7)$$

and the corresponding standard deviation:

$$\sigma_S = \sqrt{\frac{1}{R-1} \sum_{r=1}^R (A_{r,S} - \bar{A}_S)^2}. \quad (8)$$

As a complementary check, the Wilcoxon signed-rank test assesses whether paired samples have different medians. Let $d_r = A_{r,20} - A_{r,25}$ be the difference in accuracies for run r , excluding ties ($d_r \neq 0$). The test statistic is computed as:

$$W = \min(W^+, W^-), \quad (9)$$

where W^+ and W^- are the sums of signed ranks of positive and negative differences, respectively. The standardized z -score is given by:

$$z = \frac{W - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}, \quad (10)$$

where n is the number of nonzero pairs.

If the resulting p -value satisfies $p > 0.05$, we again fail to reject H_0 , confirming that the accuracies from 20- and 25-sample subsets are statistically indistinguishable.

K.2.2 ITEM-LEVEL CONSISTENCY

For each item i , its difficulty can be defined as $d_i = 1 - \bar{a} * i$. If item difficulty is consistent, correlations between runs should be high:

$$\rho * r_1, r_2 = \frac{\text{Cov}(a_{\cdot, r_1}, a_{\cdot, r_2})}{\sigma_{r_1} \sigma_{r_2}} \quad (11)$$

A high mean pairwise correlation ($\bar{\rho} > 0.8$) suggests stable item difficulty and reliable dataset behavior.

K.2.3 CRONBACH'S ALPHA

To measure whether items operate as a coherent benchmark instrument, we compute Cronbach's alpha (Cronbach, 1951):

$$\alpha = \frac{R}{R-1} \left(1 - \frac{\sum_{r=1}^R s_r^2}{s_T^2} \right) \quad (12)$$

where s_r^2 is the variance of run r and s_T^2 is the variance of the total scores aggregated across runs. Values of $\alpha \geq 0.9$ indicate excellent internal consistency (Tavakol & Dennick, 2011).

K.3 EMPIRICAL RESULTS

We perform the robustness analysis on two models: a large proprietary model, **Gemini-2.5-Flash**, and a small open-source model, **Qwen-2.5-vl-7b-Instruct**. This pairing tests stability across model scale and access paradigm. We also conduct all tests in both multiple-choice (MCQ) and open-ended evaluation formats (treated as binary: correct or incorrect). For statistical analysis, we used the following Python packages and functions: `scipy.stats`² (Virtanen et al., 2020) for wilcoxon test; `pingouin`³ (Vallat, 2018) for intraclass correlation and cronbach alpha; and `numpy`⁴’s (Harris et al., 2020) `numpy.corrcoef` for correlations.

K.3.1 MODEL EVALUATION ROBUSTNESS

Multiple Run Averages and Deviations

We analyzed the stability of model accuracy across $R = 3$ independent runs. The standard deviation of accuracy (σ_A) across runs is negligible ($< 0.6\%$), indicating that the reported scores are precise estimates unaffected by random seeding noise.

Table 26: Model Accuracy and Variance across 3 Independent Runs.

Model	Format	Mean Accuracy (A)	Std Dev (σ_A)
Qwen-2.5-vl-7b	MCQ	0.4079	0.0021
Gemini-2.5-Flash	MCQ	0.5193	0.0054
Qwen-2.5-vl-7b	Open	0.1936	0.0047
Gemini-2.5-Flash	Open	0.3124	0.0042

The consistently low standard deviations across all models and evaluation formats indicate that performance estimates are highly stable and reproducible, with minimal sensitivity to stochastic run-to-run variation. This stability suggests that observed accuracy differences meaningfully reflect underlying model capability rather than noise introduced by random initialization or decoding. Notably, although open-ended evaluation exhibits slightly higher variance than MCQ, the absolute magnitude remains negligible, confirming that free-form generation is also robustly assessed under the current protocol. These findings imply that the dataset size and scoring methodology are sufficient to yield convergent accuracy estimates without requiring extensive repeated runs. Consequently, single-run evaluations are likely to be reliable for large-scale comparisons, while multi-run analyses primarily serve to validate robustness. Overall, this strengthens the empirical credibility of the reported results and supports confident cross-model and cross-format performance interpretation.

Intra-Class Correlation (ICC)

We computed ICC(3,k) to quantify the reliability of the scoring mechanism.

Table 27: Intra-Class Correlation (ICC) Reliability Scores.

Model	Format	ICC(3,k)	Interpretation
Qwen-2.5-vl-7b	MCQ	0.988	Excellent
Gemini-2.5-Flash	MCQ	0.990	Excellent
Qwen-2.5-vl-7b	Open	0.983	Excellent
Gemini-2.5-Flash	Open	0.981	Excellent

The consistently high ICC(3,k) scores demonstrate that the evaluation exhibits near-perfect reliability across independent runs, regardless of model size or response format. This indicates that item-level difficulty ordering is preserved, and that performance fluctuations do not alter relative rankings among samples. Such strong agreement suggests that the scoring pipeline is internally consistent and not sensitive to stochastic variation in model outputs. Importantly, this rules out random guessing as a dominant factor, even in open-ended generation, where variability is typically higher. From

²<https://scipy.org/>

³<https://pingouin-stats.org/build/html/index.html>

⁴<https://numpy.org/>

an evaluation standpoint, these results imply that the benchmark provides a dependable signal for assessing model capability. Consequently, longitudinal comparisons and fine-grained ablations can be conducted with high confidence in measurement stability.

Overall, the combined evidence from multiple-run variance analysis and ICC reliability demonstrates that the evaluation is highly robust to stochastic effects. Both absolute performance (low standard deviation) and relative performance (near-perfect ICC) remain stable across independent runs, confirming that results are not driven by random seeding or decoding noise. This robustness ensures that observed performance gaps across models and formats reflect genuine capability differences. As a result, the evaluation framework provides a reliable and reproducible basis for comparative analysis and benchmarking.

K.3.2 DATASET STABILITY AND INTERNAL CONSISTENCY

Resampling Study To ensure statistical reliability, this resampling procedure is repeated over **1,000 independent trials**, each time drawing new random subsets of size $S = 20$ and $S = 25$ from the available subcategories. Repeating the test at this scale stabilizes the distribution of accuracy differences and p -values, ensuring that conclusions are not driven by a particular random draw. The consistent failure to reject H_0 across all trials therefore provides strong evidence that the observed robustness is systematic rather than incidental.

Table 28: Resampling robustness study over 1,000 trials using the Wilcoxon signed-rank test ($S = 20$ vs. $S = 25$).

Model	Format	Mean p -value	% Trials $p > 0.05$	Conclusion
Qwen-2.5-vl-7b	MCQ	0.290	100%	No Sig. Diff.
Gemini-2.5-Flash	MCQ	0.285	100%	No Sig. Diff.
Qwen-2.5-vl-7b	Open	0.294	100%	No Sig. Diff.
Gemini-2.5-Flash	Open	0.294	100%	No Sig. Diff.

The resampling results provide empirical evidence that accuracy estimates have converged at relatively small subcategory sizes. The consistently high mean (p)-values and the absence of statistically significant differences across all 1,000 trials indicate that performance metrics are insensitive to moderate reductions in sample size. This stability holds uniformly across models and evaluation formats, suggesting that neither model scale nor output structure introduces additional sampling uncertainty. Importantly, these findings imply that the variance in accuracy is dominated by model behavior rather than data sparsity at the subcategory level. As a result, the current subcategory size (≈ 50) can be considered statistically redundant, offering a substantial safety margin beyond the minimum required for reliable estimation. This supports the validity of the benchmark design and suggests that future extensions could reallocate samples across categories without compromising statistical robustness.

Item-Level Consistency and Cronbach’s Alpha We measured the mean pairwise Pearson correlation ($\bar{\rho}$) between runs and Cronbach’s Alpha to assess internal consistency.

Table 29: Item-level consistency metrics supporting SPATIALAB as a stable measurement instrument.

Model	Format	Mean Pairwise $\bar{\rho}$	Cronbach’s α
Qwen-2.5-vl-7b	MCQ	0.966	0.988
Gemini-2.5-Flash	MCQ	0.972	0.990
Qwen-2.5-vl-7b	Open	0.950	0.983
Gemini-2.5-Flash	Open	0.946	0.981

The high mean pairwise correlations (> 0.94) demonstrate item-level stability, indicating that the relative difficulty of individual questions is preserved across independent runs. This consistency shows that model responses are systematically structured rather than subject to random variation, even under stochastic decoding. The similarly high Cronbach’s α values (> 0.9) further confirm that the dataset exhibits excellent internal coherence, with items contributing reliably to the overall accuracy measure. Notably, the slight reductions observed in open-ended formats are expected given their greater expressive flexibility, yet remain well within the range of excellent reliability.

Together, these results indicate that the benchmark functions as a unified and internally consistent measurement instrument. Consequently, model performance differences can be confidently attributed to true capability variation rather than instability at the item level.

K.3.3 OVERALL DISCUSSION

The robustness analysis contributes evidence that the evaluation protocol and dataset are stable, with **ICC and Cronbach's α scores consistently exceeding 0.98**, indicating near-perfect reliability and internal consistency. The **standard deviation of accuracy across runs remains negligible** ($< 0.6\%$), confirming that reported model performance is stable and not influenced by stochastic effects such as random seeding or decoding variability. In addition, the **resampling study** demonstrates that even substantially smaller subsets ($N = 20$ and $N = 25$) yield statistically indistinguishable accuracy estimates compared to the full set, establishing that the chosen subcategory size (~ 25) is well beyond the threshold required for convergence. Collectively, these results show that both absolute scores and relative performance rankings are stable across runs and sampling regimes. This statistical robustness ensures that fine-grained comparisons across models, formats, and subcategories are meaningful. Consequently, the granular analyses reported in the paper can be interpreted as genuine model-capability differences rather than artifacts of sampling variance or evaluation noise.